



KENYATTA UNIVERSITY

SCHOOL OF PURE & APPLIED SCIENCES

DEPARTMENT OF CHEMISTRY

SIR OTONDE MICHAEL AWITI

TABLE OF CONTENTS

SIR OTONDE MICHAEL AWITI

TABLE OF CONTENTS.....	ii
Introduction.....	v
The Unit Objectives or Aims	vi
Lecture 1	1
1.0 Atomic Structure:.....	1
1 Lecture 1 Objectives or Aims	1
1.0 Introduction.....	1
1.1 Classical View of Atomic Structure	2
1.2 Historical Development of Atomic Theory	5
1.2.1 Thomson's Plum Pudding or Raisin Bun Atomic Model (1856-1940).....	5
1.2.2 Rutherford's Nuclear Model of Atom (1871-1937)	6
1.2.3 The Nature of Electromagnetic Radiation	8
1.2.4 Plank's Quantum Theory –Quantized Energy and Photons	12
1.2.4 Hydrogen Spectrum.....	13
1.2.5 Bohr Model of Atom (1913).....	16
1.2.5.1 Failures of the Bohr Model	19
Lecture 2	21
2 Lecture 2 Objectives or Aims	21
2.0 Introduction.....	21
2.1.0 Quantum Model of an Atom	23
2.2 De Broglie Equation- Dual Nature of Matter.....	23
2.2.1 Heisenberg Uncertainty Principle.....	25
2.3 Wave-Like Property of an Electron	25
2.4 Schrödinger Wave Equation.....	27
2.5 Quantum Numbers and their Properties	29
2.5.1 Relationship between quantum numbers and atomic orbitals	31
2.5.2 Electron Coordinates	31
2.5.3 The Pauli Exclusion Principle	32
2.6 Atomic Wavefunctions.....	33
2.7 Pictorial Representation of Atomic Orbitals	38
2.8 Nodal Surfaces and Electron Density Contour Plots of Atomic Orbitals	43
2.9 Electronic Configuration	43
2.9.1 Energy Levels in Hydrogen and Other Atoms	44
2.9.2 Electronic Configurations.....	47
2.9.3 Aufbau Principle.....	47
2.9.4 Order of Orbital Filling in Poly-electronic Atoms:	48
2 SUMMARY	51
Lecture 3	53
3 Lecture 3 Objectives or Aims	53
3.0 Introduction.....	53
3.1.1 Modern Periodic Table - The Long Form	54
3.1.2 Classifying the Elements into the Periodic Table.....	55
3.2 Periodic Trends of Atomic Parameters	57
3.2.1 Shielding.....	58
3.3 Periodic Trend in Effective Nuclear Charge	59
3.4 Sizes of Atoms and Ions.....	60
3.5 Periodic Trend in Atomic Radii	61
3.6 Periodic Trend in Ionic Radii	62

3.7	Periodic Trend in the First Ionization Energy	63
3.8	Periodic Trend in Electron Affinity	67
3.9	Periodic Trend in Electronegativity	68
3.10	Common Oxidation State of the Elements	69
3	SUMMARY	70
Lecture 4		71
Lecture 4 Objectives or Aims		71
4.0	Introduction.....	71
4.0	Electronic Structure and Chemical Bonding.....	72
4.1	Lewis Theory.....	72
4.1.1	Lewis Symbols and Lewis Structures.....	73
4.1.2	Ionic bonding (transfer of electrons)	74
4.1.3	Covalent bonding (sharing of electrons).....	74
4.1.4	Lewis Structures and Resonance	75
4.2	Molecular Geometry and Bonding Theories	76
4.2.1	Valence Shell Electron Pair Repulsion Theory (VSEPR)	76
4.2.3	Covalent Bonding and Orbital Overlap	77
4.3	Valence Bond Theory (VBT).....	78
4.4	Molecular Geometry and Hybrid Orbitals	79
4.5	Hybridization Involving d Orbitals	85
4.6	Multiple Bonds and Orbital Overlaps	87
4.7	Resonance Structures and Hybrid Orbitals	94
Lecture 5		97
Lecture 5 Objectives or Aims		97
5.0	Introduction.....	97
5.0	Acid and Bases.....	97
5.1	Lewis Definition.....	99
5.2	Ionic Equilibrium in Aqueous Solutions.....	100
5.3	The Strength of the Acids and Bases	100
5.4	Quantitative Measurement of pH	101
5.5	The Dissociation of Water.....	103
5.6	Important Generalization on Acids and Bases	104
5.7	Dissociation Constant of Acids and Bases.....	105
5.8	Buffer Solutions	107
5.8.1	An Equation for Buffer Solutions.....	108
5.8.2	Calculating pH Changes in Buffer Solutions	109
5	SUMMARY	112
Lecture 6		113
Lecture 6 Objectives or Aims		113
6.0	Introduction.....	113
6.0	Radioactivity.....	113
6.1	Characteristic of Radioactivity.....	114
6.2	Types of Radiation	114
6.3	Effect of Radioactive Decay on Parent Nuclide.....	115
6.3.1	α -Particle Decay	115
6.3.2	β -Particle Decay	116
6.3.3	γ -Particle Decay.....	117
6.4	Properties of Radiation.....	118
6.5	Rate of Radioactive Decay.....	118
6.7.1	Nucleus Fission.....	121

6.7.2	Nucleus Fusion	121
6.7.3	Uses of Radio Isotopes	122
6	Summary.....	122
Lecture 7		123
7.0	Redox Reactions	123
Lecture 7	Objectives or aims	123
7.0	Introduction.....	123
7.1	Redox Reaction	123
7.2	How Can We Tell When Redox Reaction is taking Place?	124
7.3	Balancing Redox Reaction	125
7.3.1	Balancing Redox Reaction by Oxidation Number Method.....	125
7.3.2	Balancing Redox Reaction the Half-Equation Method	127
7	Summary.....	130

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI



Introduction

This is a core unit in chemistry at the undergraduate level for the Bachelor of Education as well as Bachelor of Science degrees. The unit will introduce you the various basic concepts in chemistry. It will to equip you with sufficient knowledge to enable you to comprehend and relate to other concepts encountered at a later stage, as you explore more in chemistry and other related sciences.

The course requires you to have a good understanding of secondary school level chemistry. This is because it **builds on the chemistry knowledge acquired at that level**. The study unit begins with historical development of atomic structure, modern atomic structure and electronic configurations, elements arrangement in the periodic table, bonding in molecules, the structure and shapes of molecules, acids and bases, radioactivity and balancing redox reactions. This unit does not involve practical lectures, rather it emphasizes on acquisition of theoretical concepts. Practical lectures in this field of study would require equipment and skills beyond the scope of this course. However, you may be required to perform a few practicals on volumetric analysis.

SIR OTONDE MICHAEL AWITI

You are advised to attempt self test questions and perform few activities provided in the text. The purpose of the self test questions and activities is to enable you to evaluate your understanding of thoroughly the concepts presented in the lectures. Some of the answers are provided though you are advised not look at them before you are through with answering. Try to be honest with yourself as you answer the questions.

The aim of writing this first introductory unit for students studying chemistry in the university is to teach chemistry to a learner in a way he/she will find meaningful, useful and easy to learn. We have attempted to present ideas in a down-to-earth manner to come up with a module which *“one can read in bed without a pencil”*.



The Unit Objectives or Aims

At the end of this unit you should be able to;

- (a) describe the atomic structure
- (b) explain qualitatively the solutions of Schrödinger equation
- (c) explain Pauli's Exclusion Principle and write electronic configuration
- (d) discuss the periodicity of the periodic table
- (e) identify the shapes of orbitals
- (f) discuss the various types of structures of molecules through their bonding.
- (g) explain the structure of simple molecules in terms of Lewis theory, Valence Shell Electron Pair Repulsion (VSEPR) Model, Valence Bonding Theory and Hybridization of Atomic Orbitals.
- h) explain the concepts of acids and bases and calculate the pH.
- i) describe the nuclear chemistry and radioactivity
- j) balance redox reactions
- k) use ideas, concepts and skills acquired in learning chemistry to solve problems in everyday life.

GENERAL

SIR OTONDE MICHAEL AWITI

There is no single prescribed textbook and you will be provided with comprehensive notes on the relevant subject matter. However, you may consult the list below of reference books for further information. Remember, what we do as lecturers will enable you to obtain your degree; what **you** do will make you a chemist.

Reference books may be obtained from the library and /or from your lecturer.

-SIR OTONDE MICHAEL AWITI



REFERENCES

Some Useful References and Recommended Reading:

- 1 Petrucci R. H. and Harwood W. S., “General Chemistry: Principles and Modern Applications”. 7th edition. Prentice Hall. New Jersey.
- 2 McMarry J. and Fay R. C., “Chemistry”. Prentice Hall. New Jersey.
- 3 Raymond Chang, “Essential Chemistry”. McGraw-Hill Companies, Inc. Toronto
- 4 Hill G. C. and Holman J. S., “Chemistry in Context” 4th edition 1995, Thomas Nelson and Sons Ltd Surry UK.
- 5 Any other college inorganic chemistry text book.
- 6 Internet Search WWW

SIR OTONDE MICHAEL AWITI

Lecture 1

1.0 Atomic Structure:



1 Lecture 1 Objectives or Aims

At the end of this unit you should be able to;

- (a) describe the historical development of atomic structure
- (b) described the nature of electrons, protons and neutrons
- (c) explain the Thomson model
- (d) describe the nature of electromagnetic radiation
- (c) explain Rutherford model
- (d) discuss the Bohr model and hydrogen spectrum



1.0 Introduction

This lecture describes the historical development of the classical view of atomic structure. It starts with earlier view of atoms and how the understanding of atom developed. In this lecture we will learn different classical views that were put forward in history, their success and failures.

A century ago, scientists believed that atoms were solid indestructible particles like minute rebounding balls. Since then, they have built up a great deal of evidence concerning the detailed structure of atoms.

SIR OTONDE MICHAEL AWITI

Experiments involving electrolysis suggested that certain compounds contained charged particles called **ions**. The formation of these ions from atoms could be explained by the loss or gain of negatively charged particles (**electrons**). This led to the idea that atoms consisted

of a small positive charged nucleus surrounded by negative electrons. Later, scientists discovered that the nucleus is composed of two kinds of particles-**protons** and **neutrons**. In this lecture, we shall consider historical development of theory of atomic structure.

SIR OTONDE MICHAEL AWITI



SELF-TEST QUESTIONS

- i) Define the words, atom, element and molecule.
- ii) Write briefly on Dalton's Atomic theory



1.1 Classical View of Atomic Structure

The classical view of atomic structure was constructed with the body of knowledge accumulated in physics over several centuries preceding the 20th century. This body of knowledge is called *classical physics*.

SIR OTONDE MICHAEL AWITI

Known facts:

Through experience, physicists have identified a long list of particles which make up the atoms. The most fundamental particles are:

- 1 Electrons
- 2 Protons
- 3 Neutrons

Electrons

Electrons were discovered by J. J. Thomson in 1897, when conducting experiment on conductivity of electricity of gases at very low pressure. He initially called them **cathode rays**.

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

The electron is negatively charged with a charge of $-1.602189 \times 10^{-19}$ C (Coulombs). For convenience the charge of atomic and sub-atomic particles are described as multiple of this value (also known as electronic charge). Thus the charge of electron is usually referred to as -1 .

The mass of electron $m_e = 9.109534 \times 10^{-31}$ kg
 $= 5.485 \times 10^{-4}$ amu

1 atomic mass unit (amu) $= 1.660565 \times 10^{-27}$ kg

Symbol of electron is e^{-1} .

SIR OTONDE MICHAEL AWITI

Protons

Protons were discovered by J. J. Thomson, when later conducting experiment on conductivity of electricity of gases at very low pressure. He initially called them **positive rays**

The protons has a charge of +1 electronic charge or $+1.602189 \times 10^{-19}$ C

The mass of proton $m_p = 1.672648 \times 10^{-27}$ kg
 $= 1.007276$ amu

Symbol p.

Neutrons

Electrons were discovered by Chadwick in 1932. From his experiments he showed that neutrons had almost the same mass as the protons. Chadwick was able to explain the difficulty concerning the relative atomic masses of hydrogen and helium and later on the idea of **isotopes**.

SIR OTONDE MICHAEL AWITI

Neutrons have no charge. They are electrically neutral

The mass of neutron $m_n = 1.674954 \times 10^{-27}$ kg
 $= 1.008665$ amu

Symbol n.

Atomic Mass Unit

The mass of an atom is extremely small. The units of mass used to describe atomic particle is the atomic mass unit or (amu). An atomic mass unit (amu) is equal $=1.660565 \times 10^{-27}$ kg or $=1.660565 \times 10^{-24}$ g

Hence mass of proton = 1.007276 amu

neutron = 1.008665 amu

electron = 5.485×10^{-4} amu

SIR OTONDE MICHAEL AWITI

- From this comparison, we can see that the mass of proton and neutron are nearly identical.
- The nucleus (protons plus neutrons) contain virtually all the mass of the atom.
- The electrons while equal and opposite in charge to the protons have only 0.005% mass.



SELF-TEST QUESTIONS

The five main proposals in Dalton's Atomic Theory of matter were:

- 1 All matter is composed of tiny indestructible particles called atoms.
- 2 Atoms cannot be created or destroyed.
- 3 Atoms of the same elements are alike in every way.
- 4 Atoms of different elements are different
- 5 Atoms can combine together in small numbers to form molecules.
 - a) In the light of modern knowledge about atoms, isotopes, molecules and atomic structure comment on the truth of each of these proposals
 - b) Why is Dalton's Atomic Theory still useful in spite of these limitations?
 - c) Describe the contributions of the following scientists to our knowledge of atomic structure: John Dalton, J. J Thomson, R. A. Mullikan, James Chadwick and Ernest Rutherford

SIR OTONDE
MICHAEL AWITI



ACTIVITY

Visit the library the physics section, get a book on Atomic and Nuclear Physics and read on classification of the *elementary particles*. You will be excited to learn of many particles , for example: boson, fermions, leptons, mesons, baryons, photons, hyperons, nucleons, pions, kaon, muon etc

NB/ You can get most of this information from internet.

SIR OTONDE MICHAEL AWITI



1.2 Historical Development of Atomic Theory

The challenge was how these sub-atomic particles are arranged in an atom. Different atomic models were proposed. The sections that follow describe some of the models that have been proposed.



1.2.1 Thomson's Plum Pudding or Raisin Bun Atomic Model (1856-1940)

SIR OTONDE MICHAEL AWITI

Thomson didn't know exactly how positive electric charge was distributed in an atom, so he considered the case that was easiest to describe mathematically. He developed a model in which a positive charge is uniformly distributed in an electron imbedded sphere in such a way that attraction of the positive charge just offsets the repulsion among the electrons. This somehow resembles raising pudding.

SIR OTONDE MICHAEL AWITI

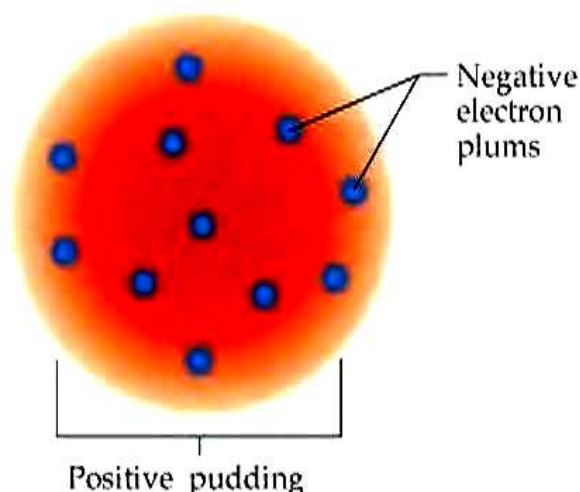


Figure 1.1. Thomson's plum pudding or raisin bun atomic model

However, this model was discarded as more information on atomic structure was discovered. Later on Rutherford proposed another Model.



1.2.2 Rutherford's Nuclear Model of Atom (1871-1937)

Rutherford used α -particles (alpha particles), doubly ionized helium atom, He^{2+} to probe the structure of matter. Based on Thomson's model of the atom Rutherford expected that most α particles would pass through atoms un-deflected. However, also he expected that any of the positively charged α -particles that come close to an electron should be deflected to some extent. By measuring such deflection he hoped to gain information about distribution of electron in an atom. Rutherford assigned his assistant, Hans Geiger and an undergraduate student Ernest Marsden an experiment to bombard a very thin foil of metal such as gold, silver and platinum with α -partilces. They found that

- i) Most of the particles went right through the foil undeflected
- ii) A few particles were deflected sharply and once in a while α -particles would bounce right back to the source.

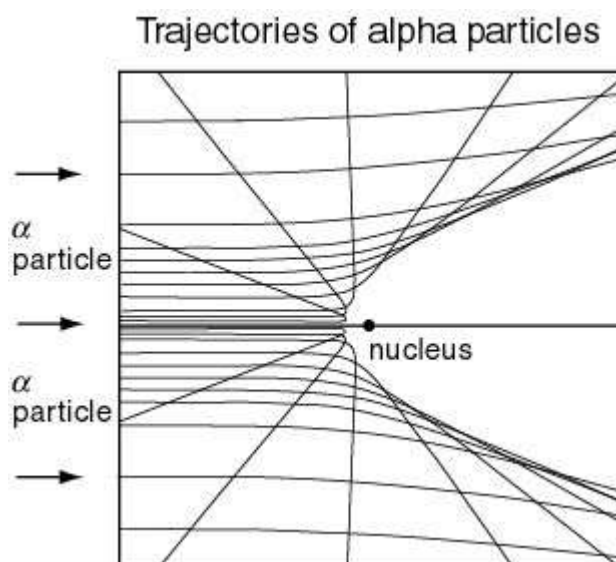


Figure 1.2 Angular Distribution of Rutherford Scattering

The deflection and bouncing back of α -particles was a big surprise to Rutherford. ***“It is about as incredible as if you had fired an inch shell at a piece of tissue paper and it came back and hit you”***.-Rutherford



SELF-TEST QUESTIONS

- i) Why do most of the α -particles pass through straight through the metal foil?
- ii) Why some α -particles do appears to rebound from the metal foil?
- iii) What factors will affect the extent of scattering of the α -particles?

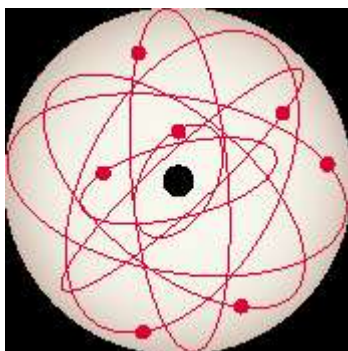


Figure 1.3 Rutherford planetary or nuclear atomic model

From these observations, Rutherford concluded that

- i) The mass of the atom and hence the positive charge was concentrated at the centre of an atom in a tiny core which he called the nucleus.
- ii) When the positively charged α -particles approached the positively charged nucleus it was repelled and therefore sharply deflected.
- iii) Electrons remained in the atom because they are strongly attracted to the positively charged nucleus.



ACTIVITY

Describe the experimental basis for believing that the nucleus occupies a very small fraction of the volume of the atom

For better understanding of the modern models, knowledge of wave mechanics is necessary. The next section reviews some aspects of wave, especially electromagnetic wave.



1.2.3 The Nature of Electromagnetic Radiation

Much of our current understanding of the electronic structure of atoms has come from analysis of the light emitted or absorbed by substances. To understand the basis for our current model of electronic structure, therefore, we must first learn more about light. The light that we can see with our eyes, *visible light*, is type of **electromagnetic radiation**. Electromagnetic radiation carries energy through space in the form of wave.

Electromagnetic radiation consists of packets (quanta) of oscillating electric and magnetic fields which are perpendicular to one another and traveling from the source in the form of a harmonic wave.

There are many types of electromagnetic radiation in addition to visible light. These different forms- such as the radio waves that carry music to our radios, the infrared radiation (heat) from the glowing fireplace, and X-rays used in hospital, may *seem* very different from one another, yet they share certain fundamentals characteristics.



ACTIVITY

List the types of the electromagnetic radiation, starting with the radiation having the longest wavelength and ending with the radiation having the shortest wavelength

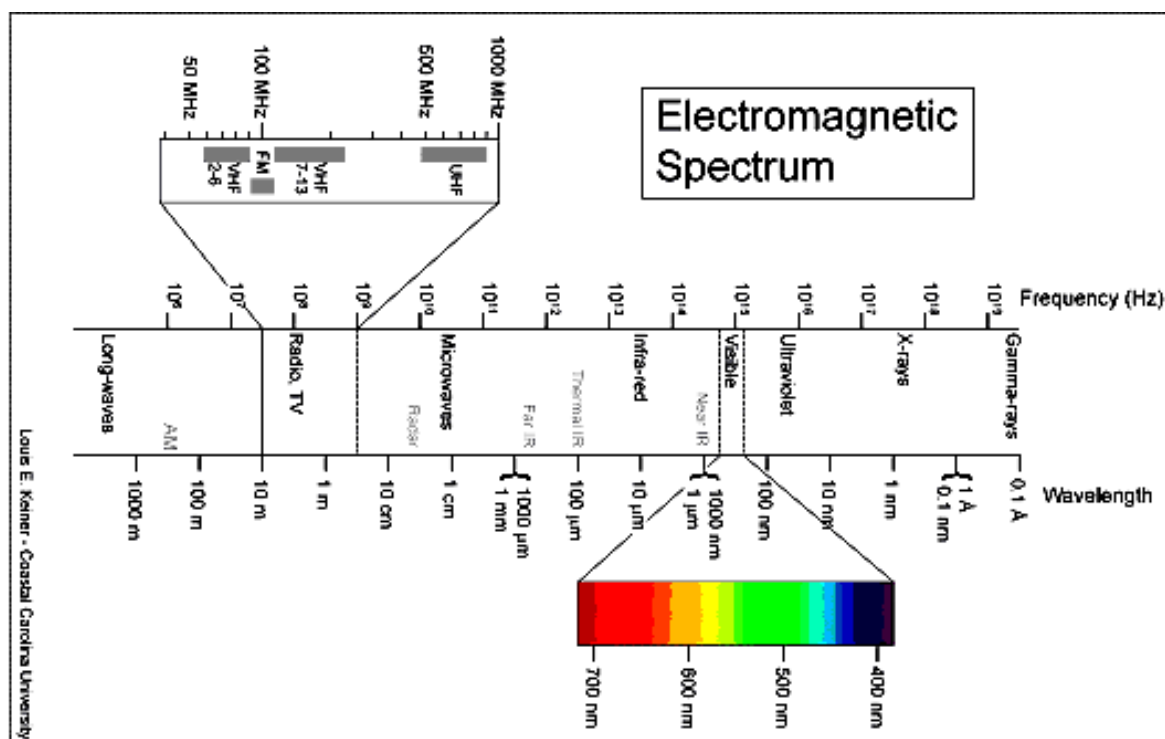


Figure 1.5 The Electromagnetic Spectrum

All types of electromagnetic radiation move through a vacuum at speed of $3.00 \times 10^8 \text{ ms}^{-1}$, the ‘speed of light’. Furthermore, they have wavelike characteristics.

SIR OTONDE MICHAEL AWITI



ACTIVITY

- i) List the colours in the visible light
- ii) Sketch a diagram of Newton's experiment that demonstrated that white light is a mixture of colours

We will review some terminology used in wave.

To understand quantum theory, we must know something about the nature of waves. A wave can be thought as a vibrating disturbance by which energy is transmitted. The speed of wave depends on the type of wave and the nature of the medium through which the wave is traveling (for example, air, water or a vacuum).

SIR OTONDE MICHAEL AWITI

The harmonic wave is characterized by properties such as frequency, ν wave length, λ and the wave number, λ^{-1} .

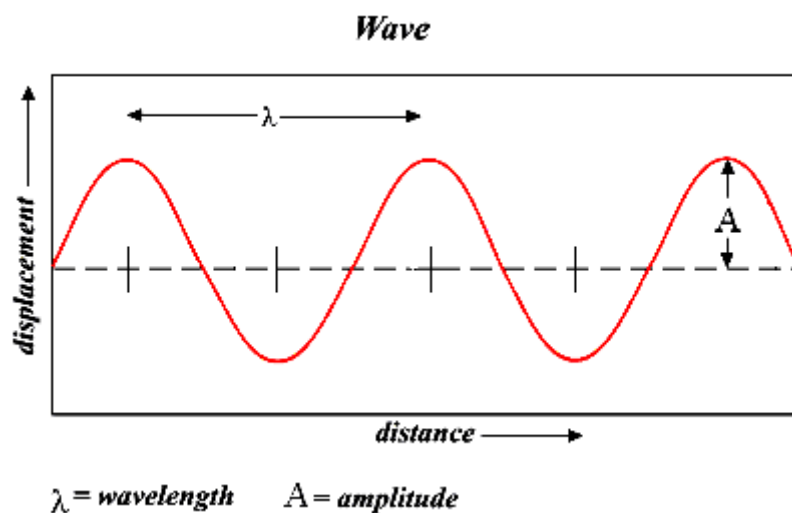


Figure 1.4 Harmonic wave

The wave length λ (Greek Lambda), is the distance between two successive crests or troughs in a wave and is expressed in meters, m. The frequency ν (Greek nu, pronounced as new) is

SIR OTONDE MICHAEL AWITI

the number of waves passing a point in unit time (usually one second). Its unit is the Hertz; Hz (i.e. cycles per second).

All electromagnetic radiation move through vacuum at a speed of $2.998 \times 10^8 \text{ ms}^{-1}$ which is called the speed of light c .

The wave length and frequency are related by:

$$c = \lambda \nu$$

where c is the velocity of the wave in the medium. In vacuum

$$c = 2.998 \times 10^8 \text{ ms}^{-1}.$$

Common wavelength units for electromagnetic radiation

Unit	Symbol	Length /m	Type of radiation
Angstrom	Å	10^{-10}	X-ray
Nanometer	nm	10^{-9}	Ultraviolet, visible
Micrometer	μ	10^{-6}	Infrared
Millimeter	mm	10^{-3}	Infrared
Centimeter	cm	10^{-2}	Microwave
Meter	m	10	TV, radio



SELF-TEST QUESTIONS

- What is the frequency of red light whose frequency is 750 nm?
- What is the wavelength in nm whose frequency is $4.5 \times 10^7 \text{ Hz}$?

The reciprocal of frequency is the period of oscillation, T . It indicates the time taken for one oscillation to pass a point. Hence,

$$T = \frac{1}{\nu}$$

The reciprocal of the wavelength, λ^{-1} , is known as the **wave number**. The wave number indicates the number of waves accommodated in unit length. Thus, since

$$c = \lambda \nu$$

then,

$$\frac{1}{\lambda} = \frac{\nu}{c} = \bar{\nu}$$

The SI unit for wave number is reciprocal metres, m^{-1} , although most of the literature values are given in cm^{-1} . The electric and the magnetic fields interact with matter leading to a spectrum.



1.2.4 Plank's Quantum Theory –Quantized Energy and Photons

Plank's quantum theory state that, atoms and molecules emit or absorb energy only in discrete quantities, like small packets or bundles.

The packets of electromagnetic radiation are known as **quanta** (singular: quantum) or photons. Each photon has energy, E , given by:

$$E = h \nu$$

where h = Planck's constant = $6.6262 \times 10^{-34} \text{ J s photon}^{-1}$.

ν = frequency of wave in Hertz

According to Plank's quantum theory, energy is always emitted in whole number multiples of $h \nu$, for example $2h \nu, 3h \nu, 4h \nu \dots$, but never fractions

SIR OTONDE
MICHAEL AWITI



SELF-TEST QUESTIONS

- 1 For radiation having $\lambda = 2.0 \times 10^{-7} \text{ m}$, calculate:
(i) E (kJmole^{-1})

(ii) v

(iii) λ^{-1}

- 2 Explain why X-ray is more energetic than microwave radiation
- 3 Briefly explain Plank's quantum theory.
- 4 Give two familiar, everyday example to illustrate the concept of quantization

SIR OTONDE MICHAEL AWITI



1.2.4 Hydrogen Spectrum

When sodium chloride is heated strongly in a Bunsen, it gives off a brilliant yellow light. Some other also emits light when they are heated in this way. Have ever noticed that compounds of one particular element always emit the same color of light when they behave like this?



SELF-TEST QUESTIONS

- 1 What colour of light is emitted by:
 - i) Potassium compounds
 - ii) Calcium compounds
 - iii) Copper compounds
- 2 What are emission spectra?
- 3 Distinguish line and continuum spectra

SIR OTONDE MICHAEL
AWITI

Some gaseous materials emit light when they are subjected to large potential differences in electric discharge tubes. Neon advertising signs work in this way and the yellow sodium street lamps are discharge tubes containing sodium vapour.

SIR OTONDE MICHAEL AWITI

If the light emitted by these substances is examined using spectroscope, it is found not to consist of a continuous range of colours like of spectrum of white light or colours in a rainbow. Instead, the light emitted by these substances is composed of separate lines of different colour. This kind of spectrum is called a line. Each element display a characteristic spectrum as first pointed out by Kirchhorff in 1859. The law governing the distribution of lines in a spectrum was discovered by Balmer and Rydbeg. The interpretation of this law in terms of atomic structure began with the work of Bohr in 1913. It revealed that a study of spectra could provide a wealth of information about the atomic structure. The study of these spectra has provided a key to the interpretation of many of the spectra of more complex atoms.

In the visible region, the spectrum of hydrogen was known to consist of 4 main lines:

Red 656.3 nm
 Blue 486.1 nm
 Violet 434.0 and 410.2 nm

Denoted as H_{α} , H_{β} , H_{γ} and H_{δ} .

Near the ultra violet there were other lines getting closer together converging to a limit of 364.6 nm

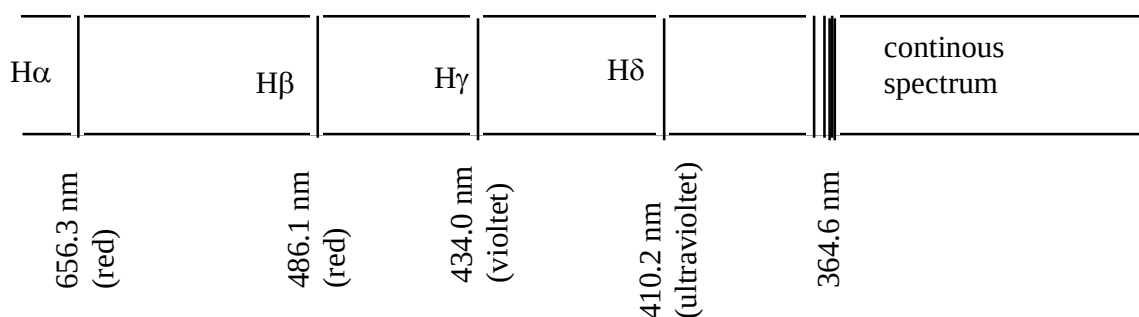


Figure 1.6 The Balmer series of atomic hydrogen excited by electric discharge. The dark lines correspond to the bright emission lines of the spectrum.

SIR OTONDE MICHAEL AWITI



SELF-TEST QUESTIONS

Explain why lines getting closer together converging to a limit and continuum

It was discovered by Balmer as early as 1885, that the wavelength (λ) of the nine lines, then known in this spectrum, could be accurately expressed by a single formula;

$$\lambda = \frac{kn^2}{n^2 - 2^2} \quad (1)$$

Where k is a constant and n is an integer. The different wavelength corresponding to the lines H_α , H_β , H_γ and H_δ etc. are obtained by putting the integers 3, 4, 5, 6 etc respectively in place of n . This set of line constitutes a spectral series and is known as the **Balmer series**.

Expressing in terms of wave numbers, equation (1) is rearranged as

$$\bar{\nu} = \frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \quad (2)$$

Where R is a constant called the Rydberg constant; $R=109677.8 \text{ cm}^{-1}$.

n was an integer which can take all values greater than 2, $n=3,4, 5 \dots \infty$.

$\bar{\nu}$ - the wave number, i.e. the number of wavelength in unit length.

The hydrogen spectrum was investigated in the far (UV) ultra violet, IR (infra red) regions and revealed that there were few other series such as Lyman (UV region), Paschen and Brackett and Pfund.

All these series were found to be in excellent agreement with the general equation

$$\bar{\nu} = \frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (3)$$

This is the universal formula for the atomic hydrogen spectrum.

Where n_1 and n_2 are integers somehow defining the energy stages of the atom.

The atomic spectrum of hydrogen atom

Series	Region	n_1	n_2
Lyman	Ultraviolet	1	2, 3, 4, 5..... ∞
Balmer	Visible	2	3, 4, 5, 6..... ∞
Paschen	Near infrared	3	4, 5, 6, 7..... ∞
Brackett	Far infrared	4	5, 6, 7, 8..... ∞
Pfund	Far infrared	5	6, 7, 8, 9..... ∞

Though equation (3) represented with great precision the entire known spectrum of atomic hydrogen, it was however an empirical formula described from experiment. The interpretation of this correct but un-derived formula was first given by theory of Niels Bohr (1913).

SIR OTONDE MICHAEL AWITI



1.2.5 Bohr Model of Atom (1913)

In order to explain the hydrogen spectrum, Bohr made the following assumptions governing the behaviour of electrons:

1. Electrons revolve in orbits of specific radius around the nucleus without emitting the radiation.
2. Within each orbit, each electron has a fixed amount of energy; electrons in orbits further from the nucleus have greater energies.
3. An electron may 'jump' from one orbit of high energy to another of lower causing the energy difference to be emitted as a photon of electromagnetic radiation such as light.
4. An electron may absorb a photon of radiation and jump from a lower-energy orbit to a higher-energy one.

The energy emitted or absorbed corresponds to the difference in the energy for the initial and final state of the system.

$$E_{final} - E_{initial} = \Delta E = h\nu$$

Of the entire possible orbit only certain ones were acceptable – namely which had a specified angular momentum. He assumed circular electron orbit with quantized electronic angular momentum.

SIR OTONDE MICHAEL AWITI

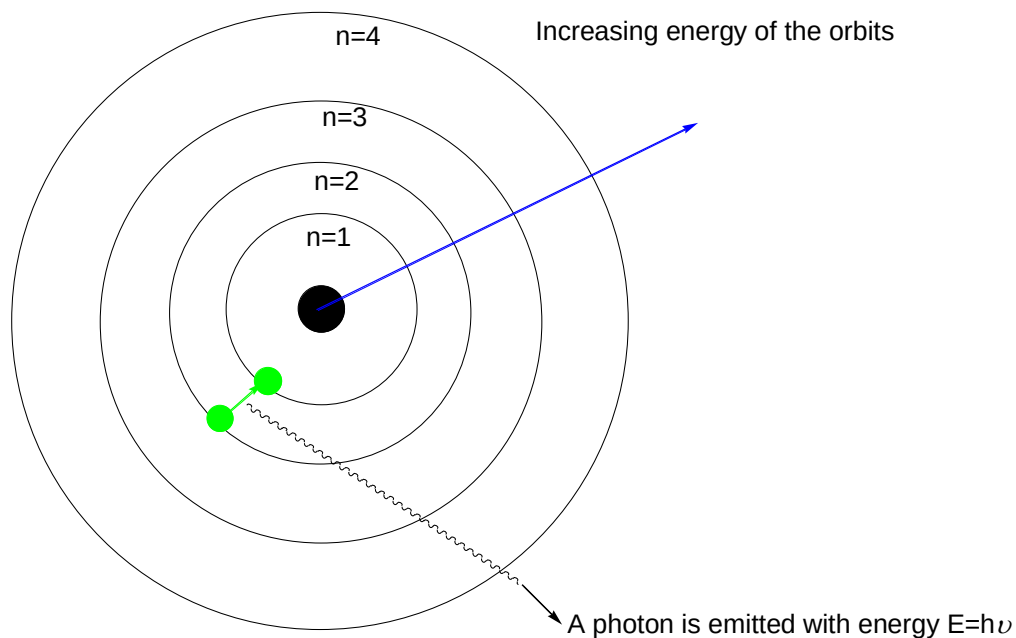


Figure 1.7. Bohr atomic model

One of the implication of these quantized energy states is that only certain photon energy are allowed when electrons jump down from high to lower levels producing the hydrogen spectrum.

SIR OTONDE MICHAEL AWITI

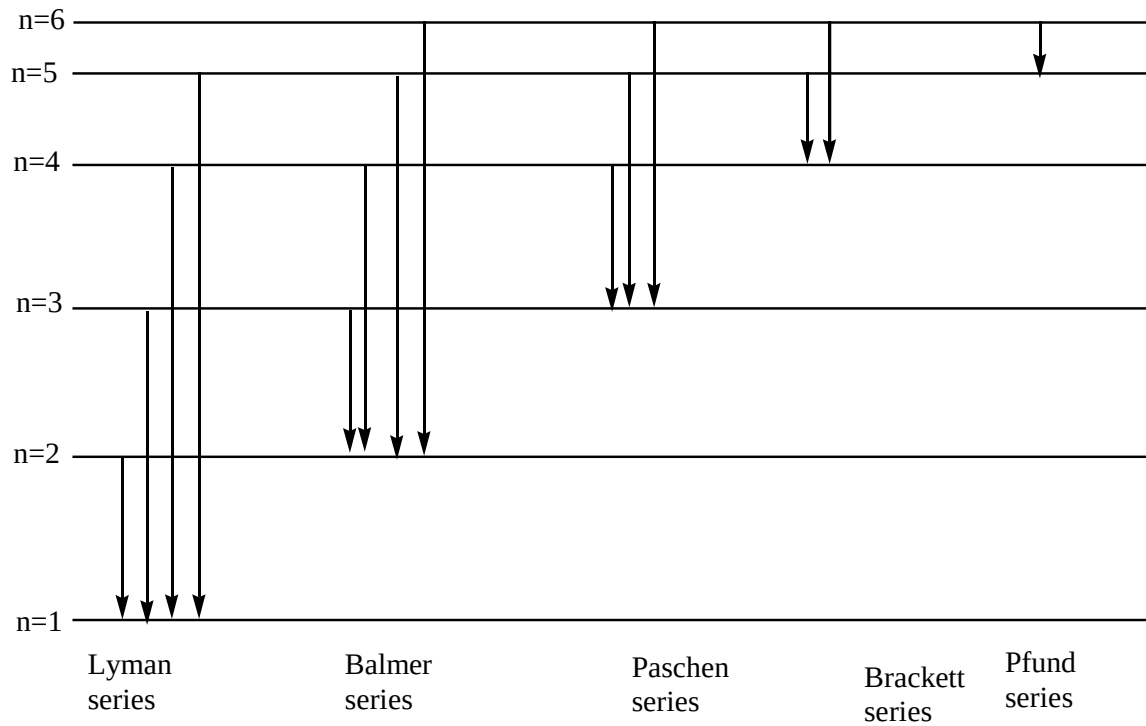


Figure 1.8. Various series in hydrogen spectrum.

For hydrogen the energy of electron in a n energy level is given by:

$$E_n = -\frac{Z^2 m e^4}{8h^2 \epsilon_0^2} \frac{1}{n^2} = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J}$$

or

$$E_n = -\frac{Z^2 m e^4}{8h^2 \epsilon_0^2} \frac{1}{n^2} = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

Where

m-mass of electron $9.1079 \times 10^{-31} \text{ kg}$

e-electronic charge $1.602 \times 10^{-19} \text{ C}$

h-Plank's constant $6.626 \times 10^{-34} \text{ J s}$

ϵ_0 -permittivity of free space $8.854188 \times 10^{-12} \text{ C}^2 \text{ s}^2 \text{ kg}^{-1} \text{ m}^{-3}$

Z-nuclear charge

$$\Delta E = E_{n_2} - E_{n_1}$$

SIR OTONDE MICHAEL AWITI

$$\Delta E = -2.18 \times 10^{18} Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ J}$$

or

$$\Delta E = -13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$$



SELF-TEST QUESTIONS

- i) Calculate the frequency and wavelength of the emitted photon when electron drops from the $n=5$ to the $n=2$ level in a hydrogen atom
- ii) Derive the expression $E_n = -\frac{Z^2 m e^4}{8 h^2 \epsilon_0^2} \frac{1}{n^2}$
- iii) Explain why elements produce their own characteristic colours when they emit photons



1.2.5.1 Failures of the Bohr Model

While the model was a major step towards understanding the quantum theory of the atom, it is not in fact a correct description of the nature of electron orbit.

Some of the shortcomings of the models are:-

1. It fails to provide any understanding why some spectral lines are brighter than others.
2. There is no mechanism for the calculation of transition probability.
3. The Bohr model treats electrons as if they were miniature planet with definite radius and momentum. This is a direct violence of the Heisenberg uncertainty principle which dictates that position and momentum cannot be simultaneously determined.

The Bohr model gives us a basic conceptual model of electrons orbit and energies. The precise details of spectra and charge distribution must be left to quantum mechanical calculations as with the Schrödinger equation.



ACTIVITY

- 1 What is an energy level? Explain the difference between a ground state and an excited state
- 2 Explain the meaning of the negative sign in the Bohr's equation
- 3 How does Bohr's theory differ from concepts of classical physics?

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

Lecture 2



2 Lecture 2 Objectives or Aims

At the end of this unit you should be able to;

- (a) explain the dual nature of matter and wave
- (b) describe qualitatively the Schrödinger equation and the quantum numbers
- (c) draw the shapes of atomic orbitals
- (d) explain the Pauli's exclusion principle and write electronic configuration



2.0 Introduction

SIR OTONDE MICHAEL AWITI

In this lecture, we will see that to explain phenomena at atomic and molecular level, classical physics is inadequate, only the quantum theory will do. This Lecture describes the current quantum model of an atom. Though, the concept requires knowledge of some mathematics and physics, we will explain it, without rigorously solving those equations.

In high school you learnt electronic configuration of atoms.

- The electrons are arranged in energy levels, labeled $n=1, 2, 3, 4$ etc.
- That the maximum number of electrons in each energy level is given by $2(n)^2$.
 - The first energy level can have a maximum of $2(1)^2=2$
 - The second energy level can have a maximum of $2(2)^2=8$
 - The third energy level can have a maximum of $2(3)^2=18$
 - And so on

No explanation was given at that time.

The table below gives the electronic configuration of the first twenty elements

Atom	Atomic number	Electronic configuration
H	1	1
He	2	2
Li	3	2 1
Be	4	2 2
B	5	2 3
C	6	2 4
N	7	2 5
O	8	2 6
F	9	2 7
Ne	10	2 8



ACTIVITY

SIR OTONDE MICHAEL AWITI

Using the knowledge learnt in high school write the electronic configuration of the next ten elements (i.e. sodium to calcium)

Atom	Atomic number	Electronic configuration
Na	11	
Mg	12	
Al	13	
Si	14	
P	15	
S	16	
Cl	17	
Ar	18	
K	19	
Ca	20	

We will now learn a more detailed method of writing electronic configuration of elements.



2.1.0 Quantum Model of an Atom

Our current understanding of the electronic structure of an atom is expressed in terms of quantum mechanics. A fundamental concept of quantum mechanics is that all matter has wave and particle-like properties.



2.2 De Broglie Equation- Dual Nature of Matter

Physicists were mystified but intrigued by Bohr's theory. They questioned why is the electron in Bohr atom restricted to orbiting the nucleus at fixed distances? In 1924 the French physicist Louis de Broglie provided a solution to this puzzle. De Broglie reasoned as follows: If light waves can behave like a stream of particles (photons), then perhaps particles such as electrons possess wave properties.

- Light can behave both as a wave and as particles (photons). Likewise all matter (e.g., electrons, protons & neutrons) can also behave as waves.

From Einstein relativity theory, the fundamental law relating energy E , rest mass m_0 and momentum p of a particle is

$$\left(\frac{E}{c}\right)^2 = p^2 + m_0^2 c^2$$

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

The rest mass of a photon is zero, so $m_0=0$ and therefore momentum is $p = \frac{E}{c}$

From Plank's quantum theory

$$E = h\nu$$

Therefore $E = h\nu = mc^2$

But $p = mc$

$$p = mc = \frac{h\nu}{c} = \frac{h}{\lambda}$$

$$\text{Implying } \lambda = \frac{h}{p}$$

λ = wavelength of the moving particle;

h : Planck's constant 6.626×10^{-34} J.s

m : mass of the moving particle.

That is, all particles in motion have a wavelength that depends on the particle's momentum.
The larger the momentum, the smaller the wavelength.

•In 1927, Davisson and Germer at Bell Laboratories showed that when a beam of electrons was directed at a nickel crystal, a diffraction pattern was observed.

SIR OTONDE MICHAEL AWITI



2.2.1 Heisenberg Uncertainty Principle

Due to the dual nature of matter, both particles and wavelike properties, it is impossible to simultaneously determine both the position and momentum of an object especially as small as an electron. The position and momentum of a particle cannot be simultaneously measured with high precision. There is a minimum for the product of the uncertainties of these two measurements. There is likewise a minimum for the product of the uncertainties of the energy and time.

$$\Delta x \Delta p \geq \frac{h}{2\pi}$$

Where Δx = uncertainty in position

Δp = uncertainty in momentum

This is not a statement about the inaccuracy of measurement instruments, nor a reflection on the quality of experimental methods; it arises from the wave properties inherent in the quantum mechanical description of nature. Even with perfect instruments and technique, the uncertainty is inherent in the nature of things.



2.3 Wave-Like Property of an Electron

An electron in an atom is viewed as a standing wave. Some common examples of standing waves are: a string attached at both ends to produce a musical tone in a guitar and violin.

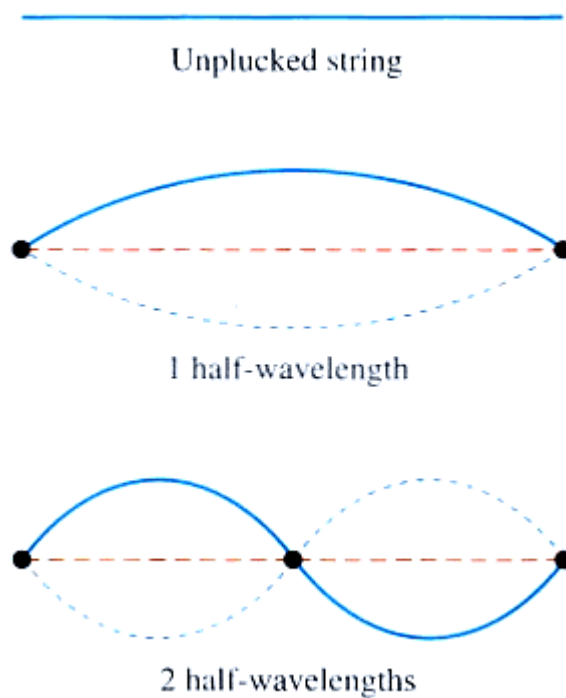


Figure 2.1 standing wave

We use the mathematical function Ψ (Greek Psi), called wavefunction to describe the wave-like behavior of an electron in a region of space called the atomic orbital.

- Ψ has positive and negative amplitudes, like any wave.

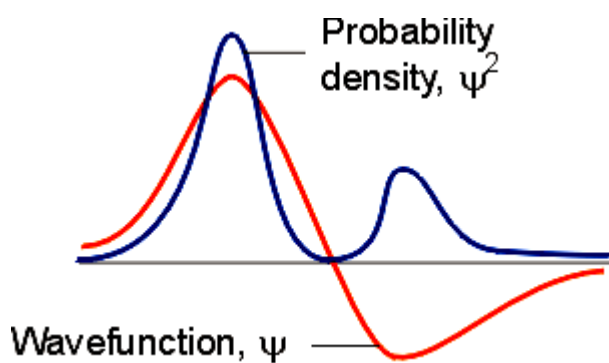


Figure 2.2

- Due to the wave-like behavior of an electron, we cannot pin down its position and momentum at the same time (Heisenberg Uncertainty Principle).
- In other words, we do not know the **exact location** of an electron and how it moves from one spot to another in an atom.

- Instead, we can only **speak of the probability of finding an electron in a given volume of space**. This is given by the function Ψ^2 .



2.4 Schrödinger Wave Equation

- Based on the wave-particle dual nature of electron, Schrödinger developed a partial differential equation to describe the behavior of an electron around an H atom. He received Nobel Prize in 1933.

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

Schrödinger equation for a hydrogen atom

- This equation shows the relationship between the wave function of the electron Ψ , and E and V , the total and potential energies of the system, respectively. The derivation of this equation is beyond the scope of this course (will be derived in SCH 200).
- There are many solutions to the Schrödinger's equation. Each solution is represented by the wavefunction, Ψ , which describes an atomic orbital. The Uncertainty Principle tells us that we cannot pin down the exact location and momentum of an electron at the same time. So chemists speak of "electron cloud" or electron density around a nucleus.
- An atomic orbital is defined as the boundary surface encloses 95% of the electron density for a particular wavefunction Ψ . Each atomic orbital (Ψ) may be uniquely defined by a set of three quantum numbers:

- n (the principal quantum number);
- l (the orbital angular quantum number) and
- m_l (the magnetic quantum number).

- Since electron behaves as if it is spinning about an axis, a spin quantum number m_s is also required to uniquely described an electron in an atomic orbital.

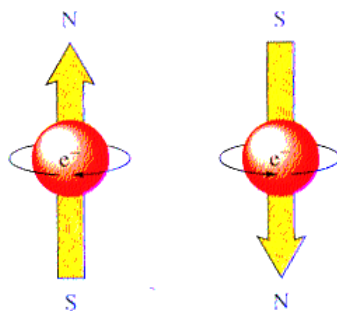


Figure 2.2 Spinning electron

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

2.5 Quantum Numbers and their Properties

Symbol	Name	Allowed Values	Properties
N	Principal Quantum Number	$n = 1, 2, 3, 4, \dots$	- determines the size and energy of an atomic orbital - As n increases, the number of allowed orbitals increases, the size and energies of those orbitals also increase.
L	Orbital Quantum Number	$l = 0, 1, 2, \dots, n-1$	Describe the shape of an atomic orbital
m_l	Magnetic Quantum Number	$m_l = -l, \dots, -(l-1), 0, 1, 2, \dots, +l$	Describe the directionality of an atomic orbital
m_s	Spin Quantum Number	$m_s = -\frac{1}{2}, +\frac{1}{2}$	Describes the orientation of the electron spin in space

Orbitals with different l values are known by the following labels:

L	0	1	2 3 4 5 ...
Label	s	p	$d f g h \dots$

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

Allowed values enable us to know the maximum number of electrons in each energy level

n	<i>l</i>	m_l and m_s					
1	0	0					
		$\uparrow\downarrow$					2 electrons in 1s orbital
							Total of 2 electrons in the first energy level
2	0	0					
		$\uparrow\downarrow$					2 electrons in 2s orbital
	1	-1	0	1			
		$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$			
							Total of 8 electrons in the first energy level; 2 in s orbital and 6 in p orbitals
3	0	0					
		$\uparrow\downarrow$					2 electrons in 3s orbital
	1	-1	0	1			
		$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$			
							6 electrons in 3p orbital
	2	-2	-1	0	1	2	
		$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	10 electrons in 3d orbital
							Total of 18 electrons in the first energy level; 2 in s orbital, 6 in p orbitals and 10 in d orbitals



ACTIVITY

Using the same argument, show the maximum number of electrons for $n=4$.

SIR OTONDE MICHAEL AWITI



2.5.1 Relationship between quantum numbers and atomic orbitals

There is a relationship between the quantum numbers and the atomic orbitals. When $l=0$, $(2l+1)=1$, and there is one value of m_l ; thus we have an s orbital. When $l=1$, $(2l+1)=3$, so there are three values of m_l ; or three p orbitals, labeled p_x , p_y , and p_z . When $l=2$, $(2l+1)=5$, and there are five values of m_l , and corresponding five d orbitals are labeled with more elaborate subscripts. This is shown in the table 2.5.1 below.

n	<i>l</i>	m_l	Number of orbitals	Atomic orbitals Designation
1	0	0	1	1s
2	0	0	1	2s
	1	-1 0 1	3	$2p_x, 2p_y, 2p_z$
3	0	0	1	3s
	1	-1, 0, 1	3	$3p_x, 3p_y, 3p_z$
	2	-2,-1, 0, 1,2	3	$d_{xy}, 3d_{yz}, 3d_{xz}, 3d_{x^2-y^2}, 3d_{z^2}$
⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮



2.5.2 Electron Coordinates

The four quantum numbers n , l , m_l and m_s enable us to label completely an electron in any orbital in any atom. For example, the four quantum numbers for a 2s orbital electron are $n=2$, $l=0$, $m_l=0$ and either $m_s=+1/2$ or $m_s=-1/2$. In practice it is inconvenient to write out all the

individual quantum numbers, and so we use the simplified notation (n, l, m_l, m_s) . For the example above, the quantum numbers are $(2, 0, 0, +\frac{1}{2})$ or $(2, 0, 0, -\frac{1}{2})$. The value of m_s has no effect on the energy, size, shape, or orientation of an orbital, but it plays a profound role in determining how electrons are arranged in an orbital.



2.5.3 The Pauli Exclusion Principle

This principle states that ***no two electrons in an atom can have the same four quantum numbers***. If two electrons in an atom should have the same n , l and m_l values (that is, these two electrons are in the same orbital), then they must have different values of m_s . In other words, **only two electrons** may exist in the **same atomic orbital**, and these **electrons must have opposite spins**.

Consider the helium atom, which has two electrons. The three possible ways of placing two electrons in the $1s$ orbital are as follows:

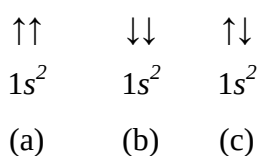


Diagram (a) and (b) are ruled out by the Pauli Exclusion Principle. In (a), both electrons have the same upward spin and would have the quantum numbers $(1, 0, 0, +\frac{1}{2})$; in (b), both electrons have the downward spin and would have the quantum numbers $(1, 0, 0, -\frac{1}{2})$. Only the configuration in (c) is physically acceptable, because one electron has quantum numbers $(1, 0, 0, +\frac{1}{2})$ and the other has $(1, 0, 0, -\frac{1}{2})$. Thus the helium atom has the following configuration:



Note that $1s^2$ is read “one s two”, not “one s squared”.



SELF-TEST QUESTIONS

- i) List the different ways to write the four quantum numbers that designate an electron in a $3p$ orbital.
- ii) List the different ways to write the four quantum numbers that designate an electron in a $5d$ orbital



2.6 Atomic Wavefunctions

As we stated earlier, the concept quantum chemistry requires knowledge of some mathematics and physics, we will explain it, without rigorously solving those equations. This section introduces you to some of those concepts. It is meant to help you appreciate that understanding of mathematics and physics theories are necessary to fully understand the structure of atom. The pictorial diagrams in section 2.7 are mathematical functions derived from those equations.

1. A wavefunction ψ describes an atomic orbital. For most calculations, it is simpler to solve the wave equation if ψ is converted from the Cartesian coordinates (x, y, z) to spherical Ψ coordinates (r, θ, ϕ). The two sets of coordinates are related by:

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

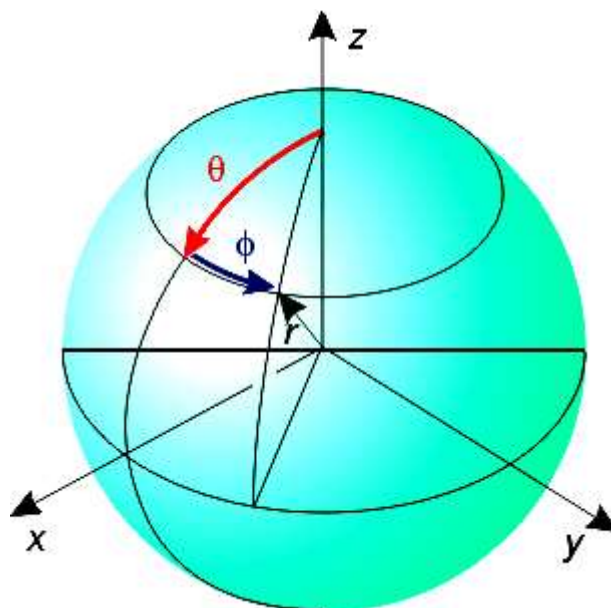


Figure 2.6.1. The relationship between Cartesian and polar coordinates (diagram taken from Atkins & Shriver).

2 In spherical coordinates, ψ can be factored into a radial wavefunction R , and an angular wavefunction A :

$$\Psi_{(x,y,z)} = R_{(r)} A_{(\theta,\phi)}$$

SIR OTONDE MICHAEL AWITI

3 **Radial wavefunction $R(r)$:**

- Describes how the electron density changes with distance from the nucleus
- s orbitals have finite electron density [$R(r) \neq 0$] at the nucleus but this drops off exponentially as the distance from the nucleus increases.
- Other orbitals (e.g., p and d orbitals) have zero electron density [$R(r) = 0$] at the nucleus.

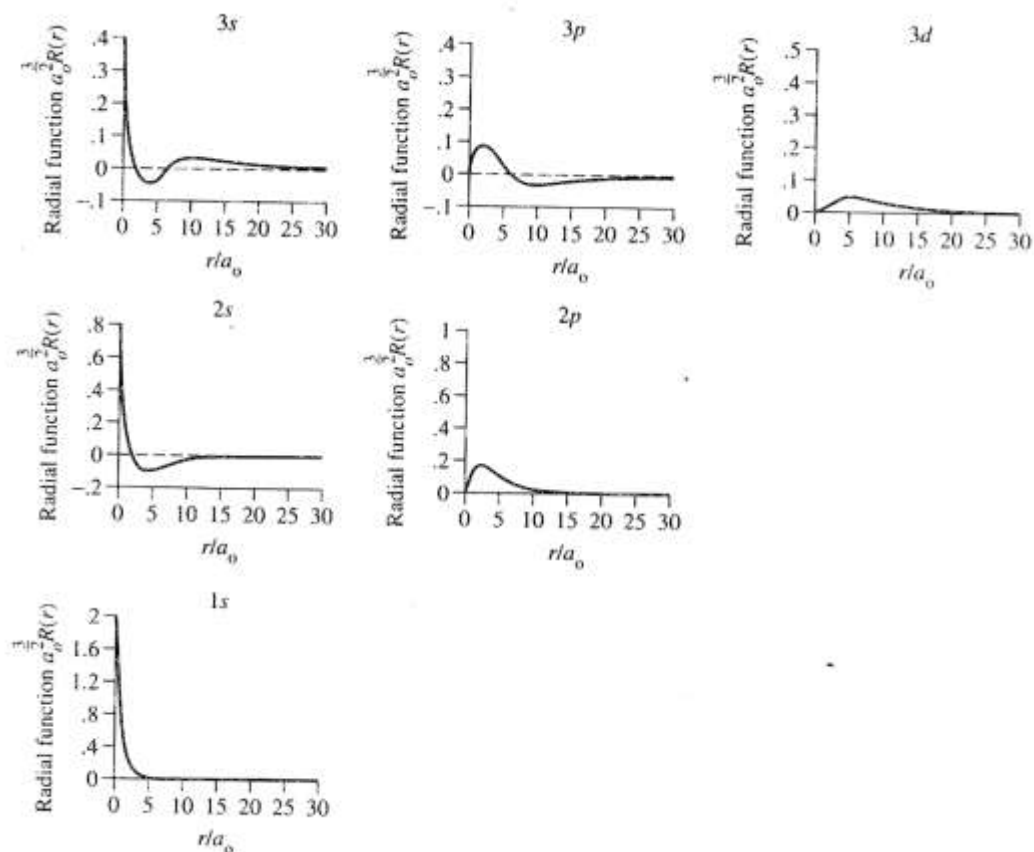


Figure 2.6.2 Plot of Radial Wavefunction (in atomic units) versus distance r (in atomic units) from the nucleus (diagram from Miessler & Tarr).

- The point at which the radial wavefunction $R(r) = 0$ (except the origin) is called a radial node.
- The sign of the radial wavefunction $R(r)$ changes (from +ve to -ve or vice versa) after passing the radial node
- The number of radial nodes for a given orbital is given by: $[n - (l - 1)]$.
- ns orbitals have $(n-1)$ radial nodes. E.g. 2s has 1 radial node.
- np orbitals have $(n-2)$ radial nodes. e.g. 2p has 0 radial node, 3p has 1 radial node.
- nd orbitals have $(n-3)$ radial nodes. E.g. 3d has 0 radial node, 5d has 2 radial nodes.
- nf orbitals have $(n-4)$ radial nodes. E.g. 4f has 0 radial node.
- 1s orbital: no radial node.
- 2s orbital: two regions of maximum probability, separated by a spherical surface of zero probability. (1 radial node)
- 3s orbital: Three regions of maximum probability, separated by two radial nodes.

SIR OTONDE MICHAEL AWITI

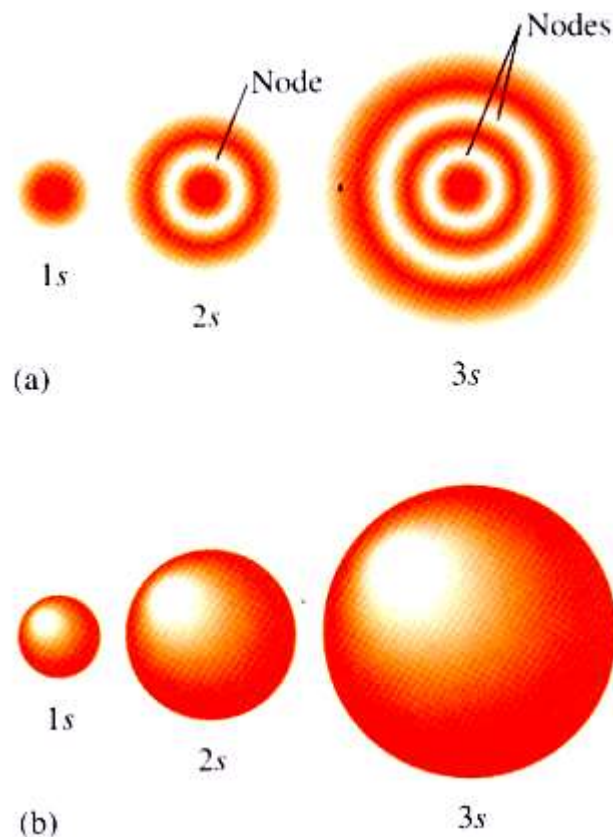


Figure 2.6.3 Boundary surface diagram of s orbitals and electron density plots showing the presence of radial nodes (diagram from Zumdahl).

4 Radial distribution function $4\pi r^2 R(r)$

- Tells us where an electron will be most likely found at a given distance from the nucleus.
- Plots of radial distribution functions for 1s, 2s and 3s orbitals show that there is at least one maximum and the fact that an electron tends to be further from the nucleus as its principal quantum number n increases. **Therefore the size of an atomic orbital gets increases as the principal quantum number n increases.**
- Plots of radial distribution functions for 3s, 3p and 3d orbitals show that s orbital has electron density closest to the nucleus than p and d orbitals. We say that an electron in a s orbital is more penetrating than that in a p or d orbital.

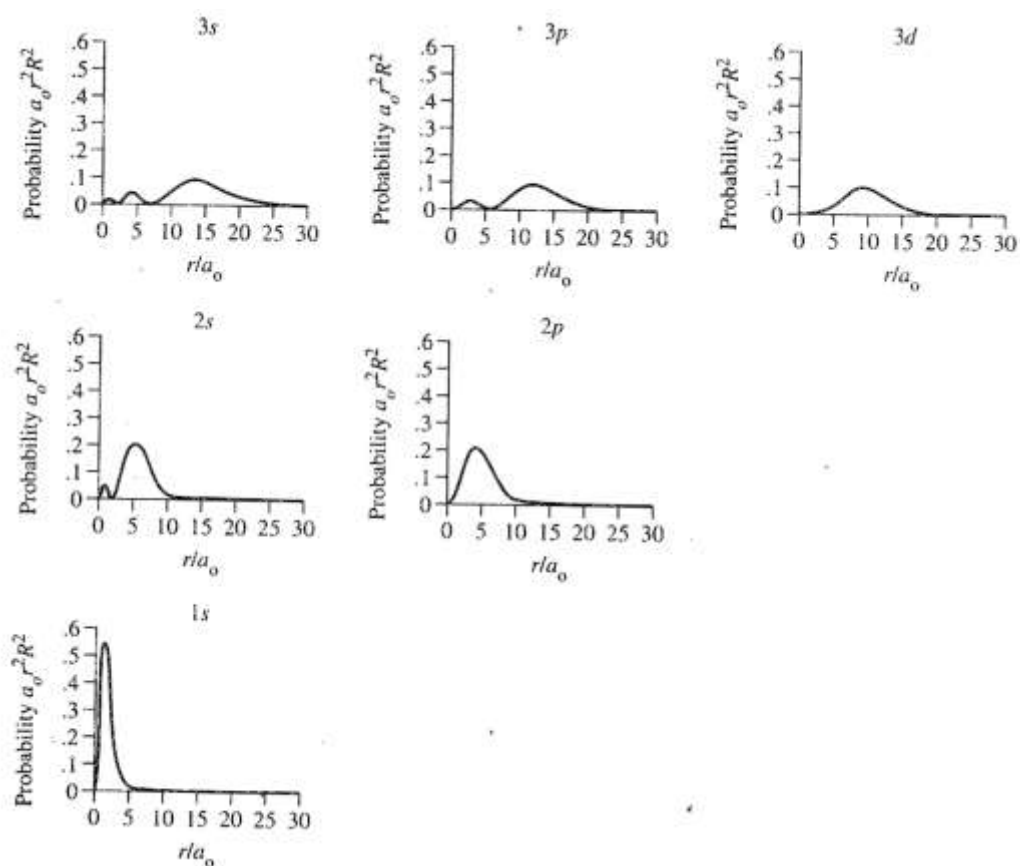


Figure 2.6.4 Plot of radial distribution function $4\pi r^2 R(r)$ for 1s, 2s, 2p, 3s, 3p, 3d orbitals of the hydrogen atom (diagram from Miessler & Tarr).

5 Angular wavefunction $A(\theta, \phi)$

- The angular wavefunction describes the shape of an atomic orbital and its orientation in space.
- The angular function $A(\theta, \phi)$ is determined by the quantum numbers l and m_l .
- **Angular node (nodal plane):** the plane(s) on which the angular wavefunction $A(\theta, \phi) = 0$. The sign of the angular wavefunction changes (from +ve to -ve or vice versa) after passing through an angular node.

SIR OTONDE MICHAEL AWITI



2.7 Pictorial Representation of Atomic Orbitals

- An atomic orbital is defined as a boundary surface which encloses 95 % of the electron density for a particular Ψ .
- Atomic orbitals can be represented pictorially by boundary surface diagrams (H&S Fig. 2.7.1, 2.7.2, 2.7.3) or electron density contour plots (see overhead).

s orbitals

- $l = 0$, $m_l = 0$, the angular wavefunction $A(\theta, \phi)$ is independent of the angle θ and ϕ . Therefore, an s orbital is spherically symmetrical about the nucleus.

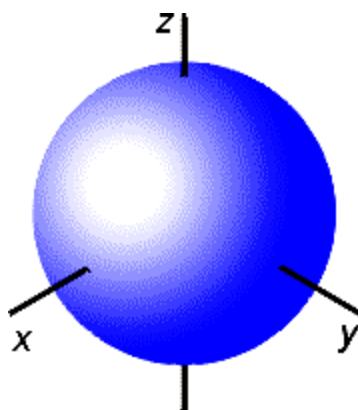
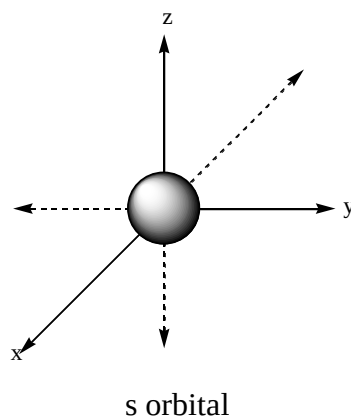


Figure 2.7.1 Boundary surface diagram of an s orbital. (Figure taken from Shriver & Atkins)

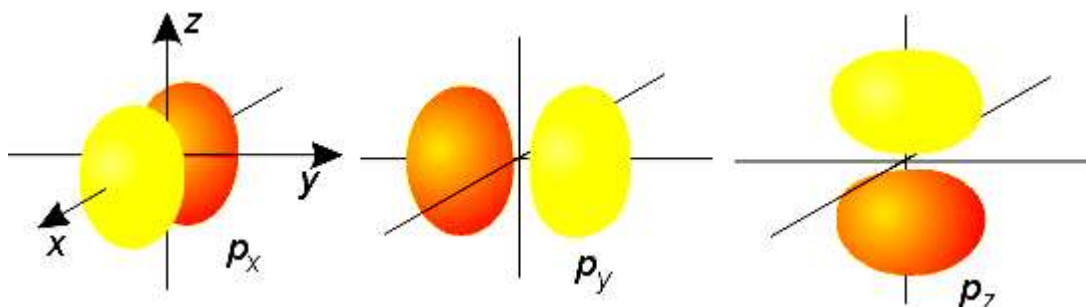
In two dimensions it is represented as:

SIR OTONDE MICHAEL AWITI



p orbitals

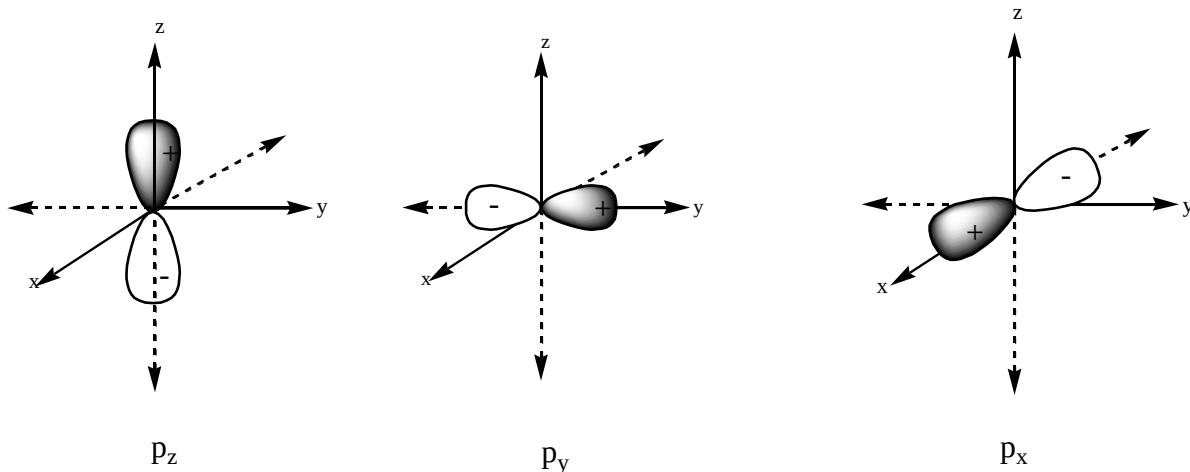
- $l = 1$; $m_l = -1, 0, +1$. 3 possible orientation of p orbitals: along the x, y and z axes.
- Dumb-bell shaped; electron density concentrated in identical lobes on either side of the nucleus.
- p_z orbital has an angular node [$A(\theta, \phi) = 0$] along the xy plane; p_x orbital has an angular node along the yz plane; p_y orbital has an angular node along the xz plane.
- The electron density contour map shows that there is no radial node for the 2p orbitals



but 1 radial node for the 3p orbitals. [see example 1]

Figure 2.7.2 Boundary surface diagram of p orbitals. The lightly shaded lobe has a positive amplitude; the more darkly shaded lobe is negative (Figure taken from Shriver & Atkins).

In two dimensions they are represented as:



d orbitals

- $l = 2$, $m_l = -2, -1, 0, +1, 2$. There are five d orbitals.
- d_{xy} , d_{yz} , d_{xz} , $d_{x^2-y^2}$ have clover leaf shape and have 4 lobes of maximum electron probability centered in the plane indicated in the orbital label. The four lobes are separated by two nodal planes through the nucleus.
- d_{xy} , and $d_{x^2-y^2}$ are both centered in the xy plane while the lobes of d_{xy} lie between the axis.
- d_{z^2} has two lobes along the z axis and a "belt" centered in the xy plane.

SIR OTONDE MICHAEL AWITI

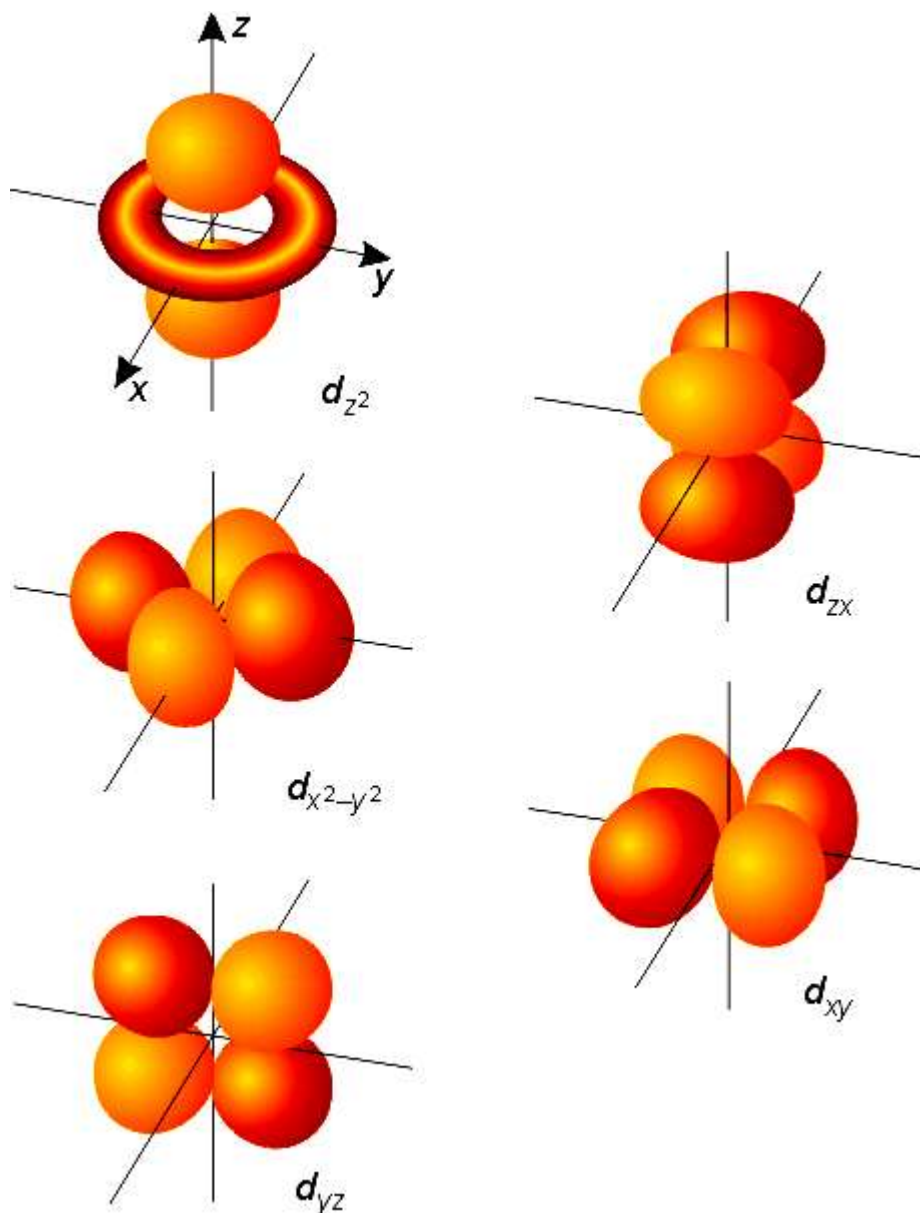
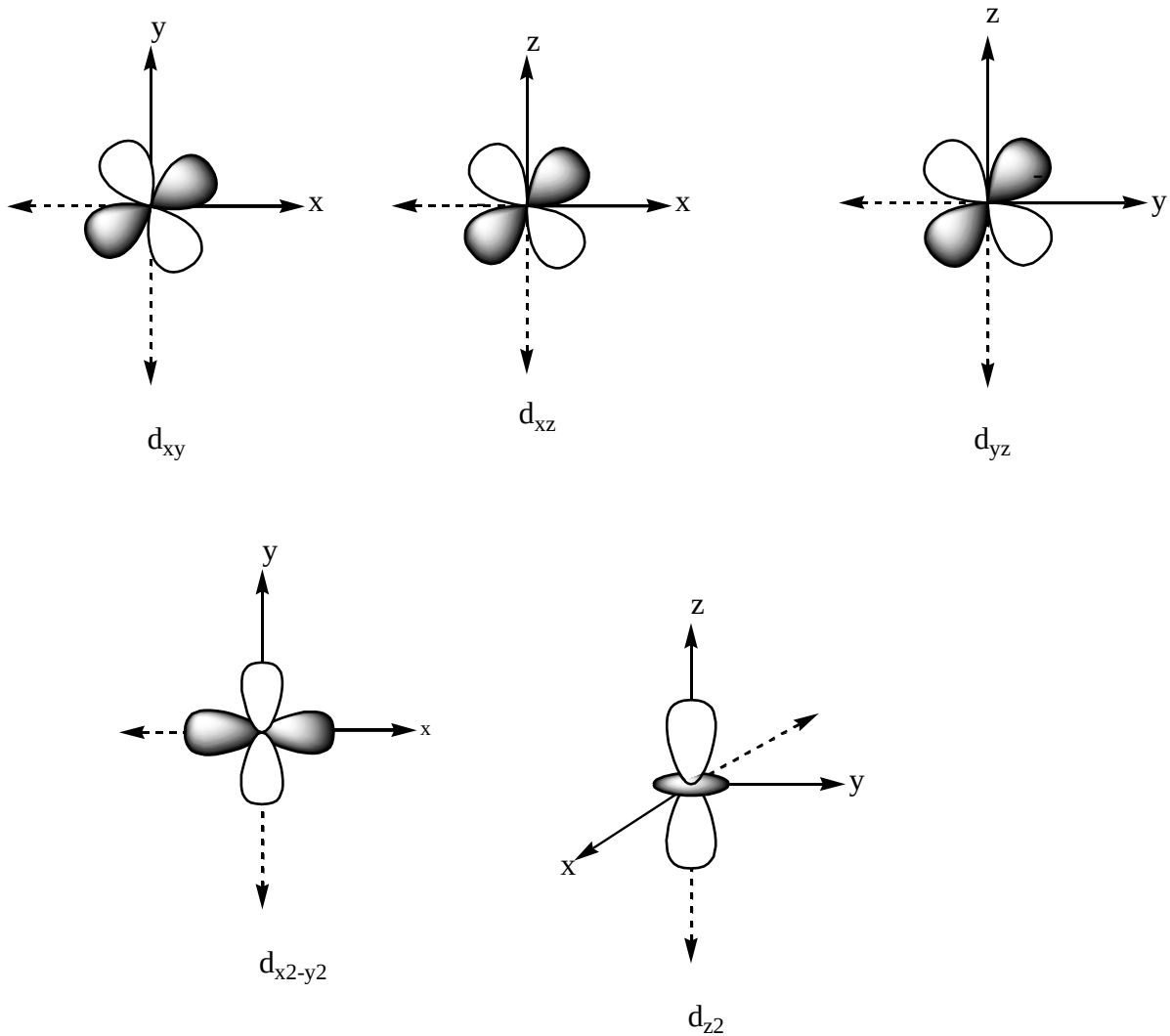


Figure 2.7.3 Boundary surface diagram of d orbitals. The lightly shaded lobe has a positive amplitude; the more darkly shaded lobe is negative (Figure taken from Shriver & Atkins).

In two dimensions they are represented as:

SIR OTONDE MICHAEL AWITI



The shaded lobe has a positive sign
Note the labelling of the axes

SIR OTONDE MICHAEL AWITI



Orbitals

2.8 Nodal Surfaces and Electron Density Contour Plots of Atomic Orbitals

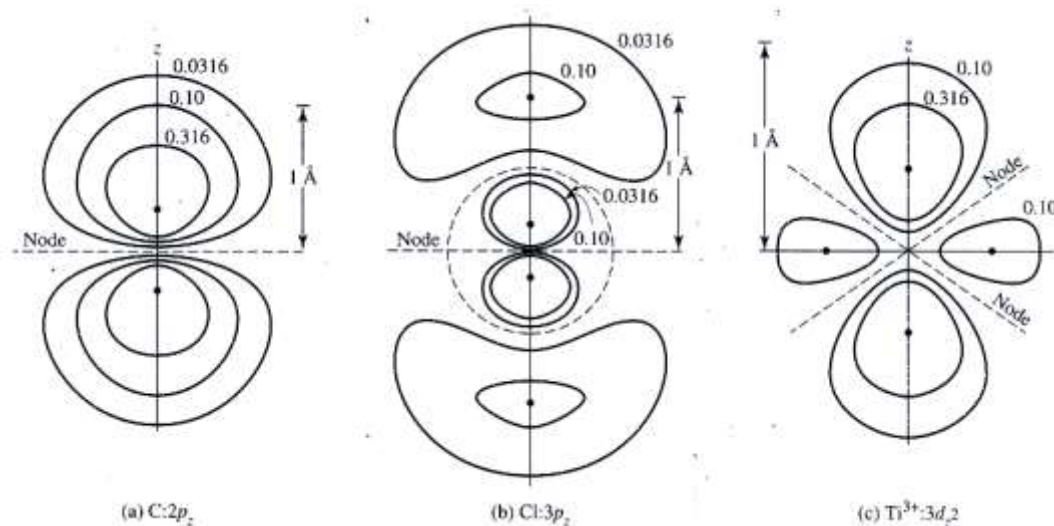


Figure 2.8. Electron density contour plots for selected atomic orbitals showing radial or angular nodes (diagram taken from Miessler & Tarr).



2.9 Electronic Configuration



2.9.1 Energy Levels in Hydrogen and Other Atoms

- 1 For H, the energy of the electron only depends on n , principal quantum number. The s , p , d , f orbitals have the same energy.
- 2 In the ground state, the electron in the H atom resides on the $1s$ orbital.
- 3 For other atoms with more than one electron, electron-electron repulsion and shielding play a role so **energy depends on angular momentum quantum number l as well as the principal quantum number n** . The energies of the atomic orbitals in multielectron atoms are arranged in the order:

$$ns < np < nd < nf$$

SIR OTONDE MICHAEL AWITI

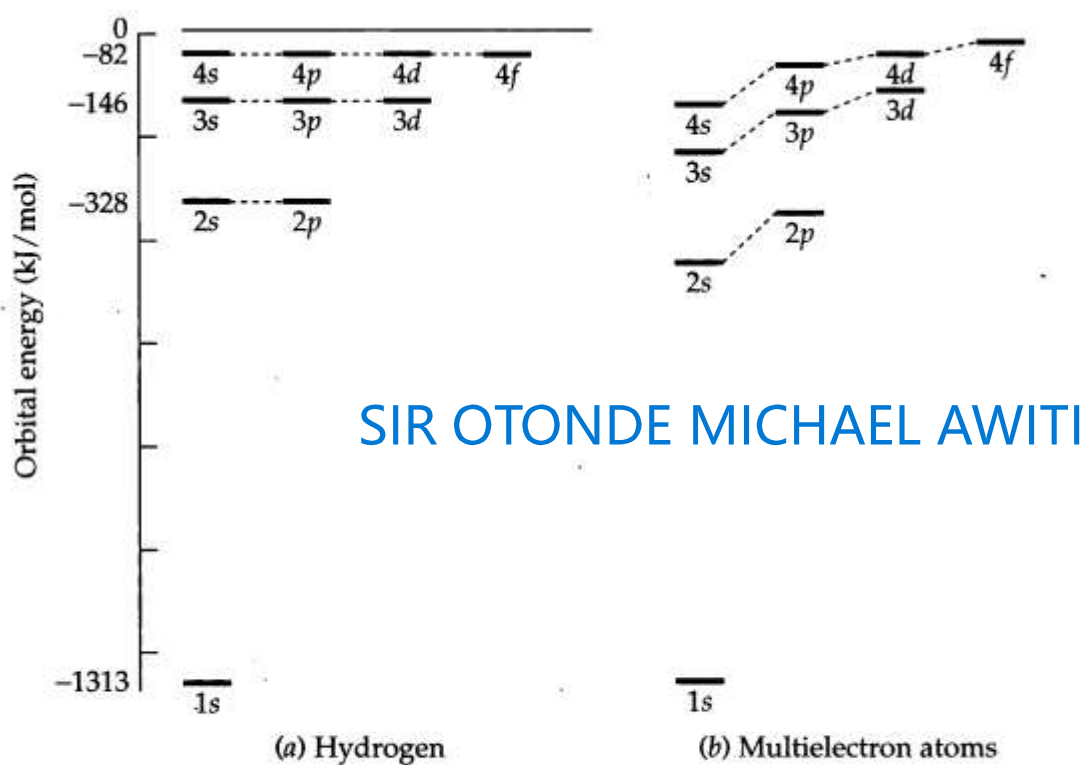


Figure 2.9.1. Orbital energy level diagrams for a hydrogen atom and multielectron atoms (diagram taken from McMurry & Fay).

- 4 This ordering can be explained by considering the ability of having electron density closest to the positively charged nucleus in different atomic orbitals.

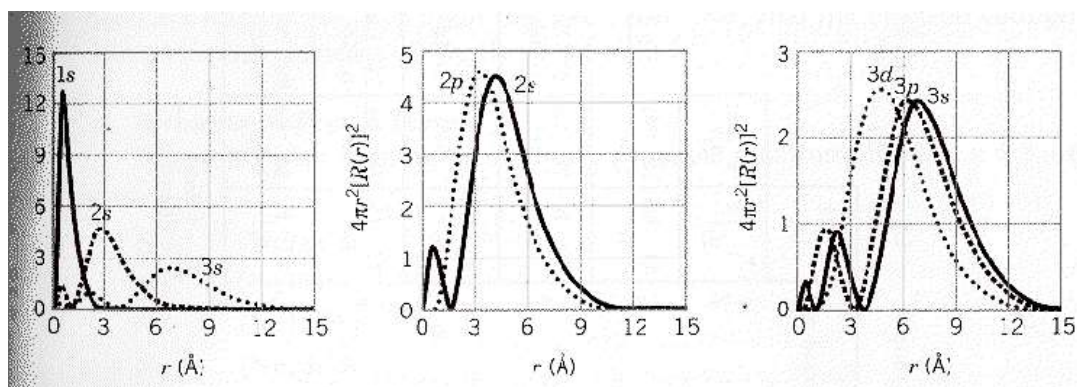
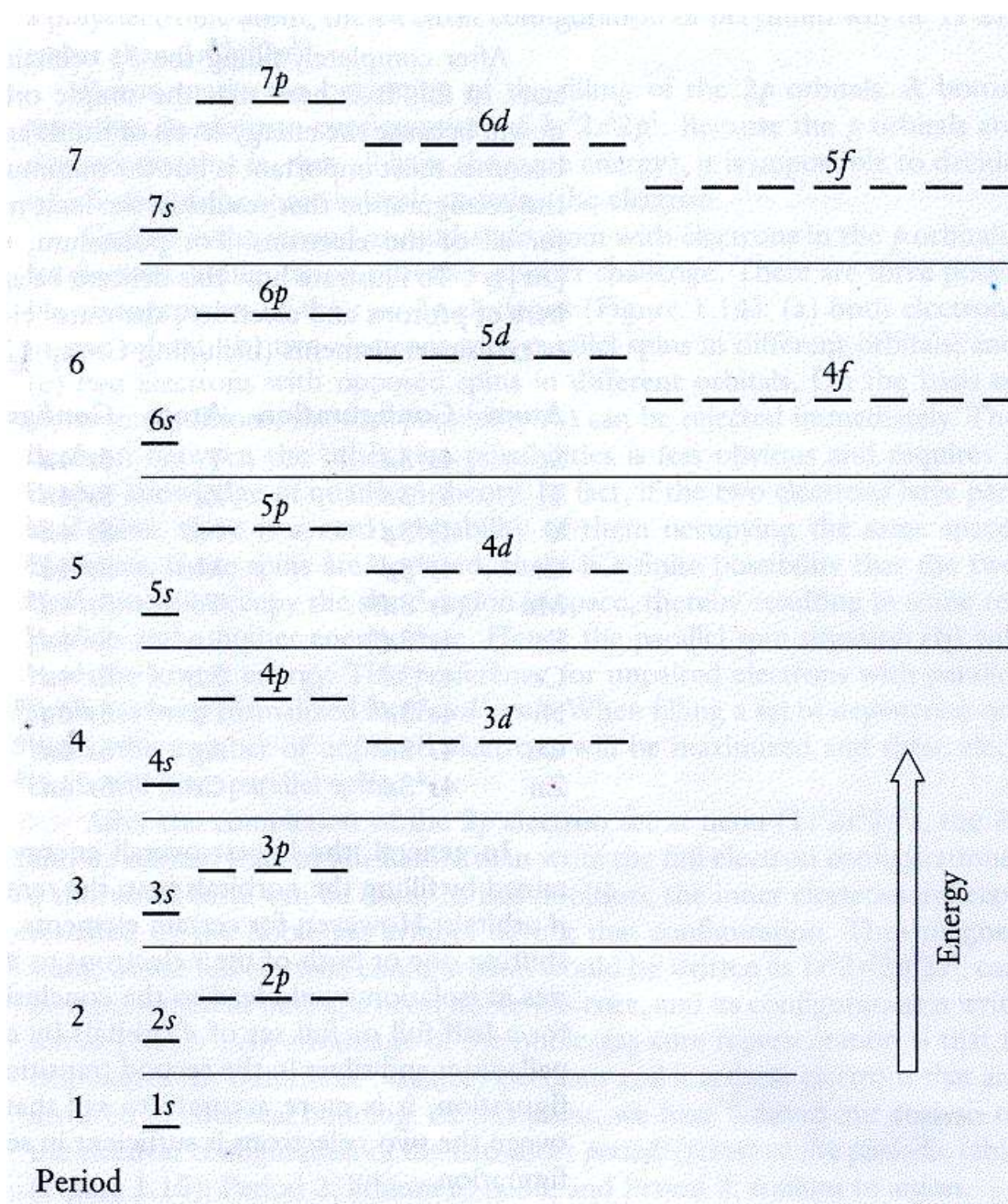


Figure 2.9.2 Plot of radial distribution functions, $4\pi r^2 [R(r)]^2$, for 1s, 2s, 2p, 3s, 3p, 3d orbitals versus radius r (diagram taken from Cotton, Wilkinson & Gaus).

5 The plot of radial distribution functions, $4\pi^2 R(r)^2$, for 3s, 3p, 3d orbitals versus radius (r) shows the first maximum for the s orbital is closest to the nucleus than the p or d orbitals, i.e., the s orbital has electron density closest to the nucleus. We say that electrons in an s orbital can penetrate closer to the nucleus than an electron in a p, d, or f orbital can.

- Since orbitals with electron density closer to the nucleus are more stable, therefore, for a given n, the energy of the orbital increases in the order: $s < p < d < f$.
- The effect is so pronounced for d and f electrons that they lie at higher energy than the



s or p of the $(n + 1)$ shell in some cases. E.g., Energy level of $3d > 4s$, $4d > 5s$; $4f > 5s$, $5p$. Figure. Relative energy level of various atomic orbitals in a multielectron atom (diagram from Rayner-Canham). Please note for the value of ns and $(n - 1)$ d orbitals, there are crossover points (e.g., $4s < 3d$; $5s < 4d$; $5s, 5p < 4f$), this becomes important when filling electrons in multi-electron atoms.



2.9.2 Electronic Configurations

- Describe how electrons distribute themselves in various orbitals in an atom.

We will learn the rules used in writing electron configurations for the elements. The process is based on the *Aufbau principle* (The German word *Aufbau* means ‘building up’)



2.9.3 Aufbau Principle

- A set of rules guiding the filling order of orbitals.
- 1 Fill in electrons in the lowest energy orbital first.
 - 2 Only two electrons with opposite spins per orbital (because Pauli exclusion principle says that no two electrons in an atom can have the same four quantum numbers).
 - 3 If two or more degenerate orbitals (i.e. orbitals with the same energy) are available, electrons go into each degenerate orbital with parallel spins until the orbitals are half-fill (Hund's rule).

SIR OTONDE MICHAEL AWITI



2.9.4 Order of Orbital Filling in Poly-electronic Atoms:

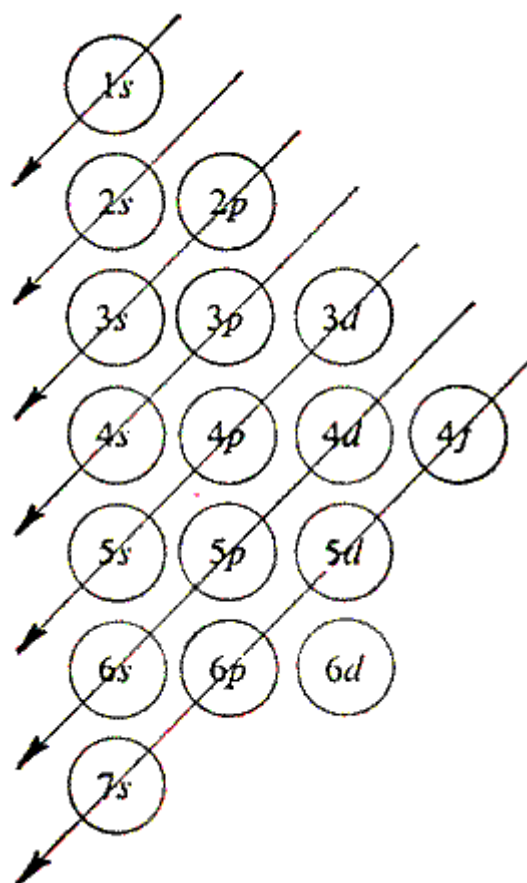


Figure 2.9.3 The order in which atomic orbitals are filled in a many electron atoms. Start with the 1s orbital and move downward, following the direction of the arrows. Thus the order goes as follows $1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \dots$



2.9.5 Filling of electrons for the first 30 elements

Filling of electrons for the first 30 elements

Atom	Atom		Electronic configuration
H	H		$1s^1$
He	He		$1s^2$
Li	Li		$1s^2 2s^1$
Be	Be		$1s^2 2s^2$
B	B		$1s^2 2s^2 2p^1$
C	C		$1s^2 2s^2 2p^2$
N	N		$1s^2 2s^2 2p^3$
O	O		$1s^2 2s^2 2p^4$
F	F		$1s^2 2s^2 2p^5$
Ne	Ne		$1s^2 2s^2 2p^6$
Na	Na		$1s^2 2s^2 2p^6 3s^1$
Mg	Mg		$1s^2 2s^2 2p^6 3s^2$
Al	Al		$1s^2 2s^2 2p^6 3s^2 3p^1$
Si	Si		$1s^2 2s^2 2p^6 3s^2 3p^2$
P	P		$1s^2 2s^2 2p^6 3s^2 3p^3$
S	S		$1s^2 2s^2 2p^6 3s^2 3p^4$
Cl	Cl		$1s^2 2s^2 2p^6 3s^2 3p^5$
Ar	Ar		$1s^2 2s^2 2p^6 3s^2 3p^6$
K	K		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
Ca	Ca		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
Sc	Sc		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$
Ti	Ti		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$
V	V		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$
Cr	Cr		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
Mn	Mn		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
Fe	Fe		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$
Co	Co		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$
Ni	Ni		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$
Cu	Cu		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$

Zn	Zn		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$



2.9.6 The Ground State Electron Configuration of the Elements*

Atomic number	Atom	Electronic configuration	Atomic number	Atom	Electronic configuration	Atomic number	Atom	Electronic configuration
1	H	$1s^1$	38	Sr	$[\text{Kr}]5s^2$	74	W	$[\text{Xe}]6s^1 4f^{14} 5d^5$
2	He	$1s^2$	39	Y	$[\text{Kr}]5s^2 4d^1$	75	Re	$[\text{Xe}]6s^2 4f^{14} 5d^5$
3	Li	$[\text{He}]2s^1$	40	Zr	$[\text{Kr}]5s^2 4d^2$	76	Os	$[\text{Xe}]6s^2 4f^{14} 5d^6$
4	Be	$[\text{He}]2s^2$	41	Nb	$[\text{Kr}]5s^2 4d^3$	77	Ir	$[\text{Xe}]6s^2 4f^{14} 5d^7$
5	B	$[\text{He}]2s^2 2p^1$	42	Mo	$[\text{Kr}]5s^1 4d^5$	78	Pt	$[\text{Xe}]6s^1 4f^{14} 5d^9$
6	C	$[\text{He}]2s^2 2p^2$	43	Tc	$[\text{Kr}]5s^2 4d^5$	79	Au	$[\text{Xe}]6s^1 4f^{14} 5d^{10}$
7	N	$[\text{He}]2s^2 2p^3$	44	Ru	$[\text{Kr}]5s^2 4d^6$	80	Hg	$[\text{Xe}]6s^2 4f^{14} 5d^{10}$
8	O	$[\text{He}]2s^2 2p^4$	45	Pd	$[\text{Kr}]5s^2 4d^7$	81	Tl	$[\text{Xe}]6s^2 4f^{14} 5d^{10} 6p^1$
9	F	$[\text{He}]2s^2 2p^5$	46	Ag	$[\text{Kr}]5s^2 4d^8$	82	Pb	$[\text{Xe}]6s^2 4f^{14} 5d^{10} 6p^2$
10	Ne	$1s^2 2s^2 2p^6$	47	Ag	$[\text{Kr}]5s^1 4d^{10}$	83	Bi	$[\text{Xe}]6s^2 4f^{14} 5d^{10} 6p^3$
11	Na	$[\text{Ne}]3s^1$	48	Cd	$[\text{Kr}]5s^2 4d^{10}$	84	Po	$[\text{Xe}]6s^2 4f^{14} 5d^{10} 6p^5$
12	Mg	$[\text{Ne}]3s^2$	49	In	$[\text{Kr}]5s^2 4d^{10} 5p^1$	85	At	$[\text{Xe}]6s^2 4f^{14} 5d^{10} 6p^5$
13	Al	$[\text{Ne}]3s^2 3p^1$	50	Sn	$[\text{Kr}]5s^2 4d^{10} 5p^2$	86	Rn	$[\text{Xe}]6s^2 4f^{14} 5d^{10} 6p^6$
14	Si	$[\text{Ne}]3s^2 3p^2$	51	Sb	$[\text{Kr}]5s^2 4d^{10} 5p^3$	87	Fr	$[\text{Rn}]7s^1$
15	P	$[\text{Ne}]3s^2 3p^3$	52	Te	$[\text{Kr}]5s^2 4d^{10} 5p^4$	88	Ra	$[\text{Rn}]7s^2$
16	S	$[\text{Ne}]3s^2 3p^4$	53	I	$[\text{Kr}]5s^2 4d^{10} 5p^5$	89	Ac	$[\text{Rn}]7s^2 6d^1$
17	Cl	$[\text{Ne}]3s^2 3p^5$	54	Xe	$[\text{Kr}]5s^2 4d^{10} 5p^6$	90	Th	$[\text{Rn}]7s^2 6d^2$
18	Ar	$[\text{Ne}]3s^2 3p^6$	55	Cs	$[\text{Xe}]6s^1$	91	Pa	$[\text{Rn}]7s^2 5f^2 6d^1$
19	K	$[\text{Ar}]4s^1$	56	Ba	$[\text{Xe}]6s^2$	92	U	$[\text{Rn}]7s^2 5f^3 6d^1$
20	Ca	$[\text{Ar}]4s^2$	57	La	$[\text{Xe}]6s^2 5d^1$	93	Np	$[\text{Rn}]7s^2 5f^4 6d^1$
21	Sc	$[\text{Ar}]4s^2 3d^1$	58	Ce	$[\text{Xe}]6s^2 4f^1 5d^1$	94	Pu	$[\text{Rn}]7s^2 5f^6$
22	Ti	$[\text{Ar}]4s^2 3d^2$	59	Pr	$[\text{Xe}]6s^2 4f^3$	95	Am	$[\text{Rn}]7s^2 5f^7$
23	V	$[\text{Ar}]4s^2 3d^3$	60	Nd	$[\text{Xe}]6s^2 4f^4$	96	Cm	$[\text{Rn}]7s^2 5f^7 6d^1$

24	Cr	[Ar]4s ¹ 3d ⁵	61	Pm	[Xe]6s ² 4f ⁵	97	Bk	[Rn]7s ² 5f ⁹
25	Mn	[Ar]4s ² 3d ⁵	62	Sm	[Xe]6s ² 4f ⁶	98	Cf	[Rn]7s ² 5f ¹⁰
26	Fe	[Ar]4s ² 3d ⁶	63	Eu	[Xe]6s ² 4f ⁷	99	Es	[Rn]7s ² 5f ¹¹
27	Co	[Ar]4s ² 3d ⁷	64	Gd	[Xe]6s ² 4f ⁷ 5d ¹	100	Fm	[Rn]7s ² 5f ¹²
28	Ni	[Ar]4s ² 3d ⁸	65	Tb	[Xe]6s ² 4f ⁹	101	Md	[Rn]7s ² 5f ¹³
29	Cu	[Ar]4s ¹ 3d ¹⁰	66	Dy	[Xe]6s ² 4f ¹⁰	102	No	[Rn]7s ² 5f ¹⁴
30	Zn	[Ar]4s ² 3d ¹⁰	67	Ho	[Xe]6s ² 4f ¹¹	103	Lr	[Rn]7s ² 5f ¹⁴ 6d ¹
31	Ga	[Ar]4s ² 3d ¹⁰ 4p ¹	68	Er	[Xe]6s ² 4f ¹²	104	Unq	[Rn]7s ² 5f ¹⁴ 6d ²
32	Ge	[Ar]4s ² 3d ¹⁰ 4p ²	69	Tm	[Xe]6s ² 4f ¹³	105	Unp	[Rn]7s ² 5f ¹⁴ 6d ³
33	As	[Ar]4s ² 3d ¹⁰ 4p ³	70	Yb	[Xe]6s ² 4f ¹⁴	106	Unh	[Rn]7s ² 5f ¹⁴ 6d ⁴
34	Se	[Ar]4s ² 3d ¹⁰ 4p ⁴	71	Lu	[Xe]6s ² 4f ¹⁴ 5d ¹	107	Uns	[Rn]7s ² 5f ¹⁴ 6d ⁵
35	Br	[Ar]4s ² 3d ¹⁰ 4p ⁵	72	Hf	[Xe]6s ² 4f ¹⁴ 5d ²	108	Uno	[Rn]7s ² 5f ¹⁴ 6d ⁶
36	Kr	[Ar]4s ² 3d ¹⁰ 4p ⁶	73	Ta	[Xe]6s ² 4f ¹⁴ 5d ³	109	Une	[Rn]7s ² 5f ¹⁴ 6d ⁷
37	Rb	[Kr]5s ¹						

*The symbol [He] is called the *helium core* and represents 1s². [Ne] is called the *neon core* and represents 1s²2s²2p⁶. [Ar] is called the *argon core* and represent [Ne]3s²3p⁶. [Kr] is called the *krypton core* and represents [Ar]4s²3d¹⁰4p⁶. [Xe] is called the *xenon core* and represents [Kr]5s²4d¹⁰5p⁶. [Rn] is called the *radon core* and represents [Xe]6s²4f¹⁴5d¹⁰6p⁶.

SIR OTONDE MICHAEL AWITI



2 SUMMARY

- 1 Planks quantum theory state that energy is absorbed or emitted by atoms or molecules in small discrete amounts (quanta), rather than over a continuous range. This is governed by the equation $E = hv$
- 2 Light and matter have the characteristic of both particles(Dual nature of light and matter)
- 3 The Schrödinger equation describes the motions and energies of subatomic particles in terms of quantum theory. It tells us the possible energy states of the electron in an atom or molecule and the probability of its location in a particular region surrounding the nucleus.
- 4 An atomic orbital is a function that defines the distribution of electron density in space. Orbitals are represented by electron density diagrams or boundary surface diagrams.
- 5 Four quantum numbers are needed to characterize completely each electron in an atom:

- The principal quantum number n , which determines the main energy level, or shell of the orbital;
- The angular momentum quantum number l , which determines the shape of the orbital;
- The magnetic quantum number, m_l , which determines the orientation of the orbital in space;
- And the electron spin quantum number m_s , which relates only to the electrons spins on its own axis.

6 Using the *Aufbau* (Building-up) principle we can determine the ground-state electron configurations of all elements based on atomic number.

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

Lecture 3



3 Lecture 3 Objectives or Aims

At the end of this unit you should be able to;

- (a) arrange elements in the periodic table according to the group and period number
- (b) discuss the periodicity of the periodic table



3.0 Introduction

We have just described the *Aufbau* principle of making probable assignments of electrons to the atomic orbitals. There are over 103 known elements. Chemists have noted that many elements show very strong similarities to one another. Recognition of periodic regularities in physical and chemical behaviours and need to organize the large volume of available information about the structure and properties of elemental substances led to the development of the **periodic table**---*a chart in which elements having similar chemical and physical properties are arranged in groups*. In this lecture you will learn how to place elements in groups and periods



ACTIVITY

- i) Read about historical development of the Periodic table: Mendeleev's Periodic Table and active rule
- ii) Read about the American and European periodic tables

In high school you learnt the short form periodic table, where elements are arranged in horizontal rows, called periods and vertical column called groups. The groups are numbered from I to VIII (Roman numerals).

There are several controversies about the details of the periodic table that are still unresolved despite the table's existence for more than a century. For instance, there are three different systems currently used to designate the vertical groups in the periodic table, namely, the American, European and the modern International Union of Pure and Applied Chemistry IUPAC format.

We will only discuss the modern periodic table in this lecture.



3.1.1 Modern Periodic Table - The Long Form

- 1 Elements are arranged with increasing atomic number in rows (called periods) and columns (called groups).
- 2 Elements in Groups 1 and 2 are also called s block elements: valence electrons fill in s orbitals.
- 3 Elements in Groups 13 to 18 are also called p block elements: valence electrons fill in p orbitals.
- 4 Elements in Groups 1, 2 and 13 - 18 are called main group elements.
- 5 Elements in Groups 3 - 11 are called transition metal because they have partially filled d orbitals. Elements in Groups 3 - 12 are called d block elements because valence electrons filled in d orbitals.
- 6 f block elements (also called Lanthanides and Actinides): valence electrons filling in f orbitals.
- 7 Elements in each group have similar valence-electron (electrons in the outermost shell) configuration and therefore similar properties.

SIR OTONDE MICHAEL AWITI

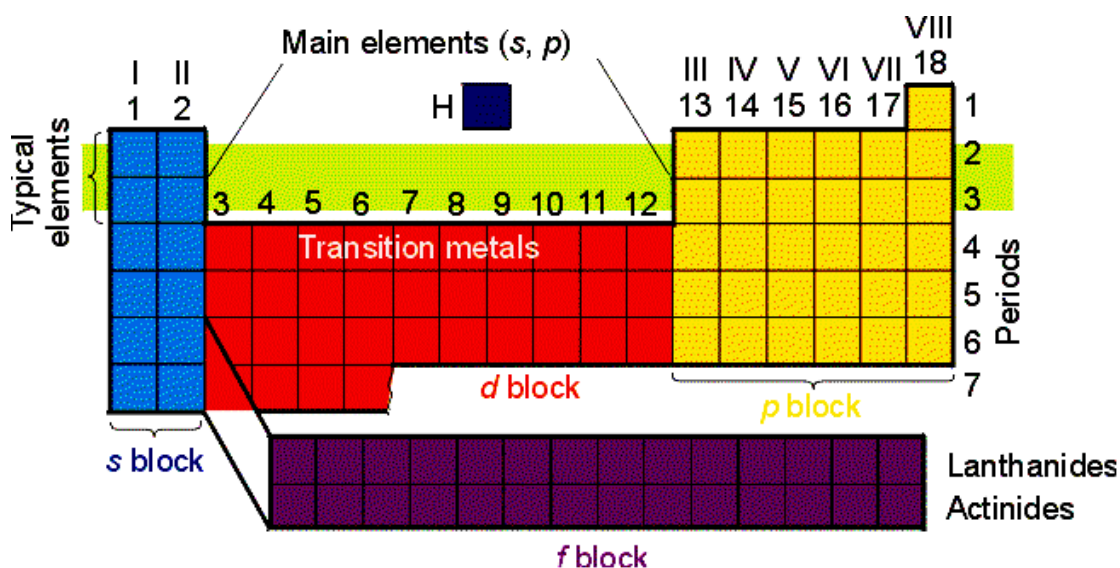


Figure 3.2. The structure of the periodic table (diagram from Atkins & Shriver).



3.1.2 Classifying the Elements into the Periodic Table

We will learn how to get the period and group numbers of the elements. This can be done in three steps:

- i) Write the electron configurations for the elements, using the *Aufbau* principle.
 - The period number is the largest principal quantum number in the electron configuration
 - To get the group number, arrange the electrons in order increasing principal quantum numbers, the group number is the sum of electrons after the last p^6 to be filled.
 - * But for period 2 and 3, the group number is obtained as “10+number of outer shell electrons”. Thus carbon and sulphur are in group 14 and 16 respectively. This takes into account the d-block elements in the short form periodic table.

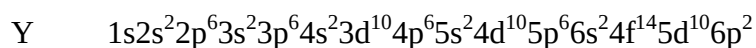
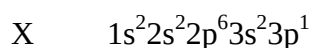


SELF-TEST QUESTIONS

- i) Elements X and Y (not their actual symbols) have atomic number 13 and 82 respectively. To which period and groups do they belong?

Solution

The electronic configuration of:



Period numbers

X the largest principal quantum number, n , is 3

Implying period 3

Y the largest principal quantum number, n , is 6

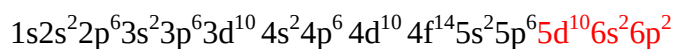
Implying period 6

Group numbers

X Electrons already arranged in ascending order according to the value of n .

Since it belongs to period 3. Group number = $10 + 3 = 13$

Y Arranging electrons according to n



The last p^6 to be filled is $5p^6$. The number of electrons after that

is $= 10 + 2 + 2 = 14$

Implying group 14

- ii) Nitrogen and Silver atomic number 7 and 47 respectively. To which period and groups do they belong?

SIR OTONDE MICHAEL AWITI

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	I	II											III	IV	V	VI	VII	VIII
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.0	9 F 19.00	10 Ne 20.18
3	11 Na 22.99	12 Mg 24.30											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
4	19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
5	37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.22	42 Mo 95.94	43 Tc 98.91	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
6	55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po 209.0	85 At (210)	86 Rn (222)
7	87 Fr (223)	88 Ra (226.0)	89 Ac (227)															

Transition Elements

1
H
1.008

Atomic number

Atomic mass

Lanthanides													
58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm 144.9	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0

Actinides													
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np 237.0	94 Pu (242)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)

Figure 3.1. The modern periodic table



3.2 Periodic Trends of Atomic Parameters

- Atomic properties such as effective nuclear charge, atomic radii, ionization energies, electron affinity and electronegativity are important in accounting for the chemical properties of an element.

SIR OTONDE MICHAEL AWITI



3.2.1 Shielding

1. In atoms with more than one electrons, the effect of electron-electron repulsion depends on where the various electrons are located in the atom.
- Electrons in the outer shell (higher n) are pushed away by electrons in the inner shell (lower n). As a result, the net nuclear charge (or effective nuclear charge, Z_{eff}) felt by an outer electron is substantially lower than the actual nuclear charge (Z). We say that the outer electrons are **shielded** from the full charge of the nucleus by the inner electrons.
 - Electrons in the same shell (same n) have an immediate effect in shielding the nuclear charge.
 - Electrons in the outer shell (higher n) do not shield the nuclear charge from the inner electrons (lower n)
 - In 1930, Slater formulated a set of rules for the **effective nuclear charge** felt by electrons in different atomic orbitals based on experimental data. The effective nuclear charge Z_{eff} can be calculated by the following equation:
 - $Z_{\text{eff}} = Z_{\text{actual}} - S$
 - Z : atomic number (number of protons);
 - S : Slater screening constant;
 - The values of S are estimated as follows:
 - Write out the electron configuration of an element in groups of : $(1s)$, $(2s, 2p)$, $(3s, 3p)$, $(3d)$, $(4s, 4p)$, $(4d)$, $(4f)$, $(5s, 5p)$ etc.
 - Electrons with higher n do not shield those in lower n ; $S = 0$.
 - **If the electron in question resides in an s or p orbitals:**
 - Each of the other electron in the same ns , np group, $S = 0.35$.
 - Each electron in the $(n - 1)$ shell, $S = 0.85$.
 - Each electron in the $(n - 2)$ and lower shell, $S = 1.00$

SIR OTONDE
MICHAEL AWITI

- If the electron in question resides in a *d* or *f* orbitals:
- Each of the other electron in the same *nd*, *nf* group, $S = 0.35$.
- Each electrons in the $(n - 1)$ shell, $S = 1.00$

Example:

1. Calculate the effective nuclear charge for the outermost electron in Oxygen.

Given: Electron configuration of O: $1s^2 2s^2 2p^4$.

2. Calculate the effective nuclear charge on a 3d electron in a Nickel atom.

Given: Electron configuration of Ni: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$.

SIR OTONDE MICHAEL AWITI



3.3 Periodic Trend in Effective Nuclear Charge

Below is a table of the values of effective nuclear charge (Z_{eff}) for *s* and *p* electrons in elements Li - Ne:

Element	Li	Be	B	C	N	O	F	Ne
Z	3	4	5	6	7	8	9	10
1s	2.69	3.68	4.68	5.67	6.66	7.66	8.65	9.64
2s	1.28	1.91	2.58	3.22	3.85	4.49	5.13	5.76
2p			2.42	3.14	3.83	4.45	5.10	5.76

1. In the same atom, each electron with different n and l values can have a different Z_{eff} because the repulsive effects due to the other electrons is different. E.g., the Z_{eff} for the B atom: $1s \gg 2s > 2p$ and the size of orbitals is: $1s \ll 2s < 2p$
2. For the same n and l value, Z_{eff} increase across a period (e.g., from Li to Ne) as Z increases. As a result, the size of the orbital decreases (electrons gets closer to the nucleus) and the size of the atoms decreases. E.g. the Z_{eff} for the $2p$ orbitals: $B < C < N < O < F < Ne$ and the size of the $2p$ orbitals is: $B > C > N > O > F > Ne$



3.4 Sizes of Atoms and Ions

- Since electrons can only be located by probability, there is no real boundary to an atom.
- Radii of atoms and ions are obtained experimentally from measuring bond lengths in molecules, metals and ionic crystals by X-ray diffraction.

1. Covalent radii r_{cov} :

- defined as the half-distance between the nuclei of two atoms of the same element joined by a covalent bond.
- E.g., Cl_2 : bond distance = 198 pm, atomic radius of Cl = $(198)/2 = 99$ pm.

2. Van der waal radius r_{vdw} :

- Defined as the half-distance between the nuclei of two non-bonded atoms or molecules.
- Obtained from the closest approach between two non-bonded atoms or molecules in the solid state.
- E.g., van der waal radius of Cl = 180 pm.
- **In general, $r_{\text{vdw}} > r_{\text{cov}}$.**

3. Ionic radius:

- defined as the distance between the nuclei of adjacent cations and anions in a purely ionic lattice.

In general, $r_{\text{anions}} > r_{\text{cov}} > r_{\text{cation}}$ for any given element.

- Cations:** $r_{\text{Na}^+} = 116 \text{ pm}$; covalent radius $r_{\text{Na}} = 154 \text{ pm}$;
- Why? From Na ($1s^2 2s^2 2p^6 3s^1$) to Na^+ ($1s^2 2s^2 2p^6$): Z_{eff} increases since no. of electrons decreases.
- Anions:** the covalent radius $r_{\text{Cl}} = 97 \text{ pm}$; $r_{\text{Cl}^-} = 167 \text{ pm}$;
- Why? From Cl ($1s^2 2s^2 2p^6 3s^2 3p^5$) to Cl^- ($1s^2 2s^2 2p^6 3s^2 3p^6$): Z_{eff} decreases since no. of electrons increases.



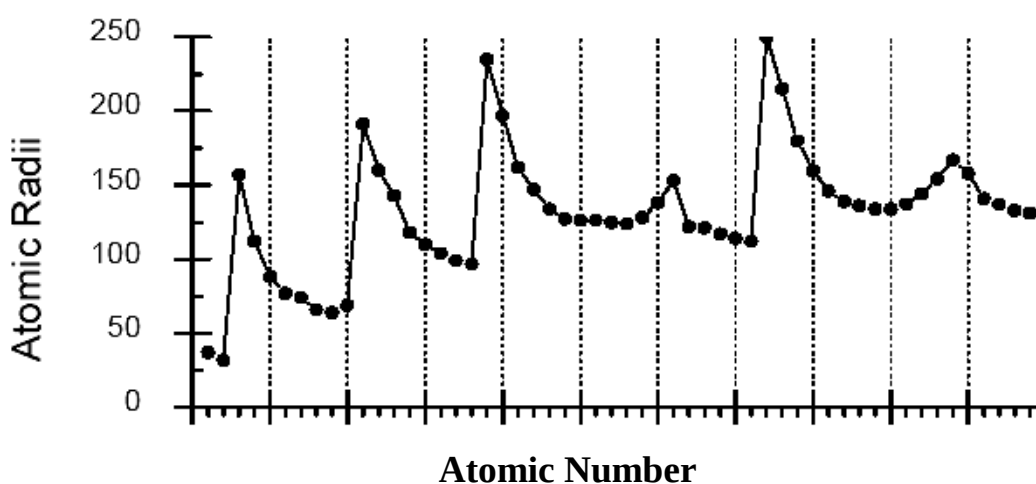
3.5 Periodic Trend in Atomic Radii

- Across a row**, the atomic radius decreases because the n value for the outer shell remains the same but Z_{eff} is increasing.
 - E.g. For row started by Na outer shell is $3s^1, 3s^2, 3s^2 3p^1, 3s^2 3p^2, \dots, 3s^2 3p^6$, atomic radius decreases (from Na to Ar) because the Z value is increasing and the added electron do not completely shield the increase in Z .
- Down a group**, the increase in Z_{eff} does not completely counteract the fact that the outer electrons are in orbitals with higher n , so radius increase down a group.
 - E.g. For group headed by Be outer shell is $2s^2, 3s^2, 4s^2, 5s^2, 6s^2, 7s^2$, atomic size increases from Be to Ra.

SIR OTONDE MICHAEL AWITI

Table.3.5 Covalent radii (pm) of some selected elements

Li 132	Be 89	B 82	C 77	N 77	O 73
Na 154	Mg 136	Al 118	Si 111	P 106	S 102
K 203	Ca 174	Ga 126	Ge 122	As 120	Se 117
Rb 216	Sr 191	In 144	Sn 140	Sb 140	Te 136



SIR OTONDE MICHAEL AWITI



3.6 Periodic Trend in Ionic Radii

Going down a group in cations and anions:

- e.g , $\text{Mg}^{2+}(2s^2 2p^6) \rightarrow \text{Ca}^{2+}(3s^2 3p^6) \rightarrow \text{Sr}^{2+}(4s^2 4p^6) \rightarrow \text{Ba}^{2+}(5s^2 5p^6)$
- e.g , $\text{F}^-(2s^2 2p^6) \rightarrow \text{Cl}^-(3s^2 3p^6) \rightarrow \text{Br}^-(4s^2 4p^6) \rightarrow \text{I}^-(5s^2 5p^6)$
- n value of outer orbital increase but not counteracted by the increasing Z_{eff} .

SIR OTONDE MICHAEL AWITI

Size of isoelectronic series (ions/atoms containing the same number of electrons):

- e.g , $O^{2-}(2s^2 2p^6) \rightarrow F^-(2s^2 2p^6) \rightarrow Ne(2s^2 2p^6) \rightarrow Na^+(2s^2 2p^6) \rightarrow Mg^{2+}(2s^2 2p^6)$
- Z_{eff} increasing $\rightarrow$ Since same no. of electrons and Z increasing.
- $r(O^{2-}) > r(F^-) > r(Ne) > r(Na^+) > r(Mg^{2+})$

Ionic radii (pm) of some selected main group ions

Li^+ 60	Be^{2+} 31				O^{2-} 140	F^- 136
Na^+ 95	Mg^{2+} 65	Al^{3+} 50			S^{2-} 184	Cl^- 181
K^+ 133	Ca^{2+} 99	Ga^{3+} 62			Se^{2-} 198	Br^-
Rb^+ 248	Sr^{2+} 113	In^{3+} 81	Sn^{4+} 71		Te^{2-} 221	I^- 216

SIR OTONDE MICHAEL AWITI

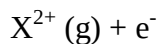
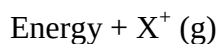


3.7 Periodic Trend in the First Ionization Energy

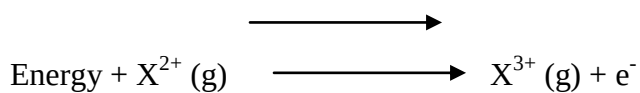
The ionization potential of an atom is the minimum energy required to remove an electron from a gaseous atom in its ground state. The magnitude of ionization potential is a measure of the effort required to force an atom to give up an electron, the higher the ionization energy, the more difficult it is to remove the electron.

For many-electron atom, the amount of energy required to remove the first electron from the atom in its ground state is called the **first ionization energy (I_1)**

The second ionization energy (I_2) and the third ionization energy (I_3) are shown in the following equations



SIR OTONDE MICHAEL AWITI



Where X is the element and (g) denotes the gaseous state.

Higher ionization potentials labelled I_2 , I_3 , I_4 and so on will correspond to the successive removal of additional electrons. The table below gives the first ionization energies of the first 21 elements.

Table 3.7; Ionization Potential (Energy) for the First 20 Elements

Z	Element	First	Second	Third	Fourth	Fifth	Sixth
1	H	1312					
2	He	2373	5248				
3	Li	520	7300	11808			
4	Be	899	1757	14850	20992		
5	B	801	2430	3660	25000	32800	
6	C	1086	2350	4620	6220	38000	47232
7	N	1400	2860	4580	7500	9400	53000
8	O	1314	3390	5300	7470	11000	13000
9	F	1680	3370	6050	8400	11000	15200
10	Ne	2080	3950	6120	9370	12200	15000
11	Na	495.9	4560	6900	9540	23400	16600
12	Mg	738.1	1450	7730	10500	13600	18000
13	Al	577.9	1820	2750	11600	14800	18400
14	Si	786.3	1580	3230	4360	16000	20000
15	P	1012	1904	2910	4960	6240	21000
16	S	999.5	2250	3360	3660	6990	8500
17	Cl	1251	2297	3820	5160	6540	9300
18	Ar	1521	2666	3900	5770	7240	8800
19	K	418.7	3052	4410	5900	8000	9600
20	Ca	589.5	1145	4900	6500	8100	11000

Ionization potentials are positive quantities.

If the ionization Energy was plotted against the atomic number, the periodicity will be clearly evident. The 1st Ionization energy generally increases across the period.

Group I elements (alkali metals) have the lowest ionization energies. This is because the elements have one valence electron that is effectively shielded by the completely filled inner shells. Consequently, it is energetically easy to remove an electron from the atom of an alkali metal to form unipositive ion (Li^+ , Na^+ , K^+ ).

On the other hand non-metals have much higher ionization energies. The ionization energies of the metalloids usually fall between those of metals and non-metals. This explains why metals readily form cations while non-metals form anions.

However, there are irregularities in the increase of the ionization energy across the periods. For example going across group 2 to 3, thus from Be to B and from Mg to Al. The group 3 elements have a single electron in the outermost sub level ($ns^2 np^1$), which is well shielded by the inner electrons and the ns^2 electrons.

Less energy is therefore needed to remove a paired s electron from the same principle energy level. This brings about the lower ionization energies in group 3 elements compared with those in group 2 in the same period.

- Periodic trends

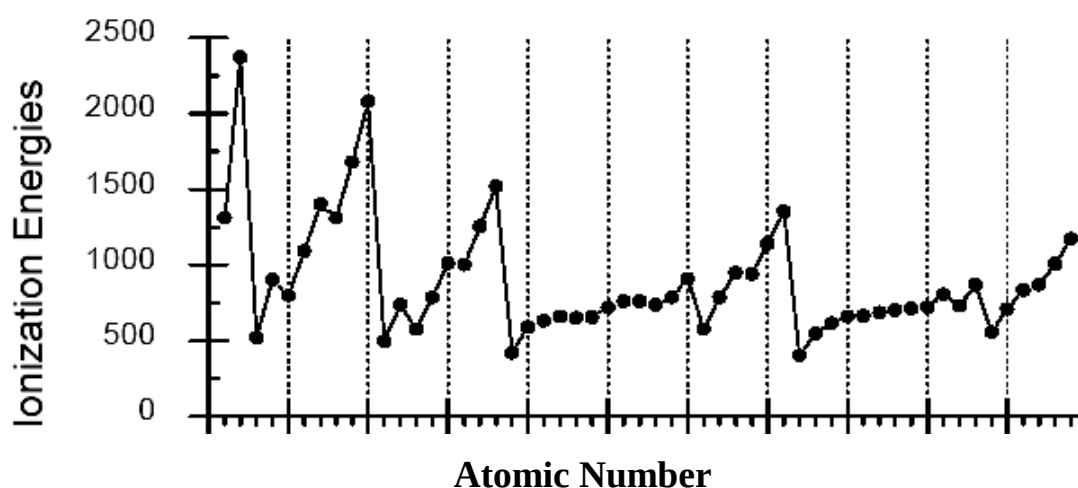


Figure. Ionization energies of the first 55 elements

- Increase across each row, e.g. from Li to Ne; Na to Ar; K to Kr and so on (*because Z_{eff} increases and the electrons in a given orbital are held more tightly, therefore, to remove the electron from the atom requires more energy*).
- Group 1 Alkaline metals (i.e. Li, Na, K, Rb, Cs and Fr) have minimum I.E. (*due to only single loosely held electron in the valence shell ns^1*).
- Group 18 Noble gases (i.e. He, Ne, Ar, Kr, Xe and Rn) have the highest I.E. (*due to filled valence shell, ns^2np^6 ; high Z_{eff} valence electrons are tightly held*).
- Minor irregularity occur from group 12 to group 13 elements also from group 15 to group 16 elements:
 - E.g. I.E. of Be (899.4 kJ/mol) > I.E. of B (800.6 kJ/mol): *due to completely filled ns^2 configuration ($2s^2$ in Be vs $2s^22p^1$ in B)*;
 - I.E. of N (1402.3 kJ/mol) > I.E. of O (1313.9 kJ/mol): *due to half filled p orbital in N ($2s^22p^3$ in N vs $2s^22p^4$ in O)*;
- I.E. decreases down a group, e.g. $\text{Li} > \text{Na} > \text{K} > \text{Rb} > \text{Cs} > \text{Fr}$; $\text{He} > \text{Ne} > \text{Ar} > \text{Kr} > \text{Xe} > \text{Rn}$.
 - *Due to electrons filling in larger shell (larger n), further away from the nucleus. Therefore, valence electrons are well shielded from the nucleus by inner shell electrons and are loosely held.*
- The ionization energy increase only slightly across a row in the transition or lanthanides and actinides series.



SELF-TEST QUESTIONS

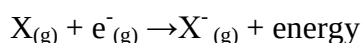
- Explain why the first ionization energy of phosphorous is higher than that of oxygen.
- Explain why the first ionization energy of magnesium is higher than that of aluminum
- Why is the third ionization energy of Mg exceptionally higher than the second ionization?



3.8 Periodic Trend in Electron Affinity

Electron Affinities

The electron affinity of an atom is defined as the energy change obtained when a neutral atom in the gaseous state captures an electron. Thus, the energy is released by the reaction represented below: -



It is the reverse of the 1st ionization potential and may be looked at as the ability of an atom to accept one or more electrons. The largest electron affinities are those of the halogens. This is as expected since the addition of one electron yields the stable octet configuration of the 18th group elements-the noble gases.

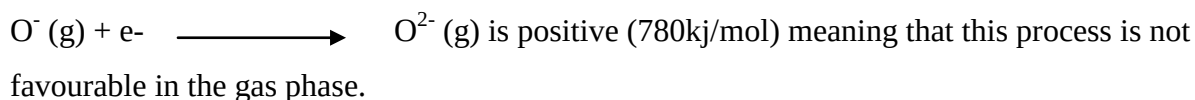
SIR OTONDE MICHAEL AWITI

X is an atom of an element. We assign a negative value to the affinity when energy is released. The more negative the electron affinity, the greater the tendency of the atom to accept an electron.

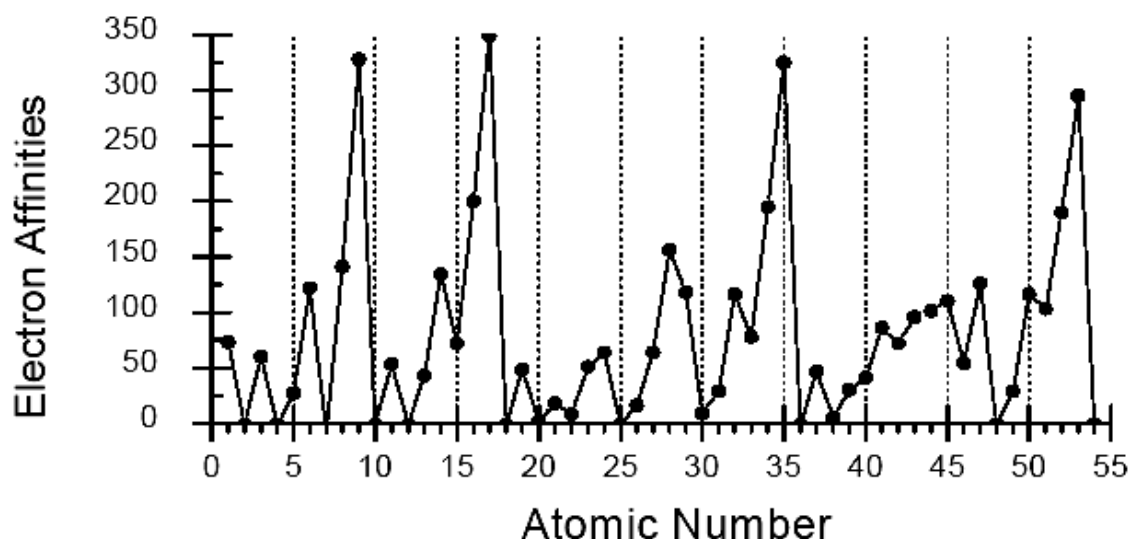
The tendency to accept electrons increases as we move from left to right across the periodic table. Thus Electron Affinity becomes more negative. The E.A of metals are generally more positive (or less negative) than those of non-metals. The values differ little within a group, but the halogens have the most negative E.A values, while the noble gases that have filled outer s and p sub shells have no tendency to accept electrons. The E.A of oxygen has a negative value, which means that the process



While that of the O⁻ is



SIR OTONDE MICHAEL AWITI



3.9 Periodic Trend in Electronegativity

SIR OTONDE MICHAEL AWITI

Electronegativity χ

The tendency of an atom to attract electrons to itself in a chemical bond is referred to as **electronegativity**. The greater the electronegativity of an atom, the more strongly the atom attracts the electrons of a bond. This concept was proposed by Linus Pauling in 1937.

Electronegativity cannot be calculated accurately or measured directly. However, we expect it to depend on the magnitude of the charge and on the distance of the bonding pair of electrons from the nucleus. There are two important trends in electronegativity within the periodic table: -

- (a) electronegativity increases across a period as the charge increases
- (b) electronegativity generally decreases from top to bottom in a group; because with each successive shell, the bonding electrons are further from the nucleus.

Because the electronegativity of an atom cannot be defined quantitatively, it cannot be given a precise value but approximate values. Table shows the electronegativity values

Table; Electronegativity χ of some elements

Group \ Period	1	2	13	14	15	16	17	18
1	H 2.2							He -
2	Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.1	O 3.5	F 4.1	Ne -
3	Na 1.0	Mg 1.2	Al 1.3	Si 1.7	P 2.1	S 2.4	Cl 2.8	Ar -
4	K 0.9	Ca 1.0	Ga 1.8	Ge 2.0	As 2.2	Se 2.5	Br 2.7	Kr 3.1
5	Rb 0.9	Sr 1.0	In 1.5	Sn 1.7	Sb 1.8	Te 2.0	I 2.2	Xe 2.4

Increase of electronegativity----->

Important application: Could use χ to estimate bond polarity

- Atoms with similar electronegativity ($\Delta\chi \leq 0.4$) form non-polar bonds.
- Atoms whose electronegativity differs by more than 2 form ionic bonds.
- Atoms whose electronegativity differs by less than 2 form covalent bonds.



3.10 Common Oxidation State of the Elements

Oxidation number is the apparent charge assigned to an atom in a molecule or in a compound.

Some oxidation numbers are fixed:

SIR OTONDE MICHAEL AWITI

Elements	Oxidation number
O	-2 except in peroxide
H	+1 except in hydride
Group 1 metals	+1
Group 2 metals	+2
Halogens	-1 except in oxygen compounds



3 SUMMARY

- 1 The *experimental* basis of the periodic table of the elements is periodic law: Certain properties recur periodically when the elements are arranged by increasing atomic number.
- 2 The *theoretical* basis is that the properties of an element are related to the electron configuration of its atoms, and elements in the same group of the periodic table have similar electron configuration.

SIR OTONDE MICHAEL AWITI

Lecture 4



Lecture 4 Objectives or Aims

At the end of this unit you should be able to;

- (a) describe bonding in terms of Lewis structures
- (b) use VSEPR to explain shapes of simple molecules
- (c) use valence bond theory and hybridization to describe the chemical bond



4.0 Introduction

We know a lot about chemical compound. We can determine their compositions and formulas. We can represent reactions of compounds by chemical equations. We can do all that without really having to consider the ultimate structure of atoms and molecules. We know the shape of molecules that is arrangements of atoms in space define the chemistry of molecules. In this lecture we will learn about the interaction between atoms called chemical bonds. Most of our discussion will centre on Lewis theory, valence bond theory and hybridization.

SIR OTONDE MICHAEL AWITI



4.0 Electronic Structure and Chemical Bonding

There are four different but related approaches to explain how atoms combine to form covalent bond:

- 1) Lewis dot structure.
- 2) Valence Shell Electron Pair Repulsion theory (VSEPR).
- 3) Valence bond theory (VBT) and hybridization of atomic orbital.
- 4) Molecular Orbital (MOT) Theory.

In this Lecture we will describe the interactions between atoms called chemical bonds. Most of our discussion will centre on the simplest method of representing chemical bonding, known as the Lewis theory. We will explore, however, another relatively simple theory, one for prediction probable shape-Valence – Shell Electron – Pair Repulsion (VSEPR) theory. The subject of chemical bonding in more depth will be examined in SCH 200-(Atomic Structure and Chemical Bonding), especially molecular orbital theory.



4.1 Lewis Theory

Some fundamental ideas in Lewis theory are:

1. Electrons, especially those of the outermost (valence) electronic shell, play a fundamental role in chemical bonding.
2. In some cases electrons are *transferred* from one atom to another. Positive and negative ions are formed and attract each other through electrostatic forces called **ionic bonds**.
3. In other cases one or more pairs of electrons are shared between atoms; this *sharing* of electrons is called a **covalent bond**.

4. Electrons are transferred, or shared, in such a way that each atom acquires an especially stable electron configuration. Usually this is a noble gas configuration, one with eight outer shell electrons, or an **octet**.



4.1.1 Lewis Symbols and Lewis Structures

A **Lewis symbol** consists of a chemical symbol to represent the nucleus and *core* (inner shell) electrons of an atom, together with dots placed around the symbol to represent the *valence* (outer shell) electrons.

Examples.

Lewis structure for: Na electron configuration $[\text{Ne}]3s^1$ is



N electron configuration $[\text{He}]2s^22p^3$ is



O electron configuration $[\text{He}]2s^22p^4$ is



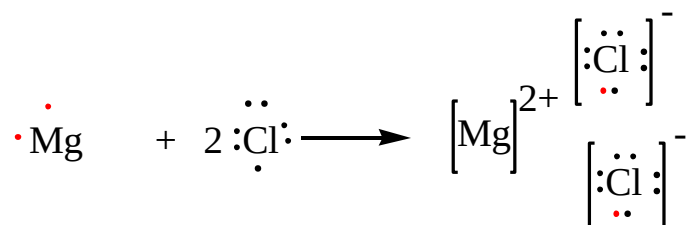
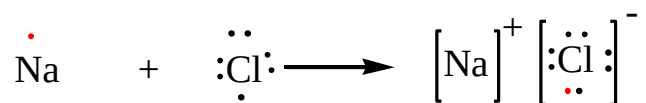
SELF-TEST QUESTIONS

- 1 Write the Lewis symbols for the following elements: H, He, Li, C, N, O, F, Ne, Cl, Na, Mg, Al, K and Ca

A **Lewis structure** is a combination of Lewis symbols that represent either the transfer or sharing of electrons in a chemical bond.

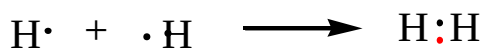


4.1.2 Ionic bonding (transfer of electrons)



4.1.3 Covalent bonding (sharing of electrons)

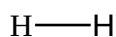
Hydrogen molecule is written showing a pair of dots between hydrogen atoms, indicating that the hydrogen share the pair of electrons in covalent bond



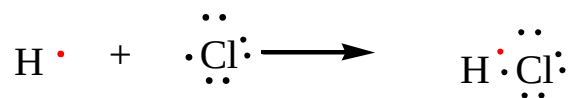
Two hydrogen
atom

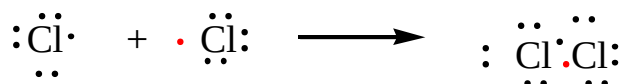
A hydrogen
molecule

The shared pair is usually represented by a line. E.g.



Atoms other than hydrogen also form covalent bonds





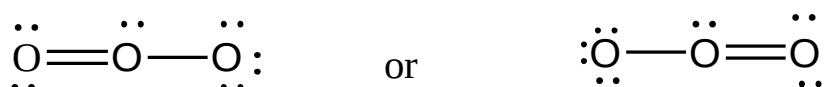
SELF-TEST QUESTIONS

- 1 Write Lewis structures of the following ionic compounds:
 - a) BaO; b) MgCl₂; c) Aluminium oxide
- 2 Write Lewis structures of the following covalent compounds:
 - a) F₂; b) H₂O; c) NH₃ d) CH₄ e) O₂ f) N₂



4.1.4 Lewis Structures and Resonance

The Lewis structure of ozone O₃ are:



Which of the two Lewis structure for O₃ is correct?

In fact, neither is correct by itself. Whenever it is possible to write more than one Lewis structure for a molecule the actual electronic structure is an average of the various possibilities called a **resonance hybrid**.

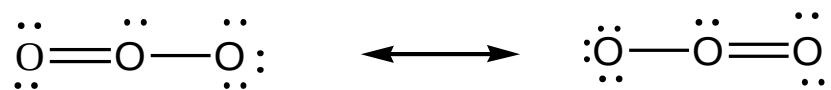
Ozone does not have:

- One O=O double bond and
- One O-O single bond as individual structure imply, rather ozone has two equivalent O-O bonds that one can think of as having a *bond order* of 1.5 midway between pure single bond and pure double bond. Both have an identical length of 1.28 Å.

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

The idea of resonance between two or more Lewis structure is indicated by drawing the individual Lewis structure and using double headed “resonance arrow” to show that both contribute to the resonance hybrid.



SELF-TEST QUESTIONS

Draw the resonance structure of carbonate ion, CO_3^{2-} .



4.2 Molecular Geometry and Bonding Theories

(Molecular geometry or shape of molecules)

Lewis structures say nothing about bond angles.



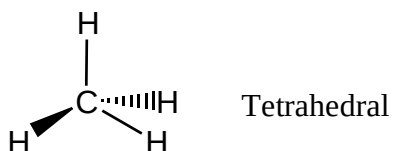
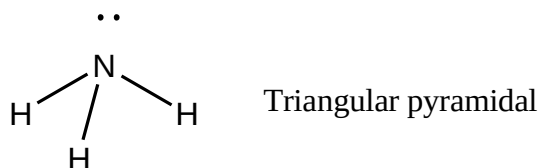
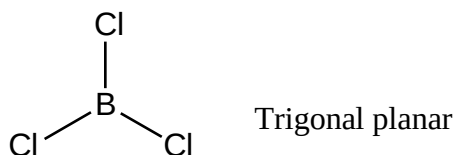
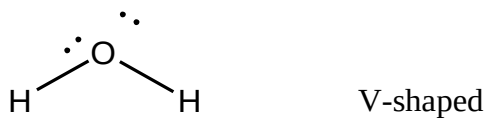
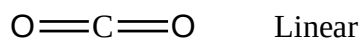
4.2.1 Valence Shell Electron Pair Repulsion Theory (VSEPR)

- ❖ A structure should be considered which let all the electron pairs of the valence shell of central atom try to get further away from each other. Repulsion between lone pairs is greater than those between bonding electrons.

The order of repulsive energy is:

$\text{Lp-Lp} > \text{Lp-Bp} > \text{Bp-Bp}$

SIR OTONDE MICHAEL AWITI



4.2.3 Covalent Bonding and Orbital Overlap

The VSEPR model is a simple method which allows us to predict molecular geometry BUT it does NOT explain why bond exist between atoms.

- How can we explain molecular geometry and basis of bonding at the same time?
 - Quantum mechanism and molecular orbitals are used.

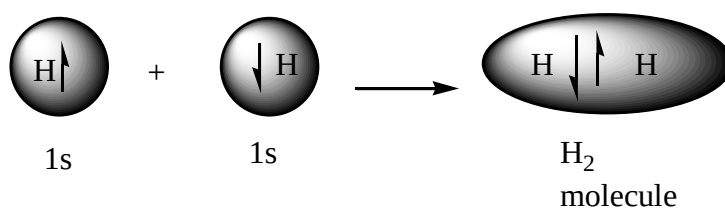


4.3 Valence Bond Theory (VBT)

- Combine Lewis idea of electron pair bonds with electron orbitals (quantum mechanic).
- Covalent bond occurs when atom shares electrons.
- Concentrate electron density between nuclei.
- The build up of electrons density between two nuclei occurs when a valence atomic orbital of one atom overlap with one of another.
- The orbital share a region of space i.e. they overlap.
- The overlap of orbital allows two electron of opposite spin to share the common space between the nuclei forming a covalent bond.

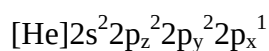


- In the hydrogen molecule for instance the H-H bond results from overlap of two singly occupied H 1s orbitals.



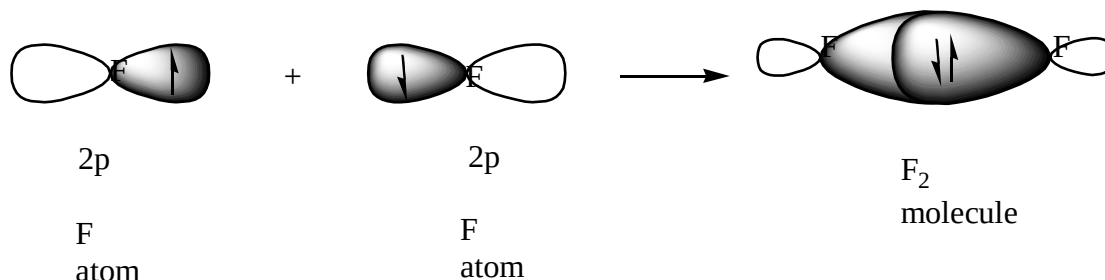
- In the valence bond model, the strength of covalent bond depends on amount of overlap. The greater the overlap, the stronger the bond. This in turn, means that bond formed by overlap other than s- orbitals have directionalities to them.

In the fluorine molecule, F_2 , for instance each atom has electron configuration:

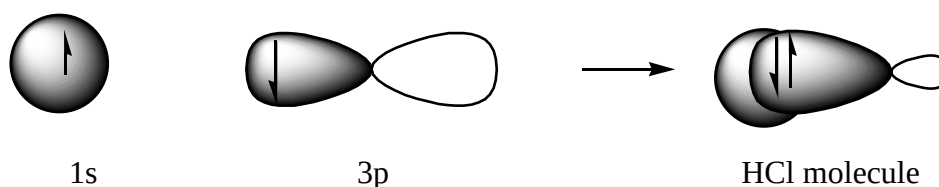


SIR OTONDE MICHAEL AWITI

Meaning that the fluorine bond results from overlap of two singly occupied two 2p orbitals. The 2p orbitals must point directly to one another for optimum overlap to occur, and the F-F form bond along the orbitals.



In HCl, the covalent bond involves overlap of 1s orbital (non-directional) with chlorine 3p orbital and formed along p-axis.



The key ideas of valence bond theory can be summarized by few statements:

- Covalent bonds are formed by overlap of atomic orbitals, each of which contains 1 electron of opposite spin.
- Each of the bonded atoms maintains its own atomic orbitals, but the electron pair in the overlapping pair is shared by both atoms.
- The greater the amount of orbital overlap the stronger the bond. This leads to a directional character to the bond when other than s – orbitals are formed.



4.4 Molecular Geometry and Hybrid Orbitals

SIR OTONDE
MICHAEL AWITI

Hybrid Orbitals

For polyatomic molecules we would like to be able to explain:

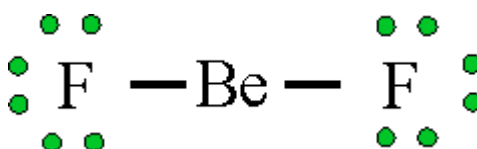
The number of bonds formed

Their geometries

sp Hybrid Orbitals

SIR OTONDE MICHAEL AWITI

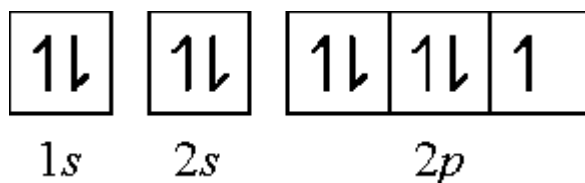
Consider the Lewis structure of gaseous molecules of BeF_2 :



- The VSEPR model predicts this structure will be linear
- What would **valence bond theory** predict about the structure?

The fluorine atom electron configuration:

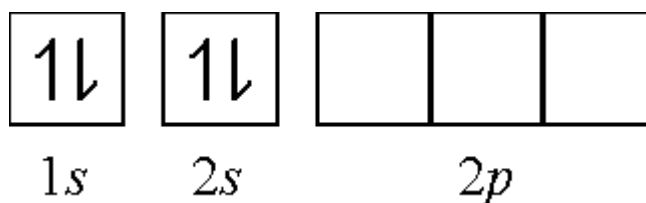
- $1s^2 2s^2 2p^5$



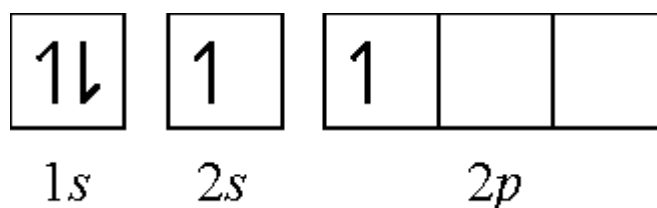
- There is an unpaired electron in a $2p$ orbital
- This unpaired $2p$ electron can be paired with an unpaired electron in the Be atom to form a covalent bond

The Be atom electron configuration:

- $1s^2 2s^2$



- In the ground state, there are no unpaired electrons (the Be atom is incapable of forming a covalent bond with a fluorine atom)
- However, the Be atom could obtain an unpaired electron by promoting an electron from the 2s orbital to the 2p orbital:

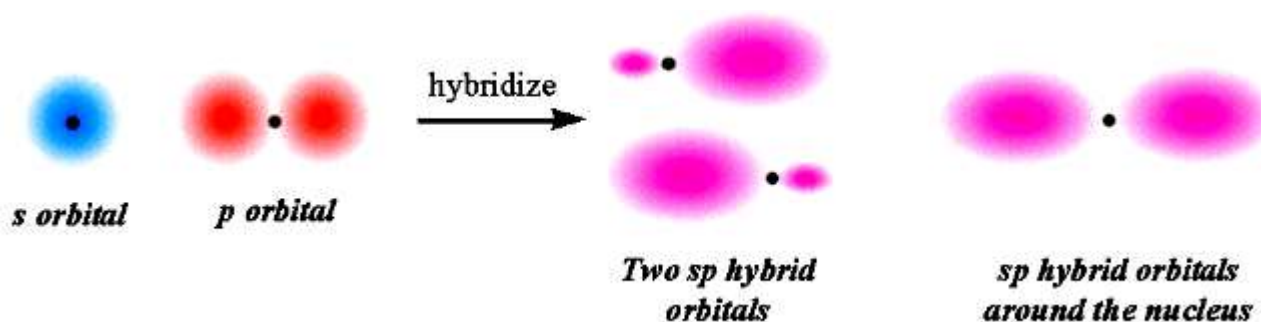


This would actually result in two unpaired electrons, one in a 2s orbital and another in a 2p orbital

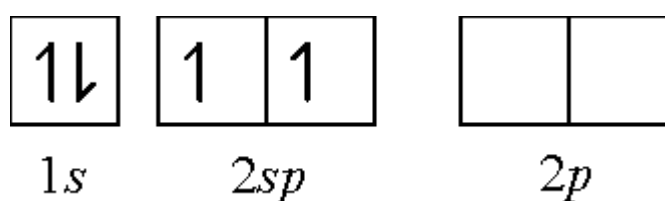
- The Be atom can now form two covalent bonds with fluorine atoms
- We would not expect these bonds to be identical (one is with a 2s electron orbital, the other is with a 2p electron orbital)

However, the structure of BeF_2 is linear and the bond lengths are identical

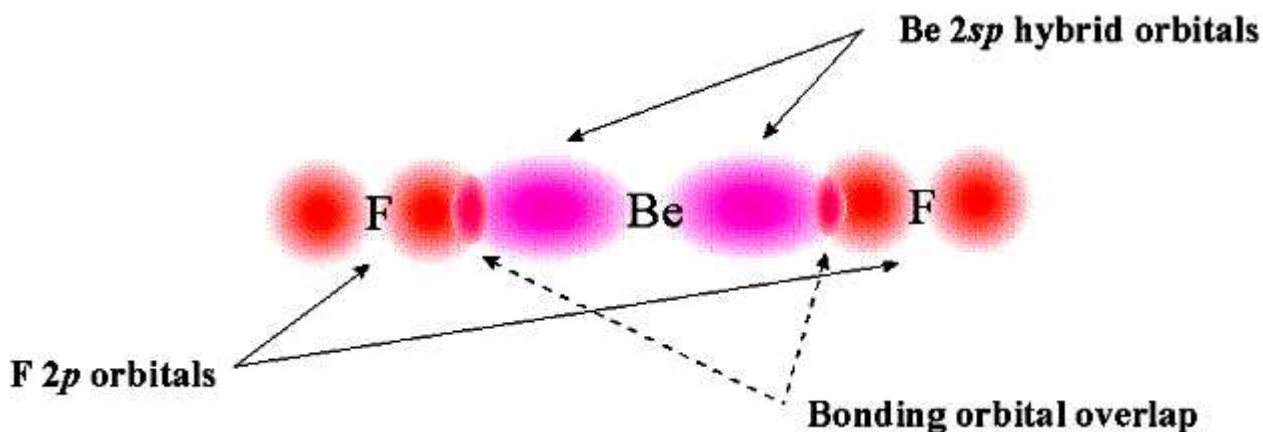
- We can *combine* wavefunctions for the 2s and 2p electrons to produce a "hybrid" orbital for both electrons
- This hybrid orbital is an "sp" hybrid orbital



- The orbital diagram for this hybridization would be represented as:



SIR OTONDE MICHAEL AWITI



Note:

- The Be $2sp$ orbitals are identical and oriented 180° from one another (i.e. bond lengths will be identical and the molecule linear)
- The promotion of a Be $2s$ electron to a $2p$ orbital to allow sp hybrid orbital formation **requires energy**.
 - The elongated sp hybrid orbitals have one large lobe which can overlap (bond) with another atom more effectively
 - This produces a stronger bond (higher bond energy) which offsets the energy required to promote the $2s$ electron

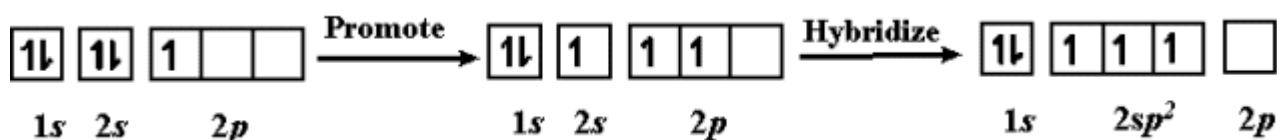
sp^2 and sp^3 Hybrid Orbitals

Whenever orbitals are mixed (hybridized):

- The number of hybrid orbitals produced **is equal to the sum of the orbitals being hybridized**
- Each hybrid orbital is identical **except that they are oriented in different directions**

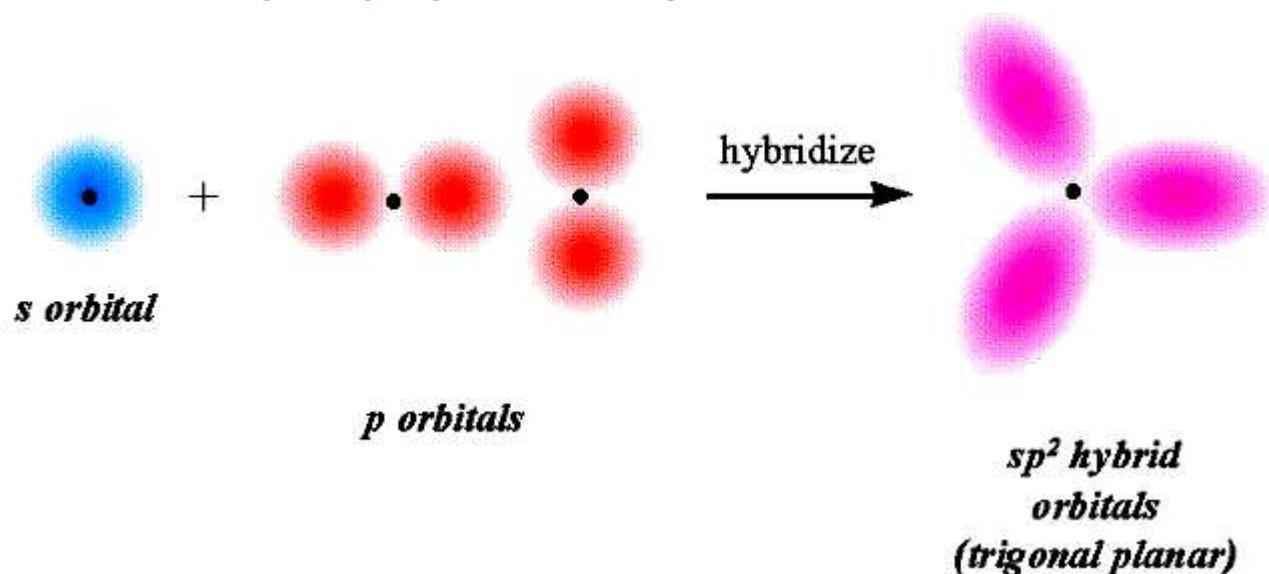
BF_3

Boron electron configuration:



- The three sp^2 hybrid orbitals have a trigonal planar arrangement to minimize electron repulsion

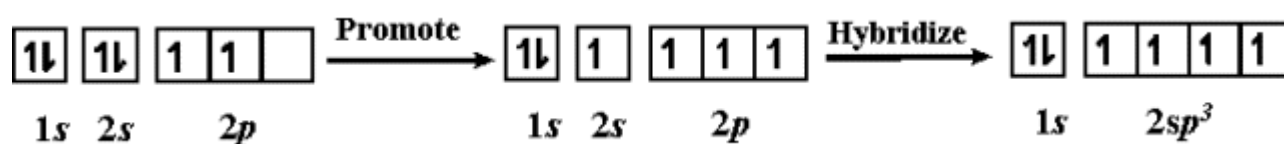
SIR OTONDE MICHAEL AWITI



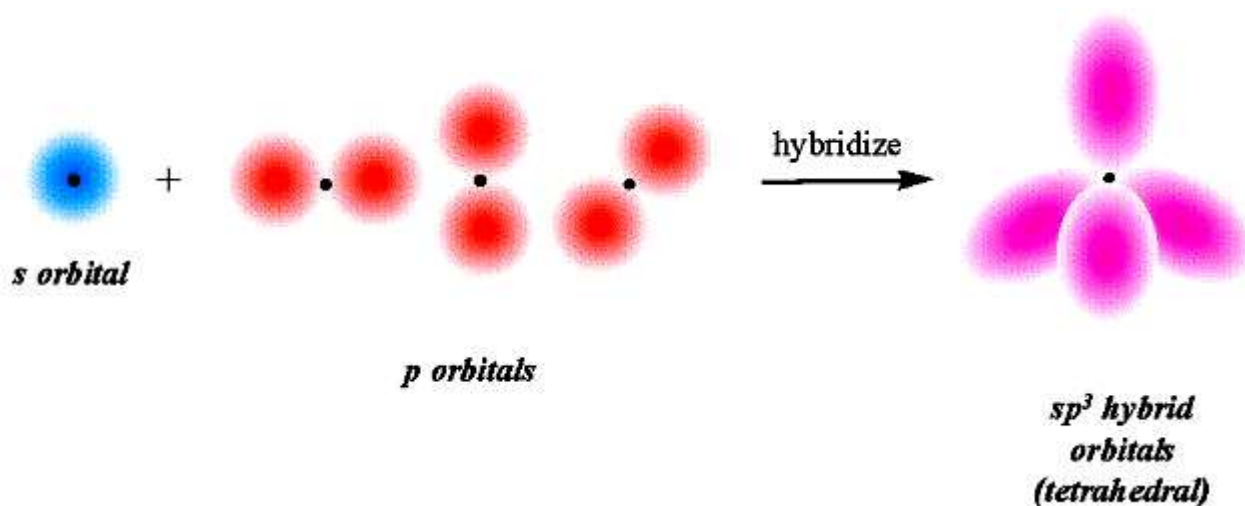
NOTE: sp^2 refers to a hybrid orbital being constructed from one s orbital and two p orbitals. Although it looks like an electron configuration notation, the superscript '2' DOES NOT refer to the number of electrons in an orbital.

- An s orbital can also mix with all 3 p orbitals in the same subshell

CH₄



SIR OTONDE MICHAEL AWITI



Thus, using **valence bond theory**, we would describe the bonds in methane as follows: each of the carbon sp^3 hybrid orbitals can overlap with the 1s orbitals of a hydrogen atom to form a bonding pair of electrons

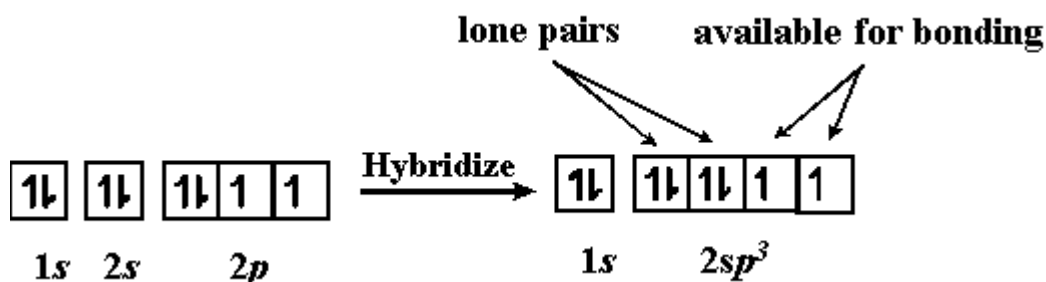
NOTE: sp^3 refers to a hybrid orbital being constructed from one s orbital and three p orbitals. Although it looks like an electron configuration notation, the superscript '3' **DOES NOT** refer to the number of electrons in an orbital.

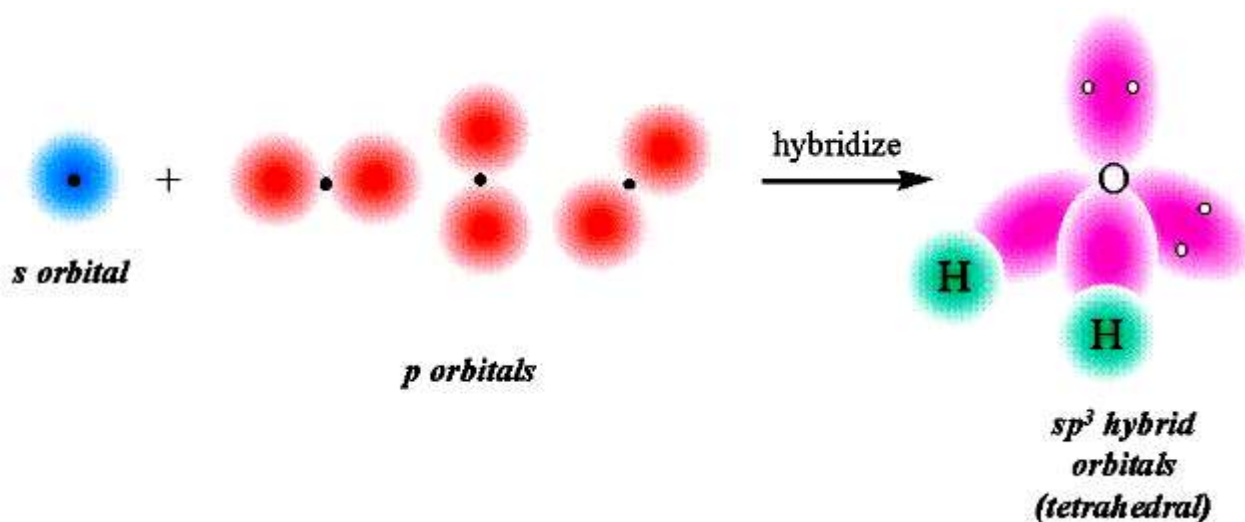
ANOTHER NOTE: the two steps often observed when constructing hybrid orbitals is to 1) promote a valence electron from the ground state configuration to a higher energy orbital, and then 2) hybridize the appropriate valence electron orbitals to achieve the desired valence electron geometry (i.e. the correct number of hybrid orbitals for the appropriate valence electron geometry)

H₂O

SIR OTONDE MICHAEL AWITI

Oxygen

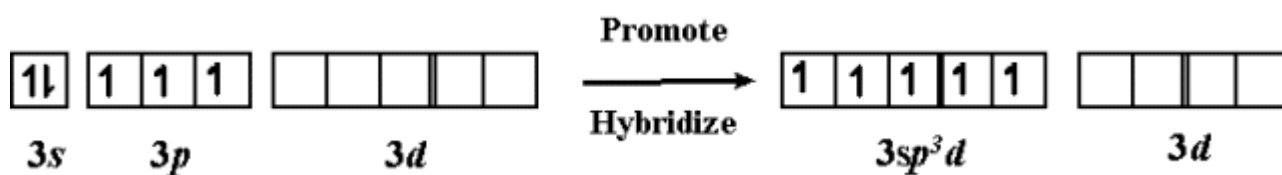




4.5 Hybridization Involving d Orbitals

Atoms in the third period and higher can utilize *d* orbitals to form hybrid orbitals

PF_5



SIR OTONDE MICHAEL AWITI

Trigonal Bipyramidal

Similarly hybridizing one s , three p and two d orbitals yields six identical hybrid sp^3d^2 orbitals. These would be oriented in an **octahedral geometry**.

- Hybrid orbitals allows us to use **valence bond theory** to describe covalent bonds (sharing of electrons in overlapping orbitals of two atoms)
- When we know the molecular geometry, we can use the concept of hybridization to describe the electronic orbitals used by the central atom in bonding

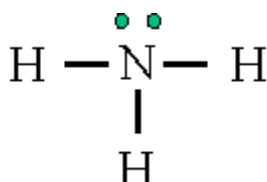
Steps in predicting the hybrid orbitals used by an atom in bonding:

1. Draw the Lewis structure
2. Determine the electron pair geometry using the VSEPR model
3. Specify the hybrid orbitals needed to accommodate the electron pairs in the geometric arrangement

NH_3

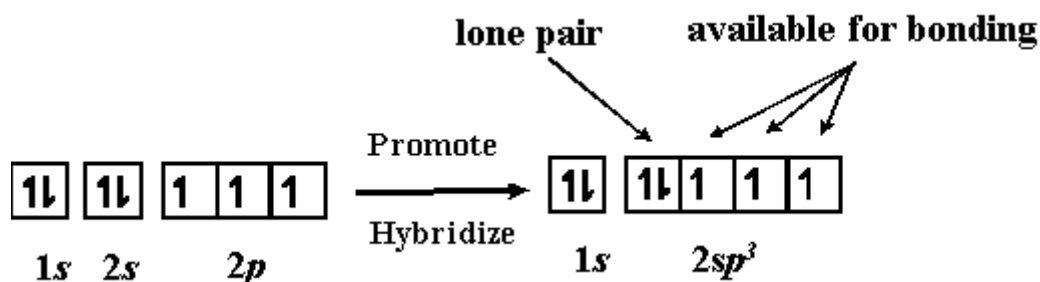
SIR OTONDE MICHAEL AWITI

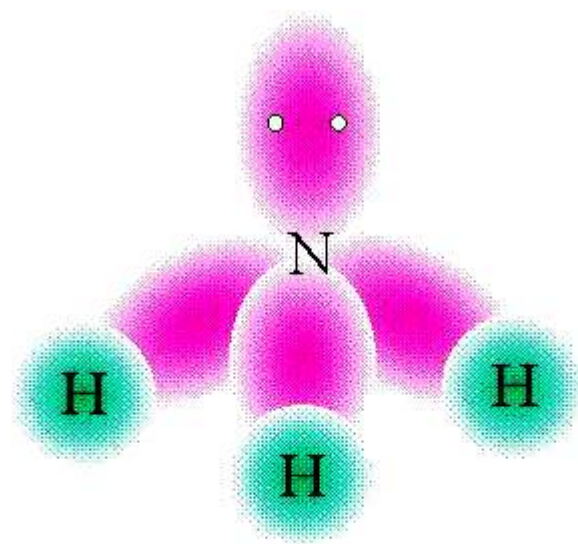
1. Lewis structure



2. **VSEPR** indicates **tetrahedral geometry** with one non-bonding pair of electrons (structure itself will be **trigonal pyramidal**)

3. Tetrahedral arrangement indicates four equivalent electron orbitals





*sp³ hybrid
orbitals
(tetrahedral)*

Valence Electron Pair Geometry	Number of Orbitals	Hybrid Orbitals
Linear	2	Sp
Trigonal Planar	3	Sp^2
Tetrahedral	4	Sp^3
Trigonal Bipyramidal	5	sp^3d
Octahedral	6	sp^3d^2

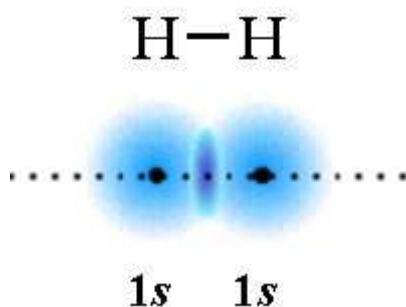


4.6 Multiple Bonds and Orbital Overlaps

Two types of bond

Sigma (σ) and pi (π) bonds

The "inter-nuclear axis" is the imaginary axis that passes through the two nuclei in a bond:

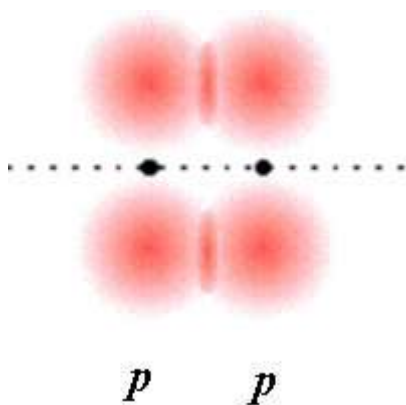


The covalent bonds we have been considering so far exhibit bonding orbitals which are *symmetrical about the inter-nuclear axis* (either an s orbital - which is symmetric in all directions, or a p orbital that is pointing along the bond towards the other atom, or a hybrid orbital that is pointing along the axis towards the other atom)

Bonds in which the electron density is symmetrical about the inter-nuclear axis are termed "sigma" or " σ " bonds

In multiple bonds, the bonding orbitals arise from a different type arrangement:

- Multiple bonds involve the overlap between two p orbitals
- These p orbitals are oriented perpendicular to the inter-nuclear (bond) axis



This type of overlap of two p orbitals is called a "pi" or " π " bond. Note that this is a single π bond (which is made up of the overlap of two p orbitals)

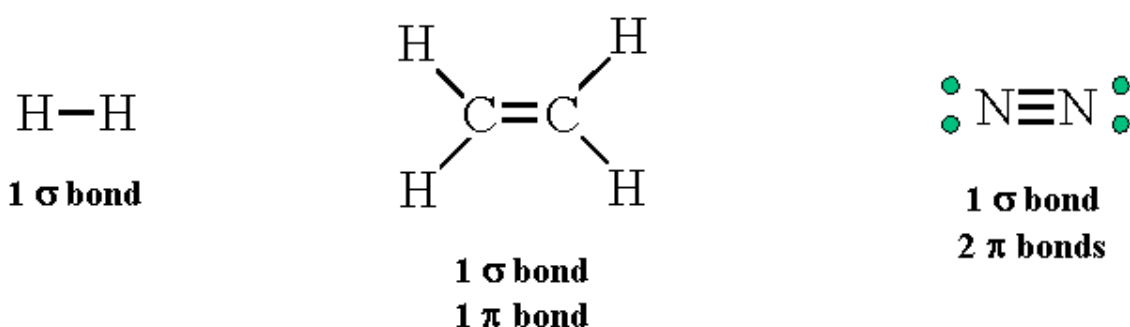
In π bonds:

SIR OTONDE MICHAEL AWITI

- The overlapping regions of the bonding orbitals lie *above* and *below* the inter-nuclear axis (there is no probability of finding the electron in that region)
- The size of the overlap is smaller than a σ bond, and thus the bond strength is typically less than that of a σ bond

Generally speaking:

- A single bond is composed of a σ bond.
- A double bond is composed of *one* σ bond and *one* π bond.
- A triple bond is composed of *one* σ bond and *two* π bonds.

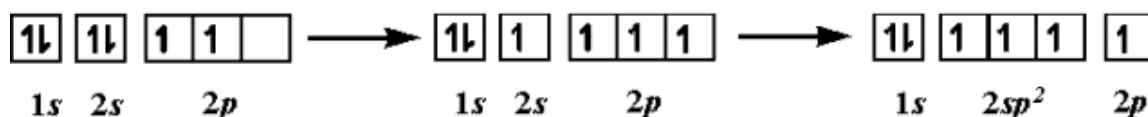


C_2H_4 (ethylene; see structure above)

- The arrangement of bonds suggests that the geometry of the bonds around each carbon is **trigonal planar**
- Trigonal planar suggests sp^2 hybrid orbitals are being used (these would be σ bonds)

What about the electron configuration?

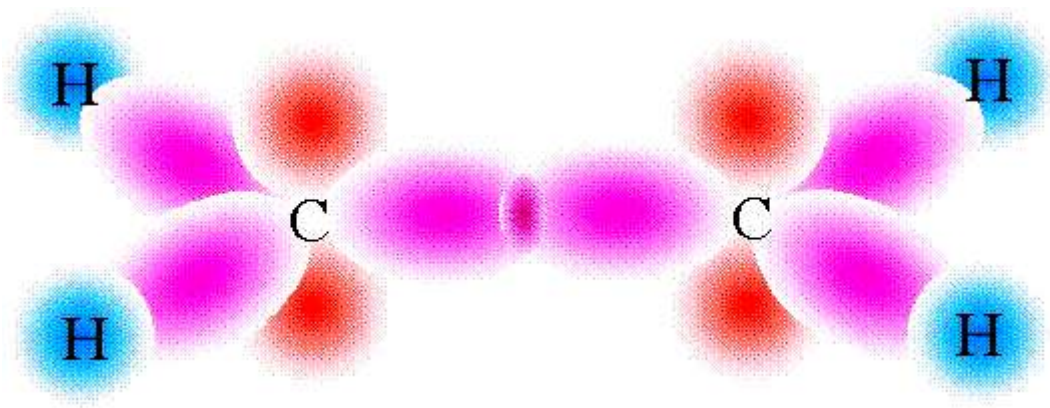
Carbon: $1s^2 2s^2 2p^2$



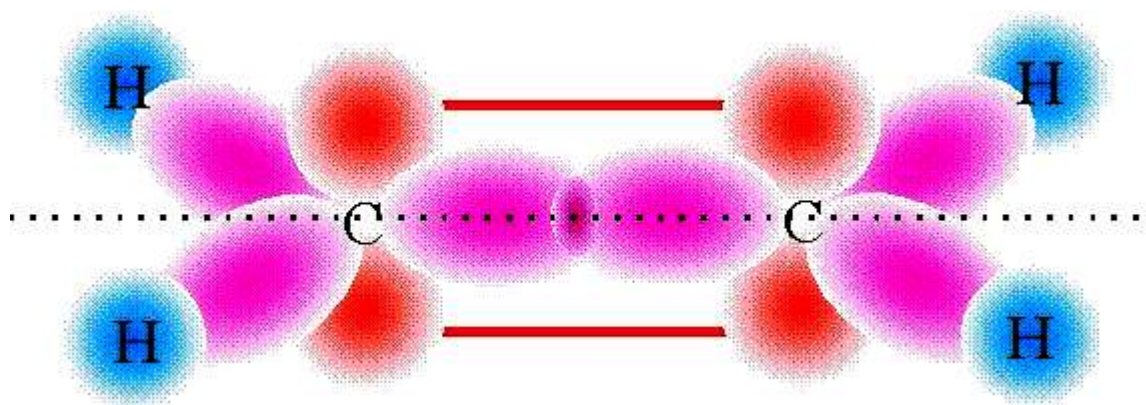
- Thus, we have an extra unpaired electron in a p orbital available for bonding
- This extra p electron orbital is oriented perpendicular to the plane of the three sp^2 orbitals (to minimize repulsion):

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI



- The unpaired electrons in the p orbitals can overlap one another above and below the internuclear axis to form a covalent bond



- This interaction above and below the inter-nuclear axis represents the *single* π bond between the two p orbitals

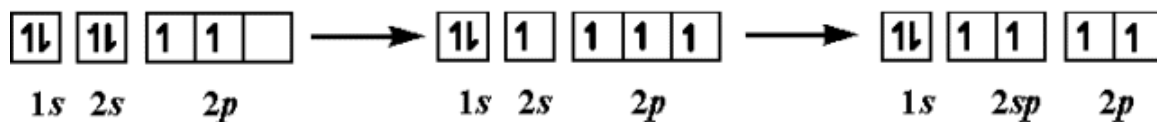
Experimentally:

- we know that the 6 atoms of ethylene lie in the same plane.
- If there was a single σ bond between the two carbons, there would be nothing stopping the atoms from rotating around the C-C bond.
- But, the atoms are held rigid in a planar orientation.
- This orientation allows the overlap of the two p orbitals, with formation of a π bond.
- In addition to this rigidity, the C-C bond length is shorter than that expected for a single bond.
- Thus, extra electrons (from the π bond) must be situated between the two C-C nuclei.

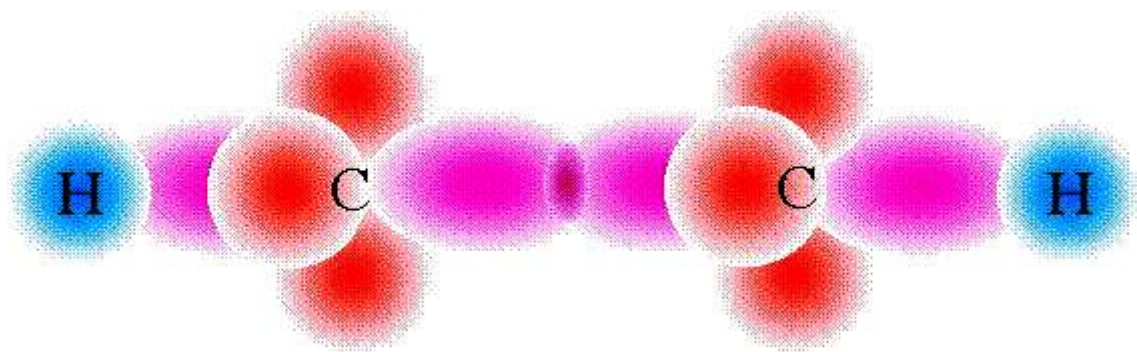
C_2H_2 (acetylene)



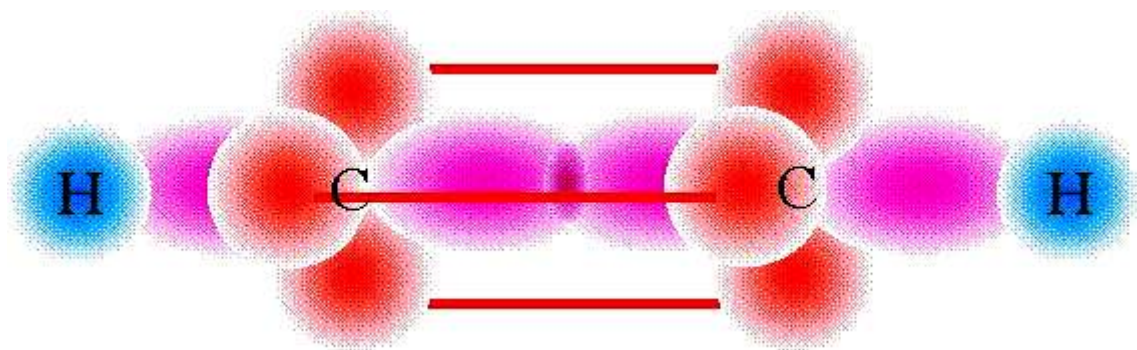
- The linear bond arrangement suggests that the carbon atoms are utilizing sp hybrid orbitals for bonding



- This leaves two unpaired electrons in p orbitals
- To minimize electron repulsion, these p orbitals are at right angles to each other, and to the inter-nuclear axis:



- These p orbitals can overlap to form two π bonds in addition to the single σ bond (forming a triple bond)



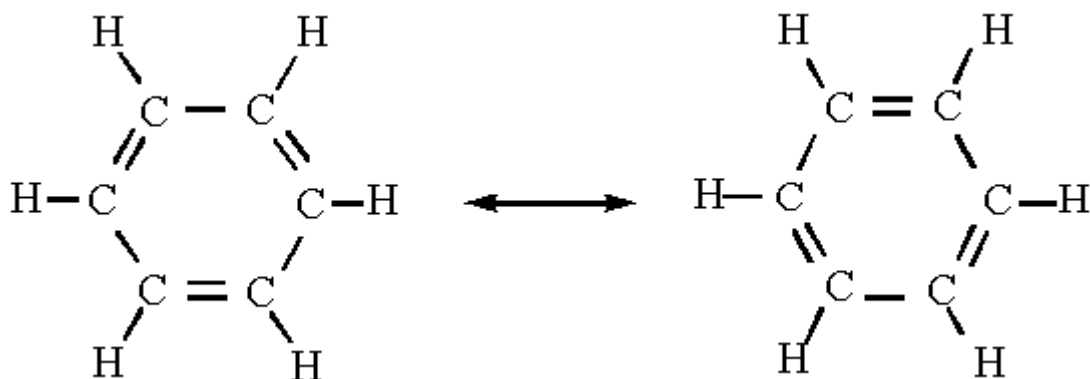
Delocalized Bonding

localized electrons are electrons which are associated completely with the atoms forming the bond in question

SIR OTONDE MICHAEL AWITI

In some molecules, particularly with resonance structures, we cannot associate bonding electrons with specific atoms

C_6H_6 (Benzene)



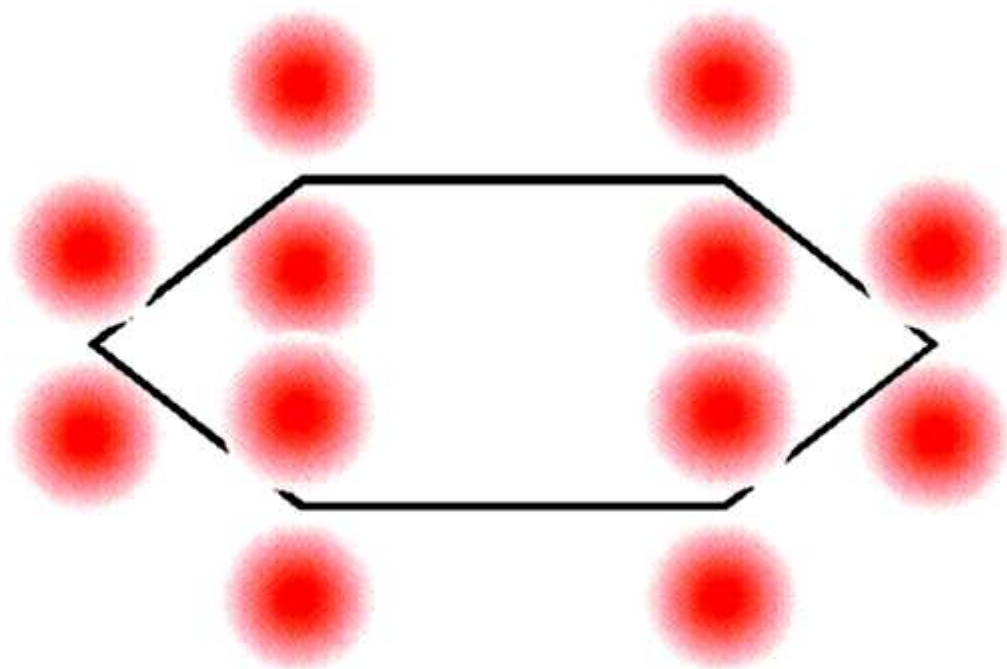
Benzene has two resonance forms

- The six carbon - carbon bonds are of equal length, intermediate between a single bond and double bond
- The molecule is planar
- The bond angle around each carbon is approximately 120°

The apparent hybridization orbital consistent with the geometry would be sp^2 (trigonal planar arrangement)

- This would leave a single p orbital associated with each carbon (perpendicular to the plane of the ring)

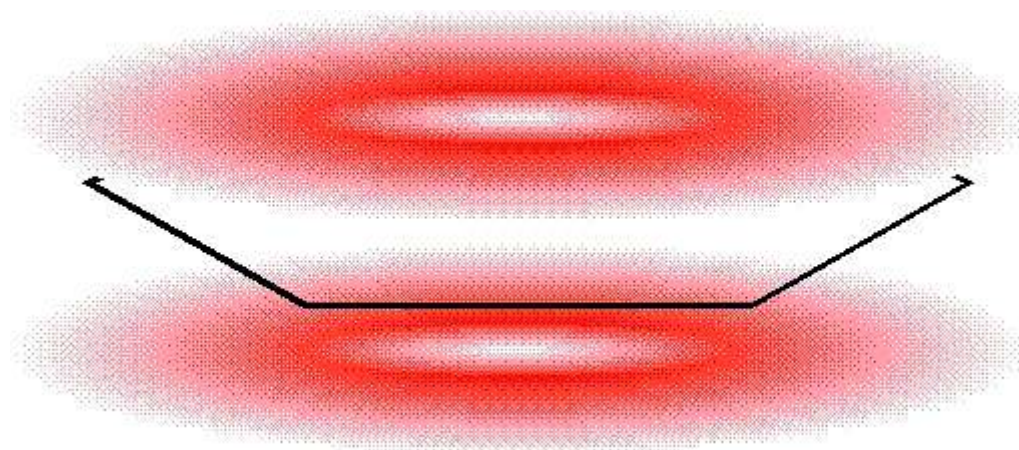
SIR OTONDE
MICHAEL AWITI



With six p electrons we could form three discrete π bonds

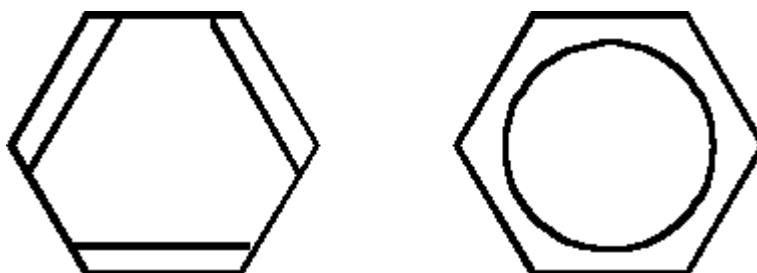
- However, this would result in three double bonds in the ring, and three single bonds
- This would cause the bond lengths to be different around the ring (which they are not)
- This would also result in one resonance structure being the only possible structure

The best model is one in which the π electrons are "smeared" around the ring, and not localized to a particular atom



- Because we cannot say that the electrons in the π bonds are localized to a particular atom they are described as being **delocalized** among the six carbon atoms

Benzene is typically drawn in two different ways:

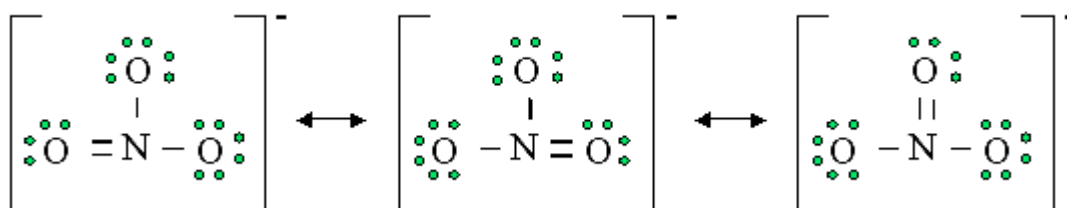


- The circle indicates the delocalization of the p bonds



4.7 Resonance Structures and Hybrid Orbitals

Structure of NO_3^-

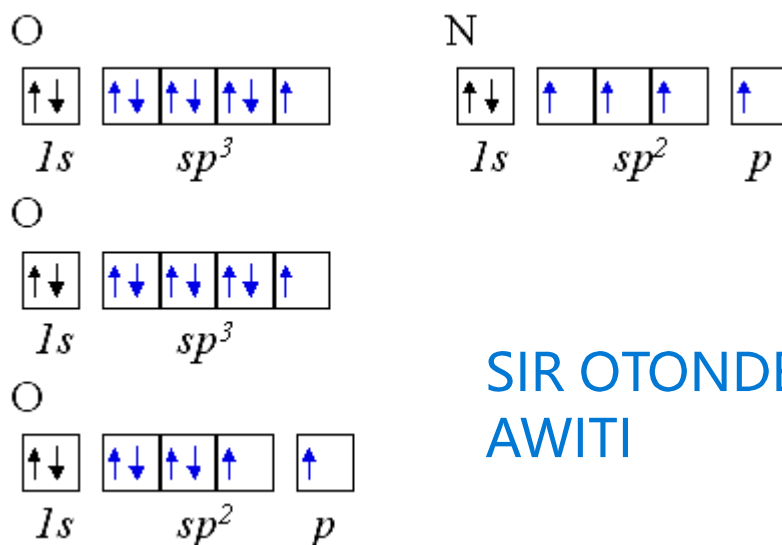


The Lewis structure of NO_3^- ion suggests that three resonance structures describe the molecular structure

- For any individual Lewis resonance structure the electronic structure for the central N atom is predicted to be sp^2 hybrid orbitals participating in σ bonds with each of the O atoms, and an electron in a p orbital participating in a π bond with one oxygen (forming a double bond)
- Two of the O atoms are predicted to have sp^3 hybrid orbitals, with one orbital participating in a σ bond with the central N atom and the other orbitals filled with non-bonding electron pairs. The other O atom is predicted to have sp^2 hybrid orbitals, with one orbital participating in a π bond with the central N and two orbitals filled with non-bonding pairs of electrons. Furthermore, this last O atom is participating in a

double bond with the central N atom and therefore should have an electron in a p orbital to participate in a π bond with the central N

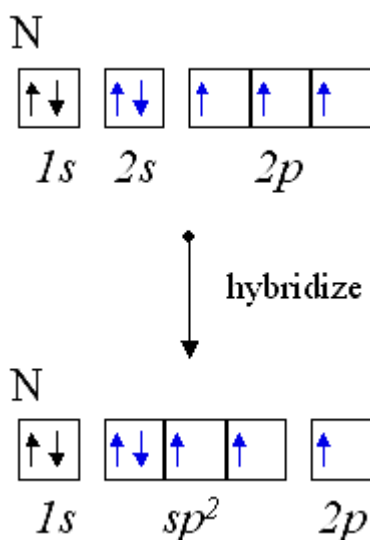
How will this arrangement look as far as the orbital diagrams?



SIR OTONDE MICHAEL
AWITI

- There are 24 valence electrons in the expected valence orbitals above
- Summing the valence electrons from the formula gives: (3×6) for O, plus 5 for N, plus 1 for ionic charge = 24

What might we expect for the electron configuration if we just started with the N atom?



SIR OTONDE
MICHAEL AWITI

- We would predict that the N can only make two σ bonds, it would have one pair of non-bonding electrons, and a p electron left over to participate in a π bond with one of the π bonds
- This is different from what the Lewis structure shows, and from our prediction of hybrid orbitals from the expected geometry
- If we look at the sp^3 O atoms above we see that they actually have 7 electrons (1 *more* than expected), while the sp^2 O atom has the expected 6. Furthermore, the N atom (in the correct sp^2 configuration) has 4 electrons (1 *less* than expected)
- The "extra" electron from the ionic charge is correctly accounted for in the summation of electrons

Thus, the correct way to determine electron configurations appears to be:

- begin by predicting the hybridization orbitals
- then determine lone pair arrangements and s and p bonding electrons for each atom
- confirm that all bonding electrons are correct and that the total of electrons is correct



SELF-TEST QUESTIONS

- What is molecular geometry? Why is the study of molecular geometry important?
- Describe the state of hybridization of arsenic in arsenic pentafluoride (AsF_5).
- What is the difference between a Lewis dot symbol and a Lewis structure
- Define the following terms: lone pairs, the octet rule, sigma bond, pi bond and bond length,

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

Lecture 5



Lecture 5 Objectives or Aims

At the end of this unit you should be able to;

- (a) State Arrhenius definition of an acid and bases
- (b) describe Arrhenius, Brønsted and Lewis definitions of acids and bases
- (c) calculate the pH of acids
- (d) explain the buffers solutions



5.0 Introduction

The ideas about acids and bases (or alkalis) date back in ancient times. In secondary school we learnt a number of things about acids and bases. In this lecture we will learn modern acid-base theories, factors affecting the strength of acids and bases, the pH scale and the calculation the pH.



5.0 Acid and Bases

Arrhenius definition of an acid is a substance that provides hydrogen ions, H^+ , when dissolved in water, and one definition of a base is a substance that provides hydroxide ions, OH^- , when dissolved in water. Arrhenius definition of a base is restricted to those substances which react with H^+ ion to form water. In order to widen the scope of acid- base reaction and

SIR OTONDE MICHAEL AWITI

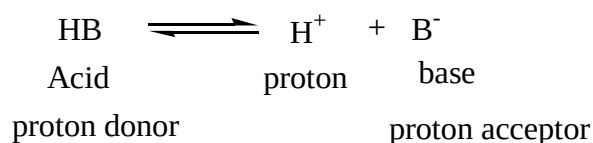
include non-aqueous systems and a wider range of bases, Brønsted and Lowry independently suggested the following definition in 1923.

An acid is a proton donor

And

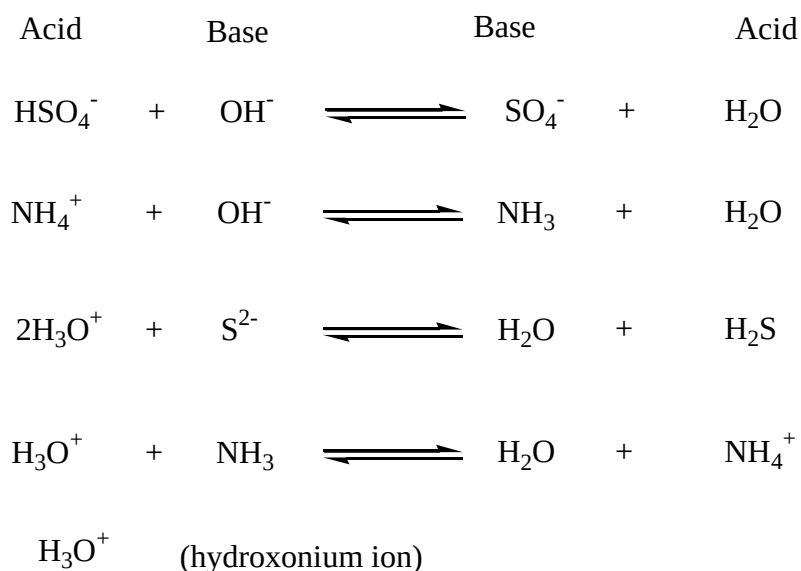
A base is a proton acceptor,

Thus the relationship between an acid and its corresponding base is



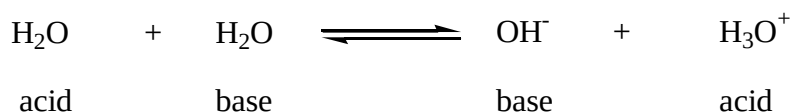
HB and B⁻ are said to be conjugate and to form conjugate acid – base pair. HB is the conjugate acid of B⁻ and B⁻ is the conjugate base of HB.

According to Brønsted – Lowry theory acid salts such as NaHSO₄ and ammonium ion (NH₄⁺) are recognized as acid. What is more the definition of a base includes: all anions, water and ammonia as well as oxides and hydroxides.



SIR OTONDE MICHAEL AWITI

NB/ On the equations above, water can act as a base and as an acid. It does this simultaneously during its dissociation.



5.1 Lewis Definition

Lewis acid - accept pair of electrons to form a covalent bond, for example BCl_3 and AlCl_3 .

Whereas

Lewis base – donate pair of electrons to form a covalent bond, for example NH_3 , CH_3NH_2 .

Soft acids - large cations with low charge density. e.g. Cs^+ and Fr^+ .

Soft bases - large anions with low charge density. e.g. I^- .

Hard acids - small cations with high charge density. e.g. Al^{3+} .

Hard bases - small anions with high charge density. e.g. F^- .

(Read about Fajan rules)

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI



5.2 Ionic Equilibrium in Aqueous Solutions

Equilibrium involving ions in aqueous solution are important in industrial analytical and biological processes. The principles and characteristics of these ionic equilibria are very similar to those in other systems in chemical equilibria.

Two fundamental types of ionic equilibrium:

1. Equilibrium between undissolved solid solute and its dissolved product in solution are solubility equilibrium.
2. The equilibrium between dissolved undissociated molecules and its dissolved ions, i.e. dissociation equilibria.

In most cases this dissociation equilibria involve either acids or bases.



5.3 The Strength of the Acids and Bases

The strength of the acid is a measure of extent of dissociation.

The strength of different acids or bases can be compared using conductivity measurements.

Strong electrolytes such as HCl and NaOH are almost completely dissociated into ions in aqueous solution. For example



Almost completely dissociated or ionized
Strong acid



Almost completely dissociated or ionized

Strong base

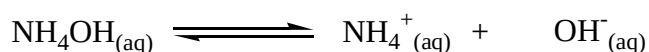
Weak acids and bases are only partially ionized. For example



Ethanoic acid

Only partially dissociated or ionized

weak acid



Only partially dissociated or ionized

Weak base



5.4 Quantitative Measurement of pH

In case of acids, relative strength can be compared by measuring the concentration of hydrogen ions or by measuring their pH.

The “p” in pH comes from German word “potenz” meaning power and “H” from $[\text{H}^+]$ concentration of hydrogen.

The pH of a solution is the negative logarithm to base 10 of the molar hydrogen ion concentration i.e.

$$\text{pH} = -\lg[\text{H}^+_{(\text{aq})}]$$

Notice that the accepted abbreviation for logarithm to base 10 is lg not log.

Some useful mathematical expressions

SIR OTONDE MICHAEL AWITI

$$\lg xy = \lg x + \lg y$$

$$\lg \frac{x}{y} = \lg x - \lg y$$

$$\lg x^n = n \lg x$$

For example $\lg_{10} 2 = 0.3010$ implying $10^{0.3010} = 2$

$[H^+]$ concentration in aqueous solution ranges from 10^{-15} to 10 mol dm^{-3} , so it is convenient to have a scale that is both negative and antilogarithmic. The negative sign produce positive pH values for almost all solutions encountered in practice.

The logarithmic scale reduces the extremely wide variation in $[H^+_{(aq)}]$ concentration to a narrow range of pH from about 15 to -1.

Some worked examples

1. What is the pH of $10^{-1} \text{ mol dm}^{-3} \text{ HCl}$ (0.1 M HCl)?

Solution

Since HCl is fully dissociated and monobasic acid

$[H^+_{(aq)}]$ in $10^{-1} \text{ mol dm}^{-3} \text{ HCl} = 10^{-1} \text{ mol dm}^{-3}$

$$\begin{aligned} pH &= -\lg[H^+_{(aq)}] = -\lg 10^{-1} \\ &= (-1)\lg 10 \\ &= (-1)(-1) = +1 \end{aligned}$$

2. What is the pH of $10^{-3} \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$ (0.001 M H_2SO_4)?

Since H_2SO_4 is fully dissociated and dibasic acid

$[H^+_{(aq)}]$ in $10^{-3} \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4 = 2 \times 10^{-3} \text{ mol dm}^{-3}$

$$pH = -\lg[H^+_{(aq)}] = -\lg(2 \times 10^{-3})$$

$$= (-1) \lg 2 \times 10^3$$

$$= (-1)(0.30 - 3) = +2.7$$

3. The pH of pure water at 25 °C is 7. What is the hydrogen ion concentration?

Solution

$$pH = -\lg[H^+_{(aq)}]$$



SELF-TEST QUESTIONS

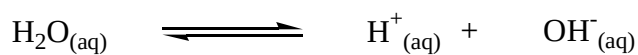
What are the pH values of the following solutions?

- a) $10^{-3} \text{ mol dm}^{-3} \text{ HCl}$
- b) $1.0 \text{ mol dm}^{-3} \text{ HCl}$
- c) 3 M HX which is only 50% dissociated



5.5 The Dissociation of Water

Water dissociate to form ions



The concentration of ions is very small as the pH of pure water shows. The equilibrium in this reaction lies far to the left. Nevertheless we can write equilibrium constant.

$$K_c = \frac{[H^+][OH^-]}{[H_2O]}$$

$$[H_2O] \text{ in water} = \frac{1000}{18} = 55.56 \text{ mol dm}^{-3}. \text{ This is a constant.}$$

SIR OTONDE MICHAEL AWITI

Thus we can incorporate the constant for $[H_2O]$ in the value of K_c .

$$K_c[H_2O] = [H^+][OH^-] \text{ as a new constant } K_w.$$

This overall constant K_w is called the ionic product for water.

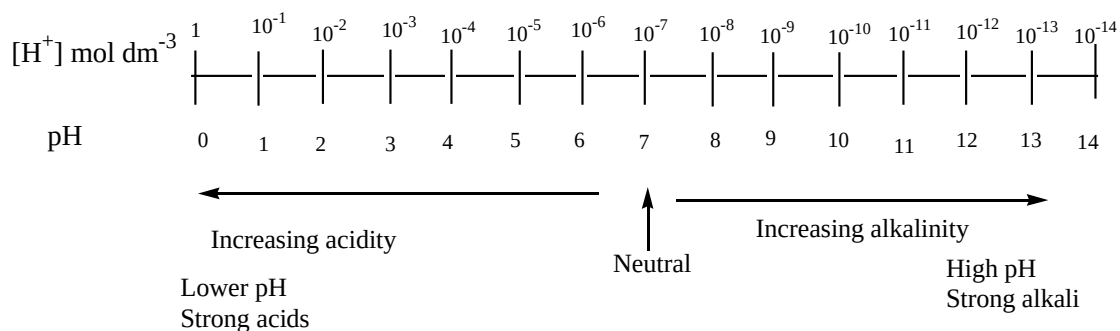
$$\text{At } 25^\circ\text{C } [H^+] = [OH^-] = 10^{-7}$$

$$\begin{aligned} \text{Hence } K_w &= [H^+][OH^-] = 10^{-7} \times 10^{-7} \\ &= 10^{-14} \text{ mol}^2 \text{ dm}^{-6} \end{aligned}$$



5.6 Important Generalization on Acids and Bases

- For neutral solutions $[H^+] = [OH^-] = 10^{-7} \text{ mol dm}^{-3}$
And $\text{pH} = 7$ at 25°C
- For acidic solution $[H^+] > [OH^-]$ and $\text{pH} < 7$
- For basic alkaline solution $[H^+] < [OH^-]$ and $\text{pH} > 7$





5.7 Dissociation Constant of Acids and Bases

In dealing with acids, K_c is usually based on a symbol K_a which is known as dissociation constant of the acid.

By eliminating aqueous state symbols, we can write



$$K_a = \frac{[H^+][A^-]}{[HA]}$$

SIR OTONDE
MICHAEL AWITI

The numerical value of K_a provides an accurate measure of the extent to which an acid is dissociated i.e. the strength of acid).

The stronger the acid, the greater the extent of dissociation, and hence the greater the K_a .

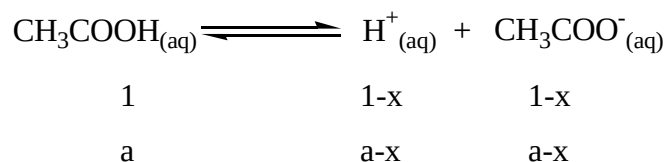
The value of K_a of some acids are given in the table below.

Acid	Equilibrium equation	K_a at 25 °C/ mol dm ³
H ₂ SO ₄	$H_{2}SO_{4(aq)} \rightleftharpoons H^+_{(aq)} + HSO_4^-_{(aq)}$	Very large
HNO ₃	$HNO_{3(aq)} \rightleftharpoons H^+_{(aq)} + NO_3^-_{(aq)}$	40
HCOOH	$HCOOH_{(aq)} \rightleftharpoons H^+_{(aq)} + HCOO^-_{(aq)}$	1.6×10^{-4}
C ₆ H ₅ COOH	$C_6H_5COOH_{(aq)} \rightleftharpoons H^+_{(aq)} + C_6H_5COO^-_{(aq)}$	6.4×10^{-5}
CH ₃ COOH	$CH_3COOH_{(aq)} \rightleftharpoons H^+_{(aq)} + CH_3COO^-_{(aq)}$	1.7×10^{-5}
H ₂ O	$H_2O_{(aq)} \rightleftharpoons H^+_{(aq)} + OH^-_{(aq)}$	1.8×10^{-6}

SIR OTONDE MICHAEL AWITI

Worked example

The pH of 0.01 mol dm⁻³ of ethanoic acid (acetic acid), CH₃COOH is 3.40 at 25 °C. What is the dissociation constant of ethanoic acid at this temperature?



$$K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$

Since the concentration of hydrogen ions arising from water is much smaller than the concentration of those from acid, we can say that;

$$[\text{H}^+] = [\text{CH}_3\text{COO}^-]$$

And

$$[\text{CH}_3\text{COOH}] = 0.01 - [\text{H}^+] \approx 0.01$$

If $[\text{H}^+] \ll 0.01$, which it can be since ethanoic acid is a weak acid

$$\text{pH} = -\lg[\text{H}^+_{(\text{aq})}] = 3.40$$

$$\text{Hence } \lg[\text{H}^+_{(\text{aq})}] = -3.40 = -(4.00 + 0.6)$$

$$\text{Therefore } [\text{H}^+] = 4 \times 10^{-4} \text{ mol dm}^{-3}$$

$$\text{And } [\text{CH}_3\text{COO}^-] = 4 \times 10^{-4} \text{ mol dm}^{-3}$$

$$\text{Now assuming } [\text{CH}_3\text{COOH}] = 0.01 \text{ mol dm}^{-3}$$

$$K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = \frac{4 \times 10^{-4} \times 4 \times 10^{-4}}{0.01}$$

SIR OTONDE MICHAEL AWITI

$$K_a = 1.60 \times 10^{-5} \text{ mol dm}^{-3}$$

By reversing this calculation, it is possible to predict the pH of solution of a weak acid. We must, however, know the molarity of acid and its dissociation constant.



SELF-TEST QUESTIONS

- 1 What is the pH of 1.0 mol dm⁻³ of benzoic acid (C₆H₅COOH), given that $K_a(\text{C}_6\text{H}_5\text{COOH}) = 6.4 \times 10^{-5} \text{ mol dm}^{-3}$.

SIR OTONDE MICHAEL AWITI



5.8 Buffer Solutions

Are solutions that resist change in pH on addition of acid or alkali.

Or

A buffer solution change pH only slightly on the addition of small amount of either acid or a base.

A buffer solution requires two components, one which is able to neutralize acids, and the other able to neutralize bases. But, of course the two components must not neutralize each other. Common buffer solutions are described either as a mixture of:

- a weak acid and its conjugate base or
- a weak base and its conjugate acid.

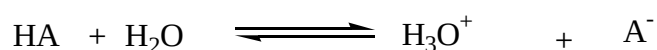
SIR OTONDE MICHAEL AWITI



5.8.1 An Equation for Buffer Solutions

The Hernderson-Hasslbebalch Equation

It is often useful to describe a buffer solution through an equation known as the Hernderson-Hasslbebalch Equation. This equation is commonly used by biochemists and molecular biologists. To derive this variation of the ionization constant expression, let's consider a mixture of a hupothetical weak acid, HA (such as CH_3COOH), and its salt, NaA (such as CH_3COONa). We start with the familiar expressions



$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

And rearrange the right side of the K_a expression to obtain

$$K_a = [\text{H}_3\text{O}^+] \frac{[\text{A}^-]}{[\text{HA}]}$$

Next, we take *negative logarithm* of each side of this equation.

$$-\lg K_a = -\lg[\text{H}_3\text{O}^+] - \lg \frac{[\text{A}^-]}{[\text{HA}]}$$

Now recalling that $pH = -\lg[\text{H}_3\text{O}^+]$ and $pK_a = -\lg K_a$

$$pK_a = pH - \lg \frac{[\text{A}^-]}{[\text{HA}]}$$

Then, we rearrange the equation to solve for pH

$$pH = pK_a + \lg \frac{[A^-]}{[HA]}$$

By recognizing the A^- is the conjugate base of the weak acid HA, we can write somewhat more general equation

$$pH = pK_a + \lg \frac{[conjugatebase]}{[acid]}$$

Example. What is the pH a buffer solution made from 0.1 M acetic acid and 0.1 M sodium acetate solution, given K_a (acetic acid) = 1.8×10^{-5} M)

Solution

$$pH = pK_a + \lg \frac{[conjugatebase]}{[acid]}$$

$$pH = -\lg(1.8 \times 10^{-5}) + \lg \frac{0.10}{0.10}$$

$$4.74$$

An important example of a buffered system is that found in blood, which needs to be maintained at a pH of 7.4.

SIR OTONDE MICHAEL AWITI



5.8.2 Calculating pH Changes in Buffer Solutions

An important example of a buffered system is that found in blood, which needs to be maintained at a pH of 7.4.

Addition of OH⁻ to a buffer

Consider what happens when we add H₃O⁺ or OH⁻ to a buffer solution. First, suppose that we add 0.01 mol of solid NaOH to 1.00 dm⁻³ of the 0.01 M acetic acid- 0.10 M sodium acetate solution. Because a neutralization reaction involving strong acids or strong bases goes essentially 100% to completion, we must take account of neutralization before calculating [H₃O⁺]. Initially, we have (1.00 dm⁻³)x(0.10 mol dm⁻³)=0.10 mol acetic acid and an equal amount of acetate ion. When we add 0.01 mol of NaOH, the neutralization reaction will alter the numbers of moles.

Neutralization reaction	$\text{CH}_3\text{COOH}_{(\text{aq})} + \text{OH}^{-}_{(\text{aq})} \longrightarrow \text{H}^{+}_{(\text{aq})} + \text{CH}_3\text{COO}^{-}_{(\text{aq})}$		
Before reaction (mol)	0.10	0.01	0.10
Change (mol)	-0.01	-0.01	+0.01
After reaction (mol)	0.09	≈0	0.11

If we assume that the solution volume remains constant at 1.00 dm⁻³, the concentrations of the buffer components after neutralization are

$$[\text{CH}_3\text{CO}_2\text{H}] = \frac{0.09\text{mol}}{1.00\text{dm}^{-3}} = 0.09\text{M}$$

$$[\text{CH}_3\text{CO}_2^{-}] = \frac{0.11\text{mol}}{1.00\text{dm}^{-3}} = 0.11\text{M}$$

Substituting these concentrations into the expression for [H₃O⁺], we can then calculate the pH:

$$[\text{H}_3\text{O}^{+}] = K_a \frac{[\text{CH}_3\text{CO}_2\text{H}]}{[\text{CH}_3\text{CO}_2^{-}]}$$

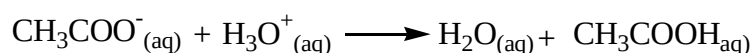
$$= 1.8 \times 10^{-5} \frac{0.09}{0.11} = 1.5 \times 10^{-5}$$

$$\text{pH} = 4.82$$

Adding 0.11 mol of NaOH changes $[\text{H}_3\text{O}^+]$ by only a small amount because the concentration ratio $[\text{weak acid}]/[\text{conjugate base}]$ changes by only a small amount, from unity to 9/11. The corresponding change in pH, from 4.74 to 4.82, is only 0.08 pH unit

Addition of H_3O^+ to a buffer

Now suppose that we add 0.01 mol of HCl to 1.00 dm³ of 0.10 M acetic acid – 0.10 M sodium acetate buffer solution. The added strong acid will convert 0.01 mol of acetate ions to 0.01 mol of acetic acid because of the neutralization



The concentrations after neutralization will be $[\text{CH}_3\text{COOH}] = 0.11 \text{ M}$ and $[\text{CH}_3\text{COO}^-] = 0.09 \text{ M}$, and the pH of the solution will be 4.66

$$[\text{H}_3\text{O}^+] = K_a \frac{[\text{CH}_3\text{CO}_2\text{H}]}{[\text{CH}_3\text{CO}_2^-]}$$

$$= 1.8 \times 10^{-5} \frac{0.11}{0.09} = 2.2 \times 10^{-5}$$

$$\text{pH} = 4.66$$

Again, the change in pH, from 4.74 to 4.66, is small because the concentration ratio $[\text{weak acid}]/[\text{conjugate base}]$ remains close to its original value



SELF-TEST QUESTIONS

- i) Explain what is meant by the strength of an acid.
- ii) What are the Lewis definitions of an acid and a base? In what way are they more general than the Brønsted definitions?



5 SUMMARY

- 1 Brønsted acids donate protons and Brønsted bases accept protons. Every Brønsted acid has a conjugate Brønsted base and vice versa.
- 2 The acidity of an aqueous solution is expressed as its pH, which is defined as the negative logarithm of the hydrogen ion concentration (in moles per litre). At 25 °C, an acidic solution has $\text{pH} < 7$, a basic solution has $\text{pH} > 7$, and a neutral solution has $\text{pH} = 7$.
- 3 Lewis acids accept pairs of electrons and Lewis bases donate pairs of electrons to form covalent bond.
- 4 A buffer solution is a combination of a weak acid and its weak conjugate; the solution reacts with small amounts of added acid or base in such a way that the pH of solution remains nearly constant. Buffer systems play a vital role in maintaining the pH of body fluids.

Lecture 6



Lecture 6 Objectives or Aims

At the end of this unit you should be able to;

- (a) discuss the nuclear chemistry
- (b) describe natural and artificial radioactivity
- (c) balance nuclear equations
- (d) describe and determine rate of radioactive decay and half-life
- (e) state uses of some radioactive isotopes



6.0 Introduction

So far we have studied chemical reactions. These originate from the electrons. We will now discuss another phenomenon that originates within the nuclei of atoms. These phenomena are referred to as nuclear chemistry.

SIR OTONDE MICHAEL AWITI



6.0 Radioactivity

Radioactivity or radioactive decay- is the spontaneous decay of the nuclei of certain atoms, accompanied by the emission of sub-atomic particles and / or high frequency electromagnetic radiation.



6.1 Characteristic of Radioactivity

All nuclei with atomic number greater than 83 are unstable and radioactive. They decompose spontaneously with emission of radiation and energy. A radioactive change is not a chemical reaction; it is a nuclear reaction that originates from the nuclei.

The decay product, if unstable will decompose further till a stable non-radioactive nuclear is formed. All radioactive series found in nature give final stable isotopes ${}_{82}^{206}\text{Pb}$, ${}_{82}^{207}\text{Pb}$, and

${}_{82}^{208}\text{Pb}$

SIR OTONDE MICHAEL AWITI



6.2 Types of Radiation

Three types of radiations emitted by radioactive substance are: alpha particles (α -particles), beta particles (β -particles) and gamma rays (γ -rays).

Alpha particles (α)

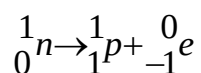
Alpha particles are positively charged Helium ions, He^{2+} i.e. Helium atom without electrons, i.e. Helium nuclei containing two protons and two neutrons but no electrons.

They are normally represented as ${}^4_2\text{He}$ or just α .

Beta particles (β)

Beta particles are negatively charged fast moving electrons. They have much smaller mass than alpha-particles. Since they are electrons, they are represented as ${}^0_{-1}e$. A beta particle is not an electron outside a nucleus, but one formed when a neutron changes to a proton within a nucleus.

SIR OTONDE MICHAEL AWITI



Gamma rays (γ)

They are not material particles but high energy electromagnetic wave. They carry no charge.

SIR OTONDE MICHAEL AWITI



SELF-TEST QUESTIONS

- i) How do nuclear reactions differ from ordinary chemical reactions?
- ii) Define nuclear fission, nuclear chain reaction and critical mass
- iii) Explain how an atomic bomb work
- iv) Explain how archeologists are able to tell the age of a fossil?
- v) State the general rules for predicting nuclear stability
- vi) How does Einstein's equation, $E = mc^2$, allow us to calculate nuclear binding energy?



6.3 Effect of Radioactive Decay on Parent Nuclide

Radioactive decay of α - and β -particles can change the number of protons and neutrons in the nuclei of an atom.



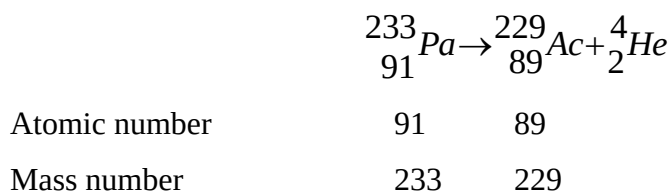
6.3.1 α -Particle Decay

When an α -particle (${}_2^4He$) is emitted from nucleus both the atomic number and proton number of the nucleus decrease. A new nuclide is formed.

SIR OTONDE MICHAEL AWITI

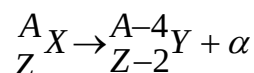
Since the α -particle consists of two protons and two neutrons the atomic number is lowered by 2 and mass number by 4 units.

Consider the nucleus of protactinium which emits α -particle to give a nuclide of actinium. This is represented by the following equation.

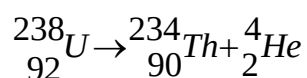


Results: Atomic number decrease by 2, mass number decrease by 4 when α -particle is emitted.

In general



For example



6.3.2 β -Particle Decay

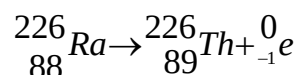
SIR OTONDE
MICHAEL AWITI

When β -particle (${}_{-1}^0e$) is emitted the atomic number increase by 1 and mass number remain constant. This is because:

- During the emission, a neutron in the nucleus spontaneously changes to a proton and emit an electron (increasing the positive charge).
- The mass number remains constant because the mass of electron loss is negligible.

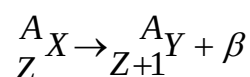
SIR OTONDE MICHAEL AWITI

Consider the nuclide of radium, which emit β -particle to give a nuclide of actinium.

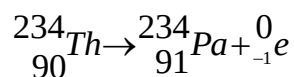


Results: Atomic number increase by 1, mass number remains constant when β -particle is emitted

In general



For example



SIR OTONDE MICHAEL AWITI



6.3.3 γ -Particle Decay

The emission of gamma ray always accompanies other radioactive emissions. They are produced when the remaining particles in the nuclei reorganizes. Generally, the gamma rays are not shown when writing nuclei equation, because, they have no effect on mass number and atomic number of the nuclide.

SIR OTONDE MICHAEL AWITI



6.4 Properties of Radiation

1 Penetrating Power

- A-particles lowest; can be stopped a paper.
- B-particles low; can be stopped a metal foil.
- γ -ray highest can only be stopped a thick lead metal.

SIR OTONDE
MICHAEL AWITI

2 Deflection in an electric field

- A-particles deflected slightly toward the negative plate
- B-particles deflected strongly toward the positive plate
- γ -rays not deflected

3 Effect of magnetic field

- A-particles deflected slightly
- B-particles deflected strongly
- γ -rays not deflected

SIR OTONDE MICHAEL
AWITI

4 Ionizing effect of radiation

- A-particles Highly ionizing
- B-particles weakly ionizing
- γ -rays do not ionize



6.5 Rate of Radioactive Decay

Radioactive decay is a random process. It is independent of temperature and pressure. However, the probability to decay in any given interval time is the same for atoms present.

This means that the rate of radioactive decay is directly proportional to the number of atoms present i.e. equal fraction of radioactive atom disintegrate in equal times. The decay rate decreases with time.

This can be expressed mathematically:

Where M is the mass remaining at time t and

M_0 is the mass at $t=0$

k is the Boltzmann constant

SIR OTONDE
MICHAEL AWITI

$$\frac{dM}{dt} \propto -M$$

$$\frac{dM}{dt} = -kM$$

$$\int_{M_0}^M \frac{dM}{M} = - \int_{t=0}^t k dt$$

$$\frac{M}{M_0} = e^{-kt} \quad \text{or} \quad \frac{M}{M_0} = \exp(-kt)$$

$$M = M_0 e^{-kt} \quad \text{or} \quad M = M_0 \exp(-kt)$$

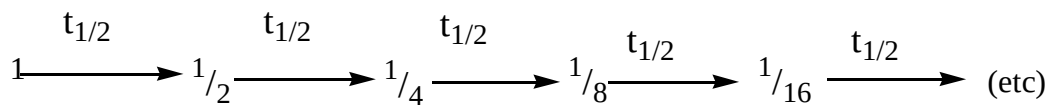


6.6 Half-Life, $t_{1/2}$

The time required to reduce the original number of radioactive atom to half its value or the time taken for half the mass of radioactive substance to decay.

SIR OTONDE MICHAEL AWITI

The fraction of the original sample which remain after successful half-life:



Expressed mathematically

$$\text{From } M = M_0 e^{-kt} \quad \text{or} \quad M = M_0 \exp(-kt)$$

At half-life

$$\text{At } t_{1/2} \quad M = \frac{M_0}{2}$$

$$\frac{M_0}{2} = M_0 e^{-kt}$$

SIR OTONDE
MICHAEL AWITI

$$\frac{1}{2} = e^{-kt}$$

$$2^{-1} = e^{-kt}$$

$$\ln 2 = kt_{1/2}$$

$$\frac{\ln 2}{k} = t_{1/2}$$

Each radioactive material has its own half-life and value ranges from a small fraction of second to thousands of years.



SELF-TEST QUESTIONS

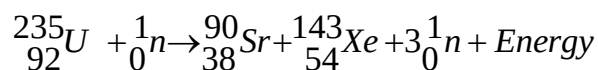
- 1 Half-life of a radioactive element is 30 days. Calculate the time required for its activity measured with G. M counter to drop from 4800 counts per minute to 300 counts per minute (ignore background counts).
- 2 State three uses of radioactive isotopes

SIR OTONDE MICHAEL AWITI



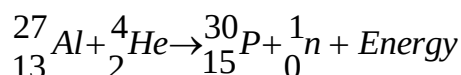
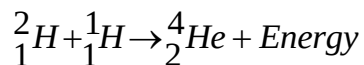
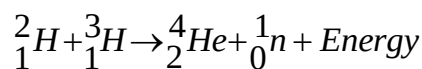
6.7.1 Nucleus Fission

In *nuclear fission* a heavy nucleus (mass number > 200) divides to form smaller nuclei of intermediate mass and one or more neutrons. For example:



6.7.2 Nucleus Fusion

In contrast to the nuclear fission, *nuclear fusion*, is the process of combining light nuclei into heavier. For example:



6.7.3 Uses of Radio Isotopes

Radioactive isotopes have many applications in science and medicine. They are used in carbon -14 dating, and determination of molecular structures. They are used to trace the path of atoms of an element in a chemical reaction or biological processes, are called, **tracers**. They are used in medical field. It must also be noted that high-energy damages living systems by causing ionization and the formation of free radical.



6 Summary

- 1 Nuclear chemistry is the study of changes in the atomic nuclei. Such changes are termed nuclear reactions.
- 2 Radioactive nuclei emit α -particles, β -particles and γ -rays.
- 3 Nuclear fission is the splitting of a large nucleus into two smaller nuclei plus neutrons.
- 4 Nuclear fusion is the combination of two light nuclei to form one heavy nucleus.

Lecture 7

7.0 Redox Reactions



Lecture 7 Objectives or aims

At the end of this unit you should be able to;

- (a) describe the redox reactions
- (b) balance a redox reactions



7.0 Introduction

SIR OTONDE MICHAEL AWITI

In high school, you learnt how to balance chemical equations. Some of those equations were redox reactions that are relatively easy to balance. However, in laboratory we often encounter more complex redox reactions, reactions that invariably involve oxoanions such as chromate (CrO_4^{2-}), dichromate ($\text{Cr}_2\text{O}_7^{2-}$), permanganate (MnO_4^-), nitrate (NO_3^-) and sulphate (SO_4^{2-}). In principle we can balance any redox equation using the basic procedures you learnt in high school, but there are some special techniques for handling redox reactions, techniques that also give us insight into electron transfer processes. In this lecture we will discuss two system approach often used for balancing redox reaction: the oxidation number method and the half-reaction method.

The next section define the redox reactions



7.1 Redox Reaction

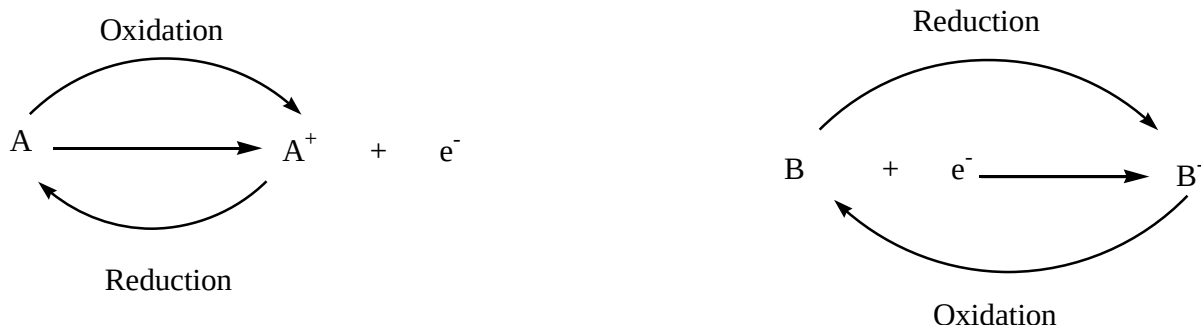
Oxidation is defined as loss of one or more electron(s) by a substance, element, compound or ion

Whereas

SIR OTONDE MICHAEL AWITI

Reduction is defined as gain of one or more electron(s) by a substance, element, compound or ion.

Thus an oxidation – reduction reaction or REDOX reaction is a process in which electrons are transferred from one substance to another.



7.2 How Can We Tell When Redox Reaction is taking Place?

We assign to atom a substance a value called oxidation number or oxidation state, which provides a measure of an atom is neutral, electron rich or electron poor. By comparing oxidation number of an atom before and after reaction, we can tell whether atom has gained or lost one or more electron(s).

NB/ Oxidation number does not necessarily imply ionic charges. They are simply convenient device to help us keep track of electrons in redox reaction. Whenever one substance lose one or more electron(s) (oxidized) another substance must gain electron(s) (be reduced).

Reducing agent

Reducing agent loses one or more electron(s).

- undergo oxidation or
- Oxidation number of atom increase.

Oxidizing agent

Reducing agent gains one or more electron(s).

- undergo reduction or
- Oxidation number of atom decrease.

SIR OTONDE
MICHAEL AWITI

SIR OTONDE MICHAEL AWITI



7.3 Balancing Redox Reaction

There are two system approach often used for balancing redox reaction:

- I. The oxidation number method.
- II. The half-reaction method.



7.3.1 Balancing Redox Reaction by Oxidation Number Method

The key to the oxidation-number method of balancing redox equation is to realize that the net charge in the total of all oxidation number must be zero. That is any increase in oxidation number for the oxidized atoms must be exactly matched by a corresponding decrease in oxidation number of the reduced atoms.

Steps used in balancing a redox reaction by the oxidation number method

1. Write the unbalanced ionic equation.
2. Balance the equation for all atoms other than H and O.
3. Assign oxidation number to all atoms.
4. Decide which atom have changed oxidation number and by how much.
5. Find the total net increase in oxidation number for the oxidized atoms and the total net decrease in oxidation number for the reduced atoms. Multiply the net increase and the net decrease by a suitable number so that the two values become equal.
6. Balance the equation for oxygen by adding H_2O to the side with less O and then balance for hydrogen by adding H^+ to the side with less H. Check your answer by making sure that the equation is balanced both for atom and for charge.

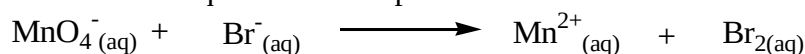
Total increase in oxidation number (oxidation)	=Total decrease in oxidation number (Reduction)	Is equivalent as saying that the total loss and gain of electron(s) are equal in redox reaction
--	---	---

Worked example

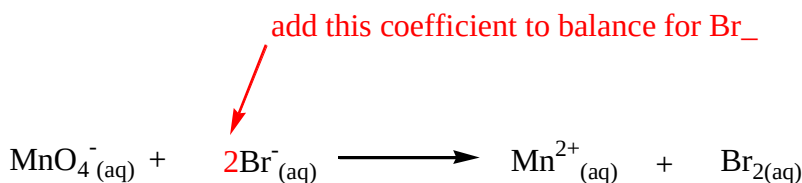
Balance the following redox reaction.

An aqueous acidic solution of the purple permanganate anion MnO_4^- is reduced by Br^- to yield the pale pink Mn^{2+} ion along with Br_2 .

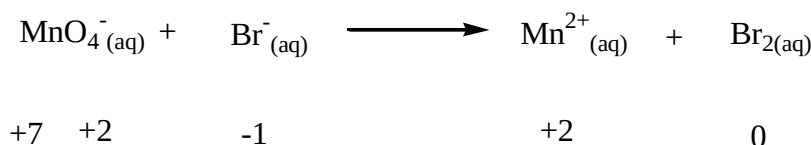
- 1 The unbalanced ionic equation for the process is:



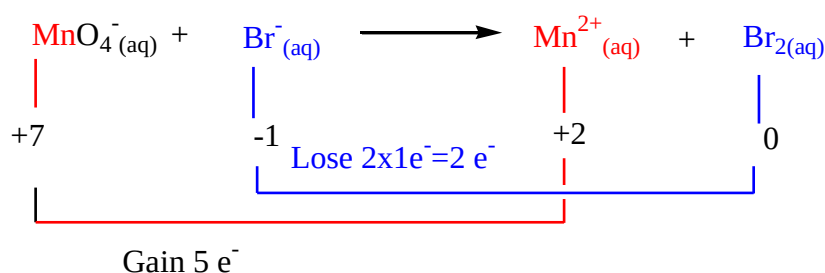
- 2 Balance the equation for all atoms other than H and O. Add a coefficient 2 for Br^- to



3 Assign oxidation numbers



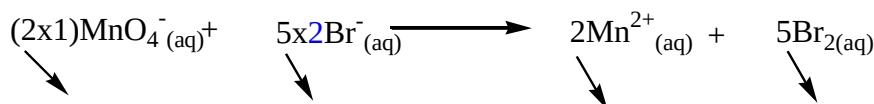
4 Decide which atom have changed oxidation number and by how much



5

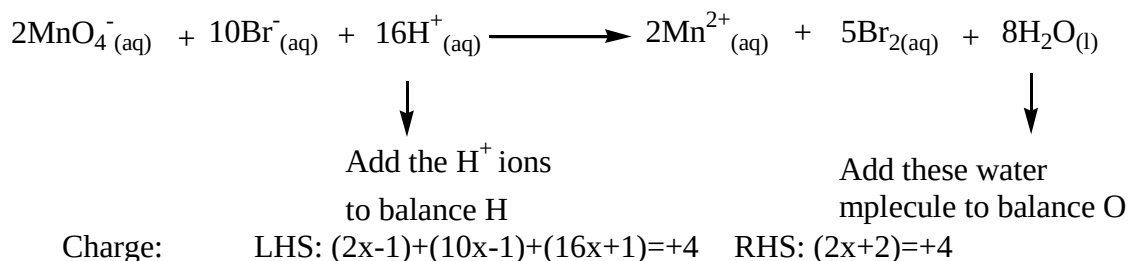
Increase in oxidation numbers $2 \times (\text{Br}^{-1} \rightarrow \text{Br}^0)$ net increase +2.

Decrease in oxidation numbers $2 \times (\text{Mn}^{+7} \rightarrow \text{Mn}^{+2})$ net decrease -5.
LCM of 2 and 5=10



Multiply by these coefficient to make the net increase and net decrease in oxidation number equal

6 Balance for O by adding H₂O to the side with less O and then balance it by adding H⁺ to the side with less H.



If everything has been done correctly, the final net ionic equation will result. The answer can be checked by noting that the equation is balanced both for atom and for charge with +\$ on both sides.



7.3.2 Balancing Redox Reaction the Half-Equation Method

The key to this method is to realize that the overall reaction can be broken into two parts or half-reaction. One half-reaction describes the oxidation part of the process and the other half-reaction describes the reduction part. Each half is balanced separately and the two halves are then added to obtain the final equation.

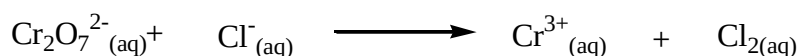
Steps used in balancing a redox reaction by half-reaction method

1. Write the unbalanced ionic equation
2. Decide which atoms are oxidized and which are reduced and write two unbalanced half-reaction for the separate process.
3. Balance both half-reaction for all atoms other than H and O.
4. Balance both half-reaction for O by adding H_2O to the side with less O and then balance for H by adding H^+ to the side with less H.
5. Balance both half-reactions for charge by adding electrons (e^-) to the side with the greatest positive charge. Then multiply the half-reactions by suitable number to make the electrons count the same in both.
6. Add the two balanced half-reaction and cancel the electrons and any other species that appears on both side of the equation. Check to see that both atoms and charge balance and that the coefficients are reduced to simplest whole number.

Worked example

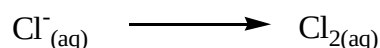
Let's look at the reaction of aqueous potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) with aqueous sodium chloride to see how this method works. The reaction occurs in acidic solution according to the unbalanced equation:

1

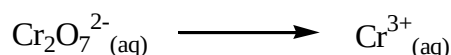


2 The first step is to decide which atoms have been oxidized and which have been reduced. In the present case, the chloride atom is oxidized (from -1 to 0), and the chromium atom is reduced (from +6 to +3). Thus, we can write two unbalanced half-reactions that show the separate steps:

Oxidation half-reaction

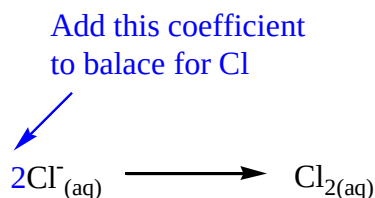


Reduction half-reaction

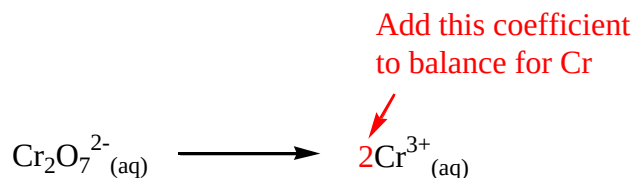


3 With the two half-reactions identified, each is balanced separately. Begin by balancing for all atoms other than H and O. The oxidation half-reaction needs a coefficient of 2 before the Cl^- , and the reduction half-reaction needs a coefficient of 2 before the Cr^{3+} .

Oxidation

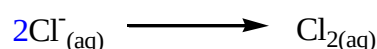


Reduction

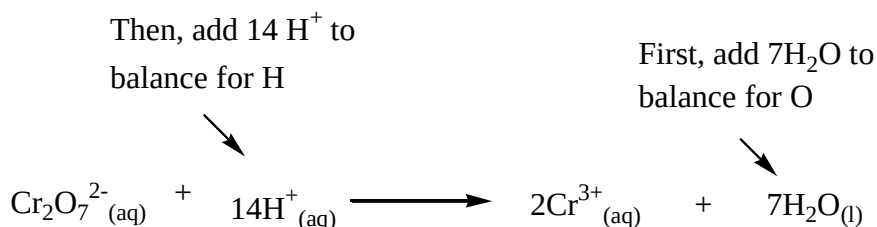


4 Next, balance both half-reactions for oxygen by adding H_2O to the side with less O, and balance for hydrogen by adding H^+ to the side with less H. The oxidation half-reaction has no H or O, but the reduction reaction needs $7\text{H}_2\text{O}$ on the product side to balance for O and 14H^+ on the reactant side to balance for H:

Oxidation

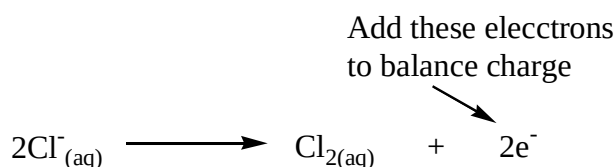


Reduction

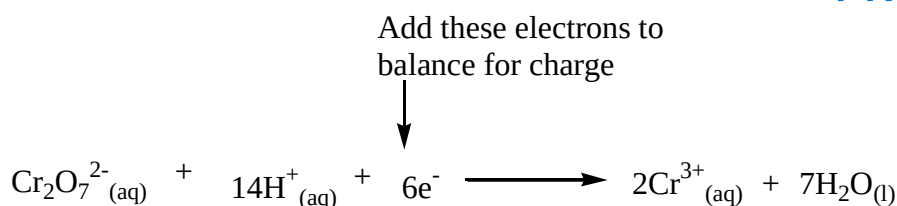


5 Now, balance both half-reactions for charge by adding electrons (e^-) to the side with the greater positive charge. The oxidation half-reaction must have 2e^- added to the *product* side, and the reduction half-reaction must have 6e^- added to the *reactant* side:

Oxidation



Reduction



SIR OTONDE
MICHAEL
AWITI

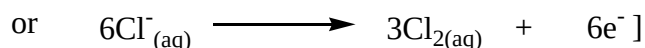
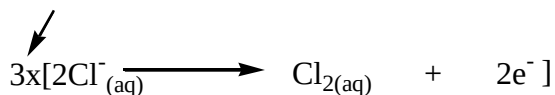
With both half-reactions now balanced, we need to multiply by suitable coefficient so that the number of electrons released in the oxidation half-reaction is the same as the number

SIR OTONDE MICHAEL AWITI

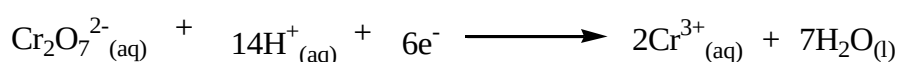
consumed in the reduction half-reaction. Since the reduction half-reaction has 6 e⁻ but oxidation half-reaction has only 2 e⁻, the entire oxidation half-reaction must be multiplied by 3.

Oxidation

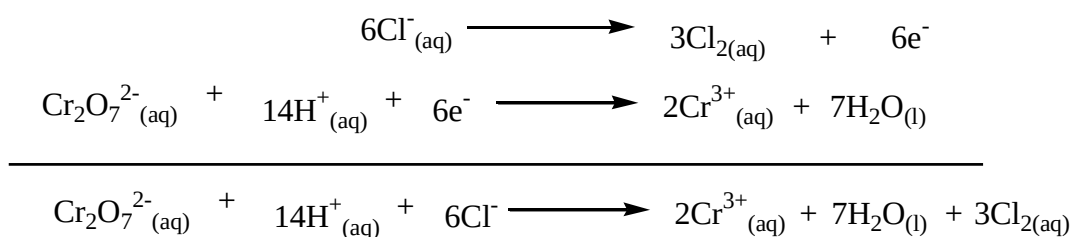
Multiply by this coefficient to equilibrate the number of electrons in the two half-reactions



Reduction



6 Adding the two half-reactions together, and canceling the species that occur on both sides (only electrons in this example), then gives the final balanced equation. Check the answer to make sure it is balanced both for atoms and for charges



Charge: $(-2) + (+14) + (6 \times -1) = +6$

Charge: $(2 \times +3) = +6$

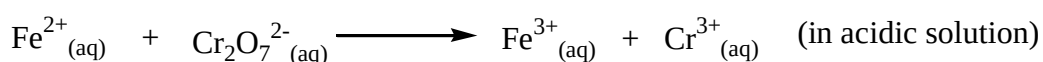
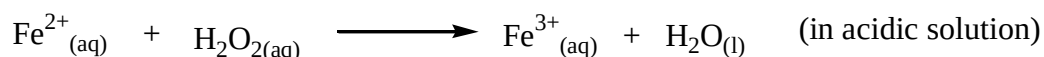
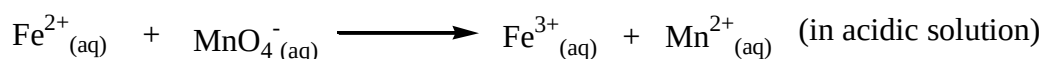
Study questions

- Assign the oxidation numbers of the underlined elements
a) $\underline{\text{C}}\text{IO}_3^-$ b) $\underline{\text{S}}\text{O}_3^{2-}$ c) $\underline{\text{Mn}}_2\text{O}_7$ d) $\underline{\text{Os}}\text{O}_4$
- Balance the following equations:

SIR OTONDE
MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

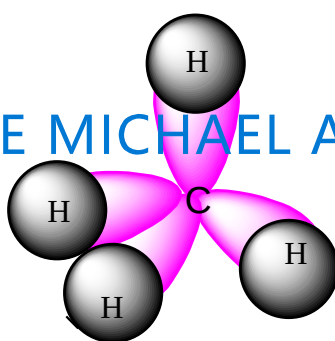
SIR OTONDE MICHAEL AWITI



7 Summary

- 1 There are two related definitions of redox, involving electron transfer, the other involving the concept of oxidation number.
- 2 In terms of electron transfer, oxidation is defined as a loss of electrons, and reduction is defined as a gain of electrons.
- 3 Electron-transfer reactions are called redox reactions.
- 4 An oxidation number is a number assigned to an atom or ion to describe its relative state of oxidation or reduction state. Using oxidation numbers it is possible to decide whether redox has occurred.
- 5 In terms of oxidation numbers, oxidation is defined as an increase in oxidation number and reduction is defined as a decrease in oxidation number.
- 6 There are two system approach often used for balancing redox reaction, the oxidation number method and the half-reaction method.

SIR OTONDE MICHAEL AWITI



SIR OTONDE MICHAEL AWITI